We have already discussed in section 6.3 that I_B is composed of three components, namely I_{pE}, I_{pC} and I_{BR}. We shall assume that the bipolar transistor has emitter efficiency almost equal to unity, and I_{pE} is negligibly small. Thus, when I_B is held constant, we can write

$$I_B \approx I_{BR} + I_{pC} = \frac{Q_{nB}}{\tau_n} + \frac{AqD_{pC}n_i^2 \coth\left(\frac{W_C}{L_{pC}}\right)}{L_{pC}N_{DC}}\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right] = \text{constant}$$

Therefore, if the CB junction is reverse biased (that is, $V_{BC} \leq 0$), the second term is almost constant and hence Q_{nB}, which is given by the area under the minority carrier concentration curve in Fig. 6.14(b), is also almost constant. On the other hand, increasingly forward biasing the CB junction increases the value of the second term in this expression for I_B and hence results in a reduction in Q_{nB}. This is reflected in the plots in Fig. 6.14(b).

This section further explains the output characteristic curves and the minority carrier concentration plots as shown in Fig. 6.14, where V_{CE} is varied in steps while keeping I_B constant.

Step 1: $V_{CE} = 0$.

This implies that $V_{BC} = V_{BE}$. Also now $n'_{pE} = n'_{pC}$. Although, it would seem that $I_{nC} = 0$, this is not the case. This is because the variation in the electron concentration in the base is not linear, as shown in plot 1 in Fig. 6.14(b). The slope of the electron concentration at $x = W_B$ implies that in addition to electrons being injected from the emitter, electrons are also injected from collector to base. Also, holes are injected from the base to the collector. Therefore, I_C is negative as shown by point 1 in Fig. 6.14(a). (Note that it is not possible to have a flat electron concentration curve in the base when $n'_{pE} = n'_{pC}$. If the distribution were flat, it would mean that there is no electron injection either from emitter or collector. Since there is a large quantity of excess charge in the base, electrons are constantly recombining. With no supply of electrons, it is not possible to maintain the steady state condition of excess stored charges in the base.)

Step 2: $V_{CE} = V_T \ln\left(\frac{1}{\alpha_R}\right)$

As already discussed in HELP DESK 6.5 and Example 6.2, the collector current is zero at this point [plot 2 in Fig. 6.14(b) and point 2 in Fig. 6.14(a)]. This implies that the electron current is exactly equal and opposite to the hole current at the BC junction, that is, $|I_{nC}| - |I_{pC}| = 0$. The situation in the collector is similar to the example in section 3.7.

HELP DESK 6.6

When a transistor is in saturation and I_B is fixed, show that for small values of V_{CE} (< 0.2 V), V_{BE} increases and V_{BC} reduces with increase in V_{CE} while for larger values of V_{CE}, V_{BE} remains almost constant and the increase in V_{CE} is equal to the reduction in V_{BC}.

When I_B is fixed at a constant value (say I_{B1}), from Eq. (6.64), we have

$$I_{B1} = \frac{I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right]}{\beta_F} + \frac{I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right]}{\beta_R} \approx \frac{I_S}{\beta_F}\exp\left(\frac{V_{BE}}{V_T}\right)\left[1 + \frac{\beta_F}{\beta_R}\exp\left(-\frac{V_{CE}}{V_T}\right)\right] \quad \text{(A)}$$

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In Eq. (A), as V_{CE} is increased, $\left[1 + \frac{\beta_F}{\beta_R}\exp\left(-\frac{V_{CE}}{V_T}\right)\right]$ reduces, therefore $\exp\left(\frac{V_{BE}}{V_T}\right)$, and consequently V_{BE} has to increase to keep their product a constant. However, when $V_{CE} > 0.2$ V, $\left[1 + \frac{\beta_F}{\beta_R}\exp\left(-\frac{V_{CE}}{V_T}\right)\right] \approx 1$. Therefore from Eq. (A), V_{BE} becomes almost constant. Equation (A) can also be rewritten as

$$I_{B1} = \frac{I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right]}{\beta_F} + \frac{I_S\left[\exp\left(\frac{V_{BC}}{V_T}\right) - 1\right]}{\beta_R} \approx \frac{I_S}{\beta_R}\exp\left(\frac{V_{BC}}{V_T}\right)\left[1 + \frac{\beta_R}{\beta_F}\exp\left(\frac{V_{CE}}{V_T}\right)\right] \quad \text{(B)}$$

In Eq. (B), $\left[1 + \frac{\beta_R}{\beta_F}\exp\left(\frac{V_{CE}}{V_T}\right)\right]$ increases with increase in V_{CE}. Therefore, $\exp\left(\frac{V_{BC}}{V_T}\right)$ and consequently V_{BC} must reduce to keep their product constant. Since V_{BE} becomes constant for $V_{CE} > 0.2$ V, and $V_{BC} = V_{BE} - V_{CE}$, any increase in V_{CE} results in an equal reduction in V_{BC}.

Step 3: $V_T \ln\left(\frac{1}{\alpha_R}\right) < V_{CE} < V_{BE}$.

The condition is depicted by point 3 and plot 3 in Figs. 6.14(a) and (b) respectively. Since $V_{BC} = V_{BE} - V_{CE}$, and $V_{BE} > V_{CE}$, V_{BC} is positive, implying a forward biased BC junction. Also, since $V_{BE} > V_{BC}$, $n'_{pE} > n'_{pC}$. The slope of the minority carrier concentration distribution curve in the base depends on $n'_{pE} - n'_{pC}$. Now,

$$(n'_{pE} - n'_{pC}) \propto \exp\left(\frac{V_{BE}}{V_T}\right) - \exp\left(\frac{V_{BC}}{V_T}\right) = \exp\left(\frac{V_{BE}}{V_T}\right)\left[1 - \exp\left(-\frac{V_{CE}}{V_T}\right)\right]$$

We have already seen in HELP DESK 6.6 that for small positive values of V_{CE}, V_{BE} increases with increasing V_{CE}. Thus, the slope of the electron concentration curve increases with increase in V_{CE}, resulting in an increased $|I_{nC}|$. From HELP DESK 6.6, we have also seen that the condition of constant I_B requires that in this range, V_{BC} is reduced with increasing V_{CE}, thereby reducing $|I_{pC}|$. As $I_C = |I_{nC}| - |I_{pC}|$, this would mean that I_C increases continuously with increase in V_{CE}. As discussed in HELP DESK 6.6, beyond $V_{CE} = 0.2$ V, V_{BE} and consequently n'_{pE} are almost constant. Since $n'_{pC} << n'_{pE}$ in this range, the slope of the electron concentration and consequently $|I_{nC}|$ becomes almost constant. Also $|I_{pC}| << |I_{nC}|$, which implies that for $V_{CE} > 0.2$ V, I_C is almost constant.

Step 4: $V_{CE} = V_{BE}$.

Therefore, $V_{BC} = V_{BE} - V_{CE} = 0$ V and $n'_{pC} = 0$ as shown in plot 4 in Fig. 6.14(b). As discussed previously, the electron concentration in the base now shows a linear dependence and since I_{pC} is negligibly small, the collector current now has a value of $I_C = (AqD_n n'_{pE}/W_B)$ shown as point 4 in Fig. 6.14(a).

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Step 5: $V_{CE} > V_{BE}$.

Now V_{BC} is negative, implying that the BC junction is reverse biased. When V_{BC} is sufficiently negative, $n'_{pC} \approx -n_{p0}$ as shown in plot 5 in Fig. 6.14(b). The collector current reaches a maximum value of $I_C = [AqD_n (n'_{pE} + n_{p0})/W_B]$ (refer to point 5 in Fig. 6.14(a)) and does not change any further with increase in V_{CE}. Since $n'_{pE} >> n_{p0}$, the increase in current from point 4 to point 5 in Fig. 6.14(a) is very small. (In real transistors, however, there is an increase in I_C due to "Early Effect," which is discussed in later section 6.8).

As in the case of the common-base configuration, the I_C versus V_{CE} characteristics obtained for a fixed value of I_B can also be replicated for various values of I_B as shown in Fig. 6.15. As expected, I_C increases with increasing I_B. The figure also shows the saturation, active and cut-off regions of operation. The region over which I_C remains almost constant for a particular I_B is the active region. In this region, $I_C = \beta I_B$. In the saturation region, I_C reduces with reducing V_{CE} so that $I_C < \beta I_B$. The region of the characteristics where $I_C < I_{CEO}$ represents the cut-off mode of operation.

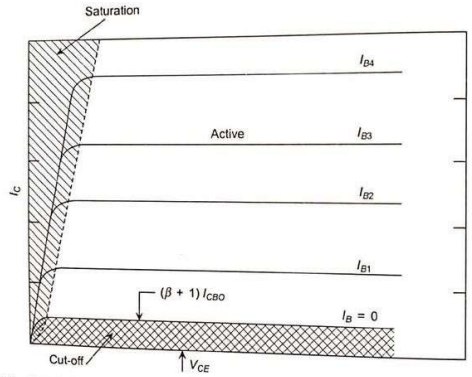


Figure 6.15 Output characteristics (I_C versus V_{CE}) for npn transistor in common-emitter configuration for various values of I_B.

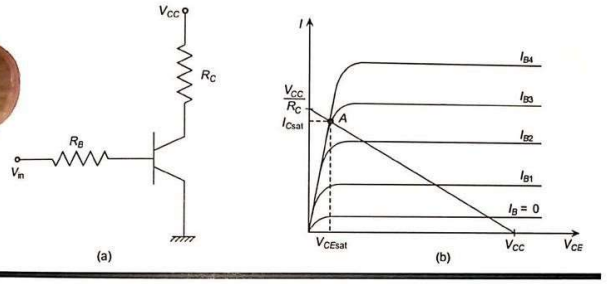
HELP DESK 6.7

Why is the term 'saturation' used to denote the region where I_C is actually less than its maximum possible value for a particular I_B?

To explain the origin of this term, let us consider the circuit shown below, where the transistor has been connected in the common-emitter configuration. In the given circuit, the input

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voltage is varied from 0 V to a large positive value, resulting in a corresponding increase in I_B. In order to obtain the operating point (I_C, V_{CE}) corresponding to a particular I_B, we have also shown the output characteristics together with a load line. As usual, the operating point is at the intersection of the load line with the characteristic curve for the particular I_B. We see that when the transistor is operating in the active region, $I_C = \beta I_B$, that is, I_C increases linearly with I_B. However, with further increase in I_B, the operating point reaches the saturation region of the characteristics (point A), corresponding to $I_C = I_{Csat}$ and $V_{CE} = V_{CEsat}$. Now any further increase in I_B does not result in any increase in I_C, and I_C has saturated at its maximum value. This is how the saturation region of the characteristics got its name.



6.8 EARLY EFFECT

For real transistors, there are some deviations from the characteristics predicted above by the simple analysis. In the common-base mode, for large reverse voltage applied across CB junction, I_C was predicted to be independent of V_{CB}, that is, $dI_C/dV_{CB} = 0$. In reality, I_C increases with an increase in V_{CB}. This is because as the reverse bias across CB junction increases, the depletion layer width also increases. This effectively reduces the neutral base width from W_B to W'_B. Hence I_C increases due to an increase in the slope of the minority carrier concentration as shown in Fig. 6.16. On the other hand, decrease in base width reduces I_B due to less recombination in the base. Consequently α increases with increase in V_{CB}. The actual output characteristics of an npn transistor in CB mode is shown in Fig. 6.17(a) reflecting this increase in I_C. For the output characteristics in the common-emitter configuration, the effect is more dominant as the collector current is seen to rise significantly with an increase in V_{CE}. This is shown in Fig. 6.17(b).

The modulation in the neutral base width with V_{CB} was first investigated by James Early and is therefore called *Early effect* after him. The dependence of I_C on V_{CB} can be modelled in the following way

$$\frac{\partial I_C}{\partial V_{CB}} = \frac{\partial I_C}{\partial W_B}\frac{\partial W_B}{\partial V_{CB}} \quad (6.68)$$

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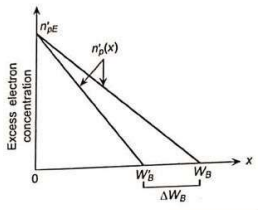


Figure 6.16 Change in the slope of the minority carrier concentration profile due to modulation of neutral base width with change in V_{CB}.

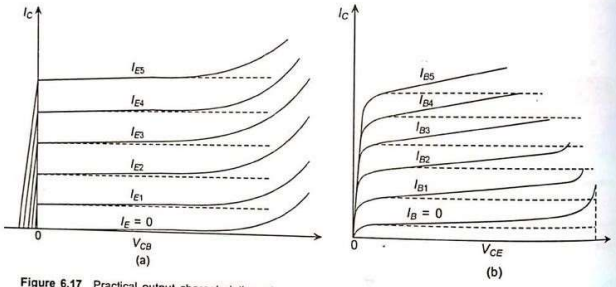


Figure 6.17 Practical output characteristics of an npn transistor in (a) common base and (b) common emitter configuration. The dashed lines show the ideal characteristics where the collector current is assumed to be constant in the active region.

Since for a large reverse bias applied across CB junction, I_C approximately equals $\alpha_F I_E$, the output conductance in common base mode (g_{cb}) with constant emitter current can be expressed as

$$g_{cb} = \left.\frac{dI_C}{dV_{CB}}\right|_{I_e = \text{constant}} = I_E \frac{\partial \alpha_F}{\partial W_B} \cdot \frac{\partial W_B}{\partial V_{CB}} \tag{6.69}$$

Assuming emitter efficiency $\gamma = 1$ and using Eq. (6.41) we can write

$$\alpha_F \approx \alpha_T \approx \left(1 + \frac{W_B^2}{2L_n^2}\right)^{-1}$$

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For small values of W_B/L_n, the common base output conductance g_{cb} can therefore be simplified as

$$|g_{cb}| \approx \frac{I_C W_B}{L_n^2} \frac{\partial W_B}{\partial V_{CB}} \tag{6.70}$$

For large values of V_{CE},

$$V_{CB} \approx V_{CE}$$

Therefore, the common emitter output conductance g_{ce} with the (input) base current constant can be expressed as

$$g_{ce} = \left.\frac{dI_C}{dV_{CE}}\right|_{I_B = \text{constant}} \approx \left.\frac{dI_C}{dV_{CB}}\right|_{I_B = \text{constant}} \tag{6.71}$$

When $\gamma = 1$, it can be shown that $\beta \approx (2L_n^2/W_B^2)$ [Refer to Eq. (6.45)]. Using this relation and $I_C \approx \beta I_B$, from Eq. (6.71), we get

$$|g_{ce}| = I_B \frac{\partial \beta}{\partial W_B} \frac{\partial W_B}{\partial V_{CB}} = \frac{2I_C}{W_B} \frac{\partial W_B}{\partial V_{CB}} \tag{6.72}$$

HELP DESK 6.8

Show that the output conductance in the common base configuration is smaller than that in the common emitter configuration.

From Eqs. (6.70) and (6.72), it can be easily seen that the ratio of g_{cb} to g_{ce} is given by

$$\frac{g_{cb}}{g_{ce}} = \frac{W_B}{L_n^2} \frac{W_B}{2} = \frac{W_B^2}{2L_n^2} \approx \frac{1}{\beta}$$

We know that usually $\beta >> 1$, therefore $g_{cb} << g_{ce}$.

From the above discussion, it can be seen that in common emitter mode, the collector current increases appreciably with V_{CE}, instead of staying constant. The Early effect is modelled in circuit simulators with the help of a parameter called the *Early Voltage* (V_A). Using Eq. (6.27), and assuming only a small change in base width, we may write

$$\frac{\partial I_C}{\partial V_{BC}} = \frac{\partial I_C}{\partial W_B} \frac{\partial W_B}{\partial V_{BC}} = -\frac{I_C}{W_B(0)} \frac{\partial W_B}{\partial V_{BC}} = -\frac{I_C}{V_A} \tag{6.73}$$

where $W_B(0)$ is the base width at $V_{BC} = 0$ V and

$$V_A = \left[\frac{1}{W_B(0)} \frac{\partial W_B}{\partial V_{BC}}\right]^{-1} \tag{6.74}$$

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Now assuming that in the active region V_{BE} is fixed for a constant I_B, from Eq. (6.73) we can write

$$\frac{\partial I_C}{\partial V_{CE}} = -\frac{\partial I_C}{\partial V_{BC}} = \frac{I_C}{V_A} \tag{6.75}$$

From Eq. (6.75), we see that if the common emitter characteristics in the active region were extrapolated, the intercept on the voltage axis would be at $-V_A$ (since V_A is usually much larger than the operating voltage V_{CE}) as shown in Fig. 6.18.

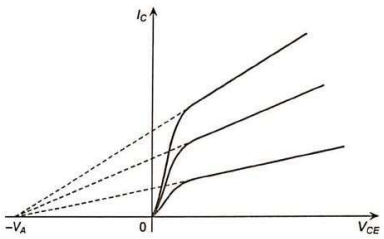


Figure 6.18 Graphical extrapolation of the *I–V* characteristics to extract the value of Early voltage (V_A).

Now, from Eq. (6.74), we have

$$\int_{W_B(0)}^{W_B} dW_B = \frac{W_B(0)}{V_A} \int_0^{V_{BC}} dV_{BC}$$

or

$$W_B = W_B(0)\left[1 + \frac{V_{BC}}{V_A}\right] \tag{6.76}$$

Now from Eq. (6.56), since I_S is inversely proportional to W_B, we have

$$I_S = \frac{I_S(0)}{1 + \frac{V_{BC}}{V_A}} \approx I_S(0)\left[1 - \frac{V_{BC}}{V_A}\right] \tag{6.77}$$

If $\gamma \approx 1$ for the bipolar transistor, so that β_F is determined by α_T, then $\beta_F \propto (1/W_B)^2$. In that case

$$\beta_F = \frac{\beta_F(0)}{\left(1 + \frac{V_{BC}}{V_A}\right)^2} \approx \beta_F(0)\left(1 - \frac{2V_{BC}}{V_A}\right) \tag{6.78}$$

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On the other hand, if $\alpha_T \approx 1$, so that β_F is determined by γ, then $\beta_F \propto (1/W_B)$. Therefore,

$$\beta_F = \frac{\beta_F(0)}{\left(1 + \frac{V_{BC}}{V_A}\right)} \approx \beta_F(0)\left(1 - \frac{V_{BC}}{V_A}\right) \tag{6.79}$$

Equations (6.77) and (6.78) or (6.79) can now be used in Ebers–Moll model to incorporate the Early effect.

6.9 LIMITATION ON THE JUNCTION VOLTAGE

If the reverse bias applied to the CB junction is continuously increased, the BJT ultimately 'breaks down' and the collector current increases sharply as shown in Fig. 6.19. This breakdown may either be due to base punch-through or avalanche multiplication at the CB junction.

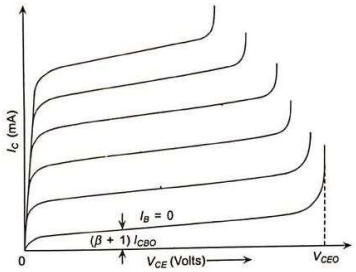


Figure 6.19 Common emitter output characteristics of a BJT depicting breakdown.

From the discussion in the previous section, it is evident that as V_{CB} is increased, the neutral base width decreases. Finally, for a particular value of V_{CB}, the depletion region at CB junction extends all the way up to EB junction and the neutral base width becomes zero. This is called *base punch-through*. At punch-through, the collector current increases sharply as the electrons are swept directly from the emitter into the collector and the transistor action is lost. This is a breakdown condition and needs to be avoided. For an n^+pn^+ transistor, assuming that the CB depletion region exists mostly in the base, the voltage applied across CB junction at punch-through can be approximated as

$$V_{PT} = \frac{qN_A W_B^2}{2\varepsilon_s} \tag{6.80}$$

In most cases, however, avalanche breakdown of the collector–base junction occurs before the punch-through condition is reached. Interestingly, it is found that the breakdown voltage in

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common emitter mode (V_{CEO}) is considerably smaller than that in the common base mode (V_{CBO}). Let us now analyze these two situations.

In the presence of a large reverse voltage applied across CB junction, the current flowing out of the collector–base depletion region is multiplied by the collector multiplication factor M. Thus, in such a situation the collector current given by Eq. (6.31) is modified as

$$I_C = (\alpha I_E + I_{CBO})M \tag{6.81}$$

As in Eq. (4.87), M can be expressed as

$$M = \frac{1}{1 - \left(\frac{V_{CB}}{V_{BR}}\right)^m} \tag{6.82}$$

where m is a constant.

Eliminating M from Eqs. (6.81) and (6.82), we get

$$V_{CB} = \left[1 - \frac{I_{CBO} + \alpha I_E}{I_C}\right]^{1/m} V_{BR} \tag{6.83}$$

From Eq. (6.83), we see that for the common base configuration, the maximum value of V_{CB} is obtained in the limiting case of $I_E = 0$. This value of voltage across the CB junction (V_{CBO}) with emitter open (that is, $I_E = 0$) is given as

$$V_{CBO} = \left[1 - \frac{I_{CBO}}{I_C}\right]^{1/m} V_{BR} \approx V_{BR} \tag{6.84}$$

However, for a nonzero I_E, the maximum voltage that can be applied across the CB junction without causing avalanche multiplication of carriers is less than V_{CBO}.

On the other hand, in the common emitter configuration, the current amplification in addition to the avalanche multiplication imposes additional restriction on the maximum voltage that can be applied across CB junction. It can be readily seen from Eq. (6.81) that if the input current (I_B) in common emitter configuration is made zero, so that $I_E = I_C$, we get

$$I_C = \frac{MI_{CBO}}{1 - \alpha M} \tag{6.85}$$

This means that I_C will increase indefinitely if $\alpha M = 1$. Since α is only slightly less than unity, for this condition to be satisfied it only needs M to be slightly greater than unity. By contrast, for the common base configuration, if the input current I_E is zero, M needs to be infinitely high for the collector current to increase in a similar manner. Thus, the maximum collector–emitter voltage that can be applied with base open (V_{CEO}) is much less than V_{CBO}.

Let us now obtain an expression for V_{CEO}. From Eq. (6.83), noting that $\alpha = \beta/(1 + \beta)$, we can write

$$V_{CB} = \left[1 - \frac{\beta}{1+\beta}\left(1 + \frac{I_B}{I_C}\right) - \frac{I_{CBO}}{I_C}\right]^{1/m} V_{BR} \tag{6.86}$$

For a large value of V_{CE}, V_{CB} is approximately equal to V_{CE}. Thus in the limiting case, for $I_B = 0$,

we get

$$V_{CEO} \approx \frac{V_{BR}}{(1 + \beta)^{1/m}} \tag{6.87}$$

Comparing Eqs. (6.84) and (6.87), we can easily verify that $V_{CEO} < V_{CBO}$.

Example 6.3

For a bipolar transistor $\beta = 80$. Assuming $m = 4$, find out the ratio of V_{CBO} to V_{CEO}.

Solution: Using Eq. (6.87), we get

$$V_{CEO} \approx \frac{V_{BR}}{\sqrt[4]{81}} = \frac{V_{BR}}{3}, \quad \text{while } V_{CBO} \approx V_{BR}$$

Therefore,

$$\frac{V_{CBO}}{V_{CEO}} = 3$$

In most bipolar transistors, the doping concentration in the collector is less than that in the base. This helps to prevent base punch-through and reduce Early effect, as the depletion width now extends more into the collector rather than into the base. Also, with a lower collector doping concentration, V_{BR} increases (refer to Eq. (4.88)), and therefore the device can withstand a larger collector bias voltage.

6.10 CAPACITANCES IN A BJT

Figure 6.9 shows the presence of stored charges in the various regions of the BJT. As in the case of diodes discussed in section 4.4.3, these charges give rise to a capacitive effect. We shall now develop a model for these capacitances.

Let us first consider $V_{BE} \neq 0$, $V_{BC} = 0$. For this condition, in Fig. 6.9, $Q_{BR} = Q_C = 0$, and the stored charges are given by Q_E and Q_{BF}. Now, from the areas of the shaded regions which represent the charges, we can write

$$Q_E = \frac{Aqp'_{nE}W_E}{2} = \frac{Aqn_i^2 W_E}{2N_{DE}}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \tag{6.88}$$

and

$$Q_{BF} = \frac{Aqn'_{pE}W_B}{2} = \frac{Aqn_i^2 W_B}{2N_A}\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \tag{6.89}$$

Now, considering the expression for I_C given by Eq. (6.55), we can write

$$Q_F = Q_E + Q_{BF} = \tau_F I_C \tag{6.90}$$

where τ_F is the *forward transit time* given by

$$\tau_F = \frac{\frac{W_E}{2N_{DE}} + \frac{W_B}{2N_A}}{\frac{D_n}{N_A W_B}} = \frac{N_A W_B W_E}{2N_{DE}D_n} + \frac{W_B^2}{2D_n} \tag{6.91}$$

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We can see that τ_F is a constant for a particular device structure and is independent of bias. We can now define the diffusion capacitance for the EB junction (C_{DE}) from Eqs. (6.90) and (6.55) as

$$C_{DE} = \frac{dQ_F}{dV_{BE}} = \tau_F \frac{dI_C}{dV_{BE}} \approx \tau_F \frac{d\left[I_S \exp\left(\frac{V_{BE}}{V_T}\right)\right]}{dV_{BE}} = \frac{\tau_F I_{CC}}{V_T} \tag{6.92}$$

Similarly, considering the reverse operation, that is, $V_{BE} = 0$, $V_{BC} \neq 0$, and following the same procedure as above, we can obtain an expression for the diffusion capacitance of the CB junction (C_{DC}) as

$$C_{DC} = \frac{\tau_R I_{EC}}{V_T} \tag{6.93}$$

where τ_R is the *reverse transit time* of the transistor.

In addition to the stored mobile charges in the neutral regions, which gives rise to the diffusion capacitances, there are the fixed charges in the EB and CB depletion regions, which gives rise to the junction capacitances. The expressions for the junction capacitances are similar to that for a diode given by Eq. (4.31). Therefore, the expressions for the EB and CB junction capacitances denoted by C_{JE} and C_{JC} respectively are given by

$$C_{JE} = C_{JE0}\left(1 - \frac{V_{BE}}{V_{biE}}\right)^{-m_E} \tag{6.94}$$

and

$$C_{JC} = C_{JC0}\left(1 - \frac{V_{BC}}{V_{biC}}\right)^{-m_C} \tag{6.95}$$

where

C_{JE0} and C_{JC0} are the values of the EB and CB junction capacitances at $V_{BE} = 0$ and $V_{BC} = 0$ respectively,

V_{biE} and V_{biC} = the built-in potentials of the EB and CB junctions, and

m_E and m_C = the factors dependent on the nature of grading of EB and CB junctions.

The complete Ebers–Moll equivalent circuit, including parasitic capacitances and series resistances, is shown in Fig. 6.20.

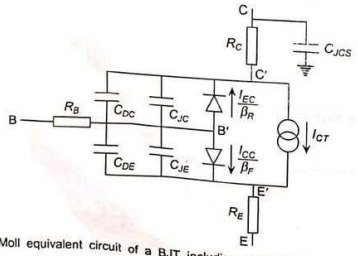


Figure 6.20 Ebers–Moll equivalent circuit of a BJT including capacitances and series resistances.

6.11 SWITCHING OF BIPOLAR TRANSISTORS

In digital circuits, the bipolar transistor is often used as an electrically controllable switch, which can respond to input changes much more rapidly than a mechanical switch. This will be clear from the circuit shown in Fig. 6.21(a). In Fig. 6.21(b), the output characteristics of the transistor connected in the common emitter configuration, together with the load line are shown. Figure 6.21(c) shows a circuit where the bipolar transistor has been replaced by a switch. When the input voltage (V_{in}) is low (almost 0 V or less) in Fig. 6.21(a), the transistor is cut-off and the operating point is shown by point A in Fig. 6.21(b). At this point, $I_C \approx 0$ and $V_{out} = V_{CE} \approx V_{CC}$. The situation corresponds to the switch in Fig. 6.21(c) being in the open circuit. On the other hand, when V_{in} is increased so that I_B is sufficiently high, the operating point moves to point B in the saturation region of the characteristics in Fig. 6.21(b). Since $I_C < \beta I_B$ in the saturation region, the minimum base current required for the transistor to go to saturation is

$$(I_{Bsat})_{min} = \frac{I_{Csat}}{\beta} = \frac{V_{CC} - V_{CEsat}}{\beta R_L}$$

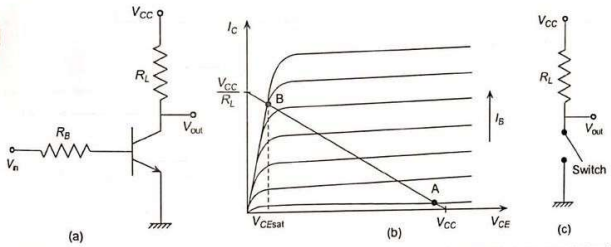


Figure 6.21 (a) A circuit using BJT as a switch, (b) common emitter output characteristics of BJT with load line, and (c) an equivalent circuit where the BJT is replaced by a switch.

At this point

$$V_{out} = V_{CE} = V_{CEsat}$$

and

$$I_C = I_{Csat} = \frac{V_{CC} - V_{CEsat}}{R_L}$$

Since V_{CEsat} is small (≈ 0.2 V), we can write $I_C \approx V_{CC}/R_L$ and $V_{out} \approx 0$. This is similar to the situation when the switch is closed in Fig. 6.21(c). Thus, we see that in Fig. 6.21(a), the transistor behaves as a switch and the flow of current through a load resistance can be controlled by a much smaller input current (since $I_B << I_C$).

Note that the given circuit also performs the operation of an inverter, which is a basic component in a digital circuit. We have already seen that when V_{in} is LOW, $V_{out} \approx V_{CC}$ (HIGH). On the other hand, when V_{in} is sufficiently HIGH, $V_{out} = V_{CEsat}$ (LOW). Although the circuit is useful as an inverter, an important aspect of its operation is the delays involved in switching from one state to the other. We shall now discuss this in detail.

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Let us consider the circuit shown in Fig. 6.22(a). The input waveform, as shown in Fig. 6.22(b), changes from $V_{in} = -V_R$ to $V_{in} = V_F$ at $t = 0$ and remains at that value till $t = t_1$, when V_{in} again changes to $-V_R$. At $t = 0^-$, the transistor was in cut-off condition with $V_{BE} = -V_R$, $I_B = 0$, and the excess minority carrier concentration in the base $Q_{nB} \approx 0$ (the small negative value is neglected) as shown in Figs. 6.22(c), (d), and (e) respectively. At $t = 0^+$, the change in the input voltage to V_F results in a forward base current. Consequently, V_{BE} changes quickly from $-V_R$ to 0, but then increases to a small positive value, as the stored charge builds up in the base. The situation is similar to the switching of a diode as discussed in section 4.4. The base current in the entire interval $0 < t < t_1$ is almost a constant and is given by

$$I_{FB} = \frac{V_F - V_{BE}}{R_B} \approx \frac{V_F}{R_B}$$

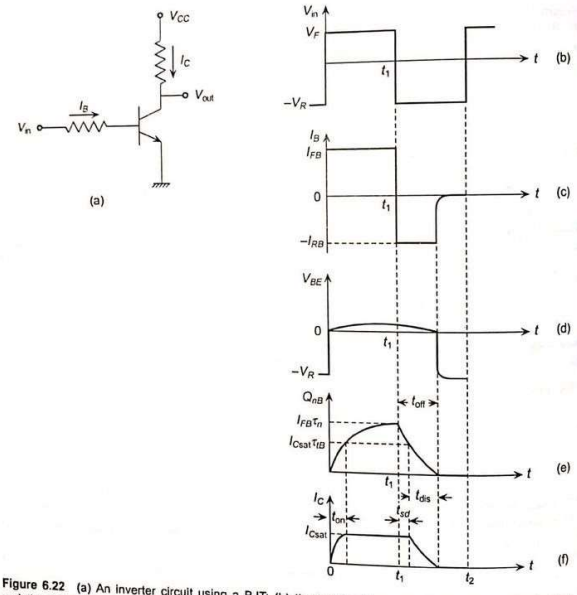


Figure 6.22 (a) An inverter circuit using a BJT; (b) the input voltage waveform; (c) variation in V_{BE}; (d) variation in I_B; (e) excess minority charge variation in base; (f) variation in I_C, showing the different time delays associated with switching of the BJT.

For this period, considering the charges stored only in the base (that is, for $\gamma = 1$), we can write

$$\frac{dQ_{nB}}{dt} = I_{FB} - \frac{Q_{nB}}{\tau_n} \tag{6.96}$$

Equation (6.96) is another form of Eq. (4.62) and essentially states that the rate of increase of base charge is equal to the difference between the rate at which charges flow in (I_{FB}) and the rate at which they recombine (Q_{nB}/τ_n). The solution of Eq. (6.96) is given by

$$Q_{nB}(t) = I_{FB}\tau_n\left[1 - \exp\left(-\frac{t}{\tau_n}\right)\right] \tag{6.97}$$

Thus, the base charge increases with time and the collector current also increases according to the relation $I_C = Q_{nB}/\tau_{tB}$ [from Eq. (6.29)] as shown in Fig. 6.22(f), where τ_{tB} is the base transit time. However, at $t = t_{on}$, I_C reaches its saturation value I_{Csat}. The corresponding charge is given by $Q_{nB} = I_{Csat}\tau_{tB}$. Therefore, substituting $t = t_{on}$ and $Q_{nB}(t) = I_{Csat}\tau_{tB}$ in Eq. (6.97), and using Eq. (6.45), we have

$$t_{on} = \tau_n \ln\frac{1}{1 - \dfrac{I_{Csat}}{\beta I_{FB}}} \tag{6.98}$$

We can see from Eq. (6.98) that a higher value of I_{FB} reduces the delay time. For example, if

$$I_{FB} = \frac{2I_{Csat}}{\beta}, \quad t_{on} = \tau_n \ln(2)$$

while if $I_{FB} = \dfrac{5I_{Csat}}{\beta}$, $t_{on} = \tau_n \ln(1.25)$.

Although the collector current is limited by the circuit, the base charge continues to increase in accordance with Eq. (6.97), and if t_1 is sufficiently long, it reaches its ultimate value of $Q_{nB} = I_{FB}\tau_n$. This is shown in Fig. 6.22(e). At $t = t_1$, when V_{in} changes to $-V_R$, the direction of the base current changes. As carriers flow out of the base, the stored charge in the base reduces. However, V_{BE}, which is related to the stored charge in the base, reduces slowly and reaches 0 V at $t = t_1 + t_{off}$. In the period $t_1 < t < (t_1 + t_{off})$, the base current is almost constant at

$$-I_{RB} = \frac{-V_R - V_{BE}}{R_B} \approx -\frac{V_R}{R_B}$$

Beyond $t = (t_1 + t_{off})$, the value of V_{BE} quickly falls to $-V_R$ and then $I_B = 0$. Now, in the period $t_1 < t < (t_1 + t_{off})$, the decay of the stored charge in the base is governed by the relation

$$\frac{dQ_{nB}}{dt} = -I_{RB} - \frac{Q_{nB}}{\tau_n} \tag{6.99}$$

From Eq. (6.99), we see that (dQ_{nB}/dt) is negative, and the rate of decay of the charges is equal to the sum of the rate at which carriers are flowing out (I_{RB}) and the rate at which they recombine (Q_{nB}/τ_n). Considering $Q_{nB} = I_{FB}\tau_n$ at $t = t_1$, the solution of Eq. (6.99) is given by

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$$Q_{nB}(t) = I_{FB}\tau_n \exp\left(-\frac{t - t_1}{\tau_n}\right) - I_{RB}\tau_n\left[1 - \exp\left(-\frac{t - t_1}{\tau_n}\right)\right] \tag{6.100}$$

As the stored charge decays, the transistor comes out of saturation and I_C begins to reduce only when $Q_{nB} = I_{Csat}\tau_{tB}$. The duration for which the transistor remains in saturation even after the input voltage has been reduced to a negative value is called the *storage delay time* (t_{sd}). Therefore, substituting $Q_{nB} = I_{Csat}\tau_{tB}$ and $t = t_1 + t_{sd}$ in Eq. (6.100), we get

$$t_{sd} = \tau_n \ln\frac{I_{RB} + I_{FB}}{I_{RB} + \dfrac{I_{Csat}}{\beta}} \tag{6.101}$$

From Eq. (6.101), we see that t_{sd} reduces with increase in I_{RB}. This is to be expected since a larger reverse base current results in a faster reduction in base charge. Now, at $t = t_1 + t_{off}$, $Q_{nB} = 0$, and therefore an expression for t_{off} can be obtained by substituting these values in Eq. (6.100). The *discharge time* (t_{dis}) is defined as the time required for the transistor to go from saturation to cut-off and is given by $t_{dis} = t_{off} - t_{sd}$. An expression for t_{dis} can now be obtained as

$$t_{dis} = \tau_n \ln\left(1 + \frac{I_{Csat}}{\beta I_{RB}}\right) \tag{6.102}$$

In all the expressions given by Eqs. (6.98), (6.101), and (6.102), we find that the delays are proportional to τ_n. Therefore, in switching transistors, the minority carrier lifetime in the base must be made as low as possible. One way of reducing the lifetime is by doping the base with impurities such as gold which give rise to energy level near the middle of the band gap. However, with reduction in τ_n, the β of the transistor reduces [refer to Eq. (6.45)]. To overcome this, the base transit time should also be reduced as far as possible, by reducing the base width. It may be mentioned here that another component of delay, which is due to the charging of the junction capacitances, has been neglected in the above analysis.

HELP DESK 6.9

Why does it take a long time for V_{BE} to change by a small voltage when the EB junction is forward biased, while V_{BE} changes almost instantaneously by a large voltage when the EB junction is reverse biased?

Let the doping concentration in the base of an npn silicon bipolar transistor be 10^{17} per cm^3. Therefore, the minority carrier concentration at thermal equilibrium at room temperature in the base is given by $n_{p0} = 2.25 \times 10^3$ per cm^3. Now, when $V_{BE} = 0.6$ V,

$$n_{pE} = n_{p0}\exp\left(\frac{V_{BE}}{V_T}\right) = 2.37 \times 10^{13} \text{ per cm}^3$$

while when $V_{BE} = -5$ V, $n_{pE} \approx 0$. Thus, to change V_{BE} from 0 to 0.6 V, n_{pE} and therefore Q_{nB} have to change by ten orders of magnitude, while when V_{BE} changes from 0 to -5 V n_{pE} changes by only three orders of magnitude. This explains why the change in V_{BE} (when the EB junction is forward biased) is a much slower process.

Example 6.4

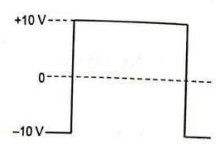
In the figure shown below, a square wave is applied as input to switch the BJT between ON and OFF states. Neglecting the effect of the junction capacitance, the total delay in one cycle may be given by $t_{delay} = t_{on} + t_{sd} + t_{dis}$. Calculate

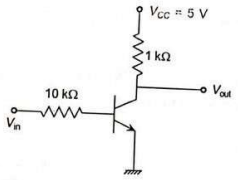
(a) t_{on}/t_{delay},

(b) t_{sd}/t_{delay}, and

(c) t_{dis}/t_{delay}.

Assume $V_{CEsat} = 0.2$ V and $\beta = 50$.





Solution: From the above figure,

$$I_{Csat} = \frac{V_{CC} - V_{CEsat}}{R_C} = \frac{5 - 0.2}{1}\text{ mA} = 4.8\text{ mA}$$

$$I_{FB} \approx \frac{V_F}{R_B} = \frac{10}{10}\text{ mA} = 1\text{ mA}$$

and

$$-I_{RB} \approx \frac{-V_R}{R_B} = \frac{-10}{10}\text{ mA} = -1\text{ mA}$$

Now, from Eq. (6.98),

$$t_{on} = \tau_n \ln\frac{1}{1 - \dfrac{I_{Csat}}{\beta I_{FB}}} = \tau_n \ln\frac{1}{1 - \dfrac{4.8}{50 \times 1}} = 0.1\tau_n$$

From Eq. (6.101),

$$t_{sd} = \tau_n \ln\frac{I_{RB} + I_{FB}}{I_{RB} + \dfrac{I_{Csat}}{\beta}} = \tau_n \ln\frac{1 + 1}{1 + \dfrac{4.8}{50}} = 0.6\,\tau_n$$

From Eq. (6.102),

$$t_{dis} = \tau_n \ln\left(1 + \frac{I_{Csat}}{\beta I_{RB}}\right) = \tau_n \ln\left(1 + \frac{4.8}{50 \times 1}\right) = 0.09\tau_n$$

Therefore, $t_{delay} = (0.1 + 0.6 + 0.09)\tau_n = 0.79\tau_n$

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Therefore

(a) $\frac{t_{on}}{t_{delay}} = \frac{0.1}{0.79} = 0.126,$

(b) $\frac{t_{sd}}{t_{delay}} = \frac{0.6}{0.79} = 0.759,$ and

(c) $\frac{t_{dis}}{t_{delay}} = \frac{0.09}{0.79} = 0.114$

We can see from the results that the storage delay time is by far the largest component of delay.

6.12 PROCESS FLOW FOR AN npn BIPOLAR JUNCTION TRANSISTOR IN INTEGRATED CIRCUIT

We shall now discuss the process flow to realize a bipolar junction transistor in an integrated circuit. This is schematically presented in Fig. 6.23. The initial substrate is p-type silicon having a resistivity of approximately 5 Ωcm. The steps followed are:

Step 1: Thermal oxide of thickness 0.5 μm is grown on the substrate.

Step 2: Photolithography is carried out to open a window in the oxide.

Step 3: Arsenic or antimony is ion-implanted (or diffused) through this window to form an n^+ region [Fig. 6.23(a)].

Step 4: SiO_2 is removed and an n-epitaxial layer is grown on the entire surface [Fig. 6.23(b)].

Step 5: Unlike in the case of discrete devices, a transistor in an integrated circuit has to be properly isolated from adjacent transistors for the circuit to function. In this particular example, a reverse-biased p-n junction is used to isolate the transistors. The isolation regions completely surround the n^+ buried layer to form islands of n-type silicon as shown in Fig. 6.23(c). This is achieved by ion-implantation followed by a high-temperature drive-in to push the boron all the way through the epitaxial layer.

Step 6: The base region is now defined by photolithography. A boron implant followed by drive-in is used to produce a base (Fig. 6.23 (d)).

Step 7: Emitter and collector contact regions are next defined by photolithography and phosphorus implantation is carried out. The n^+ diffusion in the collector is needed to produce a low-resistance ohmic contact (Fig. 6.23(e)).

Step 8: The contact windows are opened over emitter, base, and collector regions as shown in Fig. 6.23(f).

Step 9: Aluminium is evaporated over the entire surface. Afterwards, aluminium interconnections between the components of the circuit are defined. The final structure is shown in Fig. 6.23(g).

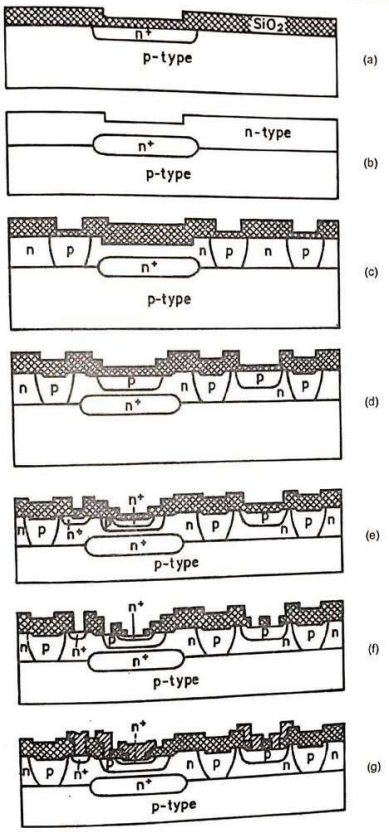


Figure 6.23 Process flow for an npn BJT.

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Along with the npn transistor, a p-n junction diode is also realized in Fig. 6.23. The base doping for the transistor is used to realize the p-n junction. The diode is isolated from the transistor by the p-tubs realized in Step 5. The impurity concentration curves for the emitter, base, and collector regions of the transistor are shown in Fig. 6.24.

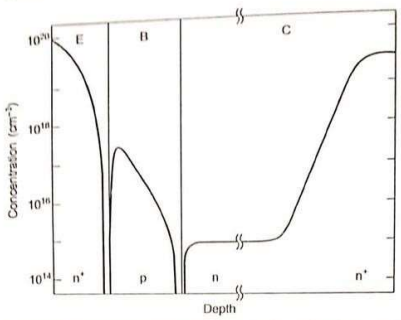


Figure 6.24 The impurity concentration distribution curves in the emitter, base, and collector regions of the transistor.

However, for modern transistors in high-speed digital circuits, the process flow is considerably modified. The main features include oxide isolation instead of junction isolation and use of doped polysilicon to realize emitter and base regions. These process modifications result in high packing density as well as reduction of parasitic sidewall capacitance and base resistance. The importance of these parameters is discussed in the next chapter. The cross-sectional diagram of such a transistor with oxide isolation, polysilicon emitter and base is given in Fig. 6.25.

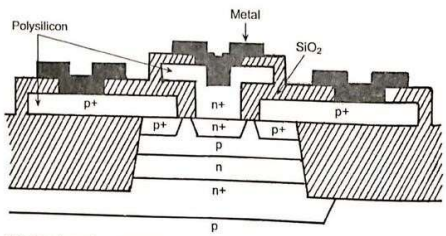


Figure 6.25 Cross-section of an improved bipolar transistor with oxide isolation, polysilicon emitter and base. (Source: P. Ashburn. Copyright © 1988 by John Wiley and Sons Inc. used by permission).

PROBLEMS

P6.1 The β of an npn transistor is estimated to be 200 considering only the effect of γ and assuming $\alpha_T = 1$. On the other hand, considering only the effect of α_T and assuming $\gamma = 1$, the β of the same transistor is found to be 100. What is the actual β of the transistor if both α_T and γ are taken into account?

P6.2 Two transistors T_1 and T_2 are identical in all respects except in their base widths. The emitter efficiencies are unity in both cases. The base width of T_1 is 1 μm and that of T_2 is 2 μm. The transistors are biased in the active region of operation. The base current of T_1 is 10 μA when the collector current is 1 mA. Determine the base current of T_2 when the collector current is 1 mA.

P6.3 The β of an n^+pn^+ transistor is 100 when the collector–base junction is reverse biased with $V_{CB} + V_{bi} = 1$ V. The neutral base width, that is, the undepleted portion of the base for this case is 1.2 μm and the depletion layer width at the CB junction is 0.3 μm. Determine the β of the transistor when $V_{CB} + V_{bi} = 9$ V. Assume $\alpha_T = 1$ and note that V_{CB} is the reverse voltage applied at the CB junction while V_{bi} is the built-in potential of this junction.

P6.4 The BJT T_1, whose parameters are given in Example 6.1, has an area $A = 10^{-3}$ cm^2. The emitter–base junction of T_1 is forward-biased and the collector base junction is left open. If the hole current injected into the emitter is $I_{pE} = 10$ μA,

(i) Determine the voltage at the collector–base junction.

(ii) Sketch the minority carrier distribution in the base.

(iii) Calculate the value of the emitter current.

(iv) If now the collector–base junction is shorted, without changing the voltage at the EB junction, what will be the value of the collector current assuming W_B does not change?

(v) Sketch the minority carrier distribution in the base for case (iv).

(vi) Calculate the base current for case (iv).

P6.5 In an n^+pn^+ transistor with uniform base doping, the neutral base width (that is, the undepleted portion of the base) is 1.2 μm when $V_{CB} = 5$ V. The depletion layer width at the collector–base junction for this case is 0.3 μm. Find the value of the collector–base voltage at which the entire base gets depleted. Neglect the built-in potential as well as the depletion layer width at the emitter–base junction.

P6.6 In an n^+pnn^+ transisistor, the n^+pn regions are identical to the BJT T_1, whose parameters are given in Example 6.1.

(a) Assuming that avalanche breakdown at the CB junction does not occur before base punch through, calculate the base punch-through voltage (V_{PT}).

(b) What is the maximum electric field at the CB junction when $V_{CB} = V_{PT}$?

(c) If the critical electric field for avalanche breakdown is 20 V/μm, what is the maximum possible value of V_{CB} for this transistor?

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REFERENCES AND SUGGESTED FURTHER READING

[1] Sze, S.M., *Physics of Semiconductor Devices*, 2nd ed., Wiley, N.Y., 1981.

[2] Roulston, David J., *Bipolar Semiconductor Devices*, McGraw Hill, New York, 1990.

[3] Streetman, B.G. and S. Banerjee, *Solid State Electronic Devices*, 5th ed., Prentice Hall Inc., New Jersey, 2000.

[4] Ashburn, P., *Design and Realization of Bipolar Transistors*, John Wiley and Sons, New York, 1988.

[5] Antognetti P. and G. Massobrio, *Semiconductor Modelling with SPICE*, McGraw Hill Book Co., Singapore, 1988.

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7 Advanced Topics in BJT

In Chapter 6, the basic operating principles of an ideal bipolar junction transistor have been discussed. This chapter analyzes the transistor action under dynamic conditions and then proceeds to obtain its high frequency characteristics. In addition, the discussion points out that the characteristics of a real transistor often differ significantly from an idealized device due to various second order effects.

7.1 OPERATION OF THE BJT AT HIGH FREQUENCIES

In Chapter 6, we have mostly discussed the operation of the bipolar transistor under static conditions. However, one of the important applications of the BJT is in small signal amplifiers, where a small signal voltage or current is superimposed on the dc values. The term 'small signal' is used when the peak value of the ac signal component is much smaller than the dc value. In this section, we shall study the operation of the bipolar transistor under small signal conditions with the help of models specially suited for this purpose.

7.1.1 Charge Control Model

In the previous chapter, we have considered the BJT to be a current-controlled or a voltage controlled device. We shall now describe the BJT as a charge-controlled device. The charge-control approach is useful in the modelling of terminal currents of the bipolar junction transistor for small-signal or large-signal transient operation. In this approach, the currents are modelled in terms of the stored charge and transit time.

As in the case of the Ebers-Moll model discussed in section 6.6, we shall partition the stored charge in the base into two components, namely Q_{BF} and Q_{BR}. As shown in Fig. 7.1, Q_{BF} and Q_{BR} are the stored charges in the base for the forward ($V_{BE} > 0$, $V_{BC} = 0$) and reverse ($V_{BE} = 0$, $V_{BC} > 0$) modes of operation respectively.

Let us first consider the forward mode of operation, so that $Q_{BR} = 0$. We have already seen in section 6.6 that when the transistor is in the active mode, the base recombination current can be

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expressed as $I_{BR} = Q_{BF}/\tau_B$, τ_B being the minority carrier lifetime in the base. Then the total base current in the forward mode of operation (I_B^F) can be expressed as

$$I_B^F = I_{BR} + |I_{pE}| = \frac{Q_{BF}}{\tau_B} + \left(\frac{|I_{pE}|}{I_{BR}}\right) = \frac{Q_{BF}}{\tau_{eff}^F} \tag{7.1}$$

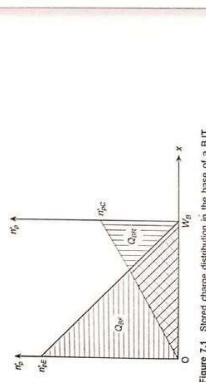


Figure 7.1 Stored charge distribution in the base of a BJT.

where τ_{eff}^F = effective lifetime for the forward operation given by

$$\tau_{eff}^F = \frac{\tau_B}{1 + \left|\frac{I_{pE}}{I_{BR}}\right|} \tag{7.2}$$

Since both I_{pE} and I_{BR} are proportional to $[\exp(V_{BE}/V_T) - 1]$, τ_{eff}^F is a constant independent of applied bias. Also, τ_{eff}^F is approximately equal to τ_B if $I_{BR} \gg I_{pE}$. The collector current in the forward mode of operation (I_C^F) can now be expressed using Eq. (6.29) as

$$I_C^F = \frac{AqD_p n_{pE}'}{W_B} = \frac{Aqn_{pE}' W_B}{2}\left(\frac{W_B^2}{2D_p}\right) = \frac{Q_{BF}}{\tau_B} = \frac{Q_{BF}}{\tau_{tB}^F} \tag{7.3}$$

where $\tau_{tB}^F = (W_B^2/2D_p)$ is the forward base transit time of the electrons injected from the emitter.

The emitter current in the forward mode (I_E^F) can now be written as

$$I_E^F = I_C^F + I_B^F = \frac{Q_{BF}}{\tau_B^*} + \frac{Q_{BF}}{\tau_{eff}^F} = Q_{BF}\left(\frac{1}{\tau_{tB}^F} + \frac{1}{\tau_{eff}^F}\right) \tag{7.4}$$

Let us now consider the reverse mode of operation, where $V_{BE} = 0$, implying $Q_{BF} = 0$. Expressions for the terminal currents in the reverse mode of operation can now be obtained by replacing Q_{BF} with Q_{BR} and the forward parameters τ_{tB}^F and τ_{eff}^F with the parameters for reverse

operation, namely τ_{tB}^R and τ_{eff}^R in Eqs. (7.1), (7.3), and (7.4). Also the emitter and collector terminals are interchanged in reverse operation.

For a more general analysis, considering electron injection from both emitter (forward mode) and collector (reverse mode) into the base, we can write

$$I_B = I_B^F + I_B^R = \frac{Q_{BF}}{\tau_{eff}^F} + \frac{Q_{BR}}{\tau_{eff}^R} \tag{7.5}$$

$$I_C = I_C^F + I_C^R = \frac{Q_{BF}}{\tau_{tB}^F} - Q_{BR}\left(\frac{1}{\tau_{tB}^R} + \frac{1}{\tau_{eff}^R}\right) \tag{7.6}$$

$$I_E = I_E^F + I_E^R = Q_{BF}\left(\frac{1}{\tau_{tB}^F} + \frac{1}{\tau_{eff}^F}\right) - \frac{Q_{BR}}{\tau_{tB}^R} \tag{7.7}$$

Note that in the above equations, the superscripts F and R refer to the forward and the reverse modes of operation respectively.

Equations (7.5)–(7.7) express the terminal currents in a BJT for a static situation, that is, when the junction voltages and the currents do not change with time. Let us now incorporate the time-varying situation in our analysis. If v_{BE} (the instantaneous base–emitter voltage) varies with time, clearly Q_{BF} is also going to vary with time, giving rise to a component of base current ($= dQ/dt$). Ignoring the stored charge in the emitter, the instantaneous base current in the forward mode, i_B^F can be written as

$$i_B^F = \frac{Q_{BF}}{\tau_{eff}^F} + \frac{dQ_{BF}}{dt} \tag{7.8}$$

The above equation is the same as Eq. (6.96), which was used while discussing the switching of transistors. Now, if the time rate of change of voltages and currents is small, the so-called quasi-static approximation holds. Thus, even though Q_{BF} changes, the slope of the electron distribution at any instant in the base is almost fixed. Therefore,

$$i_C^F = \frac{Q_{BF}}{\tau_{tB}^F} \tag{7.9}$$

Using Eqs. (7.8) and (7.9), we have

$$i_E^F = \frac{Q_{BF}}{\tau_{tB}^F} + \frac{Q_{BF}}{\tau_{eff}^F} + \frac{dQ_{BF}}{dt} = \frac{Q_{BF}}{\tau_F^*} + \frac{dQ_{BF}}{dt} \tag{7.10}$$

where $\dfrac{1}{\tau_F^*} = \dfrac{1}{\tau_{tB}^F} + \dfrac{1}{\tau_{eff}^F}$.

It should be noted that when the junction voltage varies with time, the stored charge in the base Q_{BF} has two components—one associated with the dc bias $\overline{Q_{BF}}$ and the other with the small signal ac variation q_{BF}, so that $Q_{BF} = \overline{Q_{BF}} + q_{BF}$. We can represent the ac component as $q_{BF} = q_{BFO}e^{j\omega t}$, so that

$$Q_{BF} = \overline{Q_{BF}} + q_{BFO}e^{j\omega t} \tag{7.11}$$

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Substituting Eq. (7.11) in Eqs. (7.9) and (7.10), the ac components of the collector and emitter currents can be expressed as

$$i_c = \frac{q_{BF}}{\tau^F_{tB}} \tag{7.12}$$

$$i_e = \frac{q_{BF}}{\tau^*_F}(1 + j\omega\tau^*_F) \tag{7.13}$$

The ac common base current gain $\alpha(\omega)$ can now be expressed from Eqs. (7.12) and (7.13) as

$$\alpha(\omega) = \frac{i_c}{i_e} = \frac{\tau^*_F}{\tau^F_{tB}(1 + j\omega\tau^*_F)} = \frac{\alpha_F}{1 + j\omega\tau^*_F} = \frac{\alpha_F}{1 + j\dfrac{\omega}{\omega_\alpha}} \tag{7.14}$$

where $\alpha_F = (\tau^*_F/\tau^F_{tB})$ is the dc common base current gain in normal active mode [as can be seen from Eqs. (7.3) and (7.4)]
and

$$\omega_\alpha = \frac{1}{\tau^*_F}$$

At $\omega = \omega_\alpha$, $\quad |\alpha(\omega)| = \dfrac{\alpha_F}{\sqrt{2}}$.

Therefore, $f_\alpha = \omega_\alpha/2\pi$ is defined as the *α-cutoff frequency* of the transistor. At this frequency, the real (conductive) part of the emitter current becomes equal to the imaginary (capacitive) part.

Similarly, an ac common emitter current gain $\beta(\omega)$ and a *β-cutoff frequency* (ω_β) can also be defined as

$$\beta(\omega) = \frac{i_c}{i_b} = \frac{\beta_F}{1 + j\dfrac{\omega}{\omega_\beta}} \tag{7.15}$$

where

$$\beta(\omega) = \frac{\alpha(\omega)}{1 - \alpha(\omega)},$$

$$\beta_F = \frac{\alpha_F}{1 - \alpha_F}, \text{ and}$$

$$\omega_\beta = \omega_\alpha(1 - \alpha_F).$$

Evidently the β-cutoff frequency is much smaller than the α-cutoff frequency.

A more important factor to evaluate the performance of the transistor at high frequencies is the *cut-off frequency* (f_T). This is defined as the frequency at which $|\beta| = 1$ for short-circuit load. From Eq. (7.15), when $\omega >> \omega_\beta$, we see that

$$|\beta(\omega)| \approx \frac{\beta_F\omega_\beta}{\omega} = \frac{\beta_F f_\beta}{f} \tag{7.16}$$

Therefore, from the defintion of f_T, we have

$$f_T \approx \beta_F f_\beta = \alpha_F f_\alpha \tag{7.17}$$

Since $\alpha_F \approx 1$ in a well-designed transistor, we can see from Eq. (7.17) that f_T is nearly equal to f_α. Figure 7.2 shows how the current gain varies with the operating frequency. Since $\alpha_F = (\tau^*_F/\tau^F_{tB})$ and $\omega_\alpha = (1/\tau^*_F)$, from Eq. (7.17), we have

$$f_T \approx \frac{\alpha_F\omega_\alpha}{2\pi} = \frac{1}{2\pi\tau^F_{tB}} \tag{7.18}$$

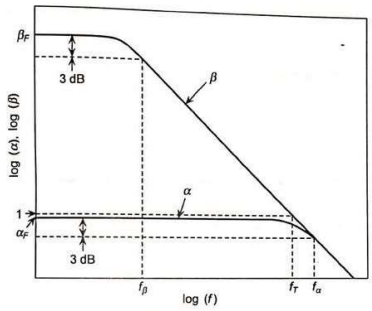


Figure 7.2 Variation of α and β with frequency.

However in reality, f_T is considerably lower. Considering the stored charge in the emitter, it can be shown that $f_T = (1/2\pi\tau_F)$, where τ_F is the forward transit time defined by Eq. (6.90). The cut-off frequency f_T is further reduced due to the presence of parasitic capacitance and series resistance. A more realistic expression of f_T can be obtained by considering the small-signal equivalent circuit, presented in the next sub-section.

7.1.2 Small Signal Equivalent Circuit

A complete equivalent circuit of a BJT based on the Ebers–Moll model is shown in Fig. 6.20. For a transistor biased in the normal active mode, this equivalent circuit can be simplified as shown in Fig. 7.3(a). The small-signal equivalent circuit of a BJT, which is derived from Eq. 7.3(a), is shown in Fig. 7.3(b).

In Fig. 7.3(b), C_π is the parallel combination of the base–emitter junction depletion capacitance C_{JE} and the diffusion capacitance C_{DE}. We have already discussed C_{DE} in section 6.10. From Eq. (6.92), we have

$$C_{DE} = \frac{dQ_F}{dV_{BE}} = \tau_F\frac{dI_C}{dV_{BE}} = \tau_F g_m \tag{7.19}$$

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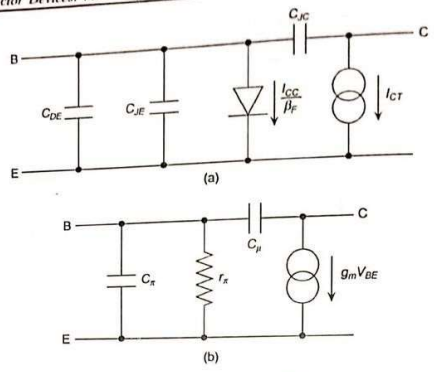


Figure 7.3 (a) Small-signal equivalent circuit of a BJT and (b) simplified equivalent circuit when the BJT is operating in normal active mode.

where

τ_F = forward transit time and

$g_m = (dI_C/dV_{BE})$ is the transconductance of the BJT.

Since, in the forward active region, $I_C \approx I_S \exp(V_{BE}/V_T)$, the transconductance can be expressed as

$$g_m = \frac{I_C}{V_T}$$

Thus, we see that the transconductance and therefore C_{DE}, is actually dependent on the dc operating current of the bipolar transistor. We can now express C_π as

$$C_\pi = C_{JE} + C_{DE} = C_{JE} + \tau_F g_m = C_{JE} + \frac{\tau_F I_C}{V_T} \tag{7.20}$$

In Fig. 7.3(b), $C_\mu = C_{JC}$, where C_{JC} is the base–collector junction depletion capacitance. Note that the diffusion capacitance associated with the BC junction has been neglected since this junction is reverse biased in the normal active mode.

The small-signal base current i_b and collector current i_c due to small-signal input voltage v_{be} can be expressed as

$$i_b = v_{be}\left.\frac{\partial I_B}{\partial V_{BE}}\right|_{V_{ac}} = \frac{v_{be}}{r_\pi} \tag{7.21}$$

$$i_c = v_{be}\left.\frac{\partial I_C}{\partial V_{BE}}\right|_{V_{ac}} = g_m v_{be} \tag{7.22}$$

where

$$\frac{1}{r_\pi} = g_\pi = \frac{\partial I_B}{\partial V_{BE}} = \frac{I_B}{V_T}$$

Since $g_m = I_C/V_T$, we see that

$$g_m r_\pi = \frac{I_C}{I_B} = \beta_F$$

Therefore, the currents between the BE and CE terminals are modelled using r_π and the current source $g_m v_{be}$ respectively. The current source between the BC terminals has been neglected in this case as the BC junction is reverse biased.

Using Fig. 7.3(b), and shorting the C and E terminals, we obtain the short-circuit current gain as a function of frequency in the following manner:

$$\beta(\omega) = \frac{i_c}{i_b} = \frac{g_m - j\omega C_\mu}{g_\pi + j\omega C_\pi + j\omega C_\mu} \tag{7.23}$$

Assuming $g_m >> \omega C_\mu$, which is valid for most cases, we have

$$\beta(\omega) = \frac{i_c}{i_b} = \frac{(g_m/g_\pi)}{1 + j\omega\left(\dfrac{C_\pi + C_\mu}{g_\pi}\right)} \approx \frac{\beta_F}{1 + j\omega\left(\dfrac{C_\pi + C_\mu}{g_\pi}\right)} \tag{7.24}$$

At high frequencies, we can neglect the unity term in the denominator of Eq. (7.24). Therefore, the cut-off frequency f_T, at which the magnitude of β is unity, is given by

$$2\pi f_T \approx \left[\frac{C_\pi + C_\mu}{\beta_F g_\pi}\right]^{-1} = \left[\frac{C_{JE} + \tau_F g_m + C_\mu}{g_m}\right]^{-1} = \left[\tau_F + \frac{V_T}{I_C}(C_{JE} + C_{JC})\right]^{-1} \tag{7.25}$$

For the sake of completeness, the product of the collector series resistance and C–B junction capacitance (RC) delay should also be included. Thus, the expression for the cut-off frequency can further be modified as

$$2\pi f_T \approx \left[\tau_F + R_C C_{JC} + \frac{V_T}{I_C}(C_{JE} + C_{JC})\right]^{-1} \tag{7.26}$$

The variation of f_T with I_C is shown in Fig. 7.4. At low values of I_C, the third term in Eq. (7.26) dominates and hence f_T increases with an increase in current till it reaches a peak value ($f_{T\max}$). However, f_T reduces again when the collector current becomes very high. This is due to an increase in τ_F due to the base-widening at high collector current (Kirk Effect). This is discussed in detail later in section 7.2.3.

Even though f_T is often used as figure of merit for the ac performance of BJT, it is not a realistic criterion for judging the practical circuits. This is because it is measured with a short-circuit load. Also it does not take into account the base resistance (R_B). Hence, an alternative figure of merit, $f_{\max}$ is often quoted. This is defined as the frequency at which the power gain falls to unity and is given by [1]

$$f_{\max} = \left[\frac{f_T}{8\pi C_{JC} R_B}\right]^{1/2} \tag{7.27}$$

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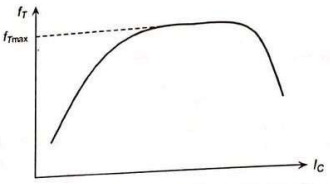


Figure 7.4 Variation of f_T with collector current.

7.1.3 Design of High Frequency Transistors

In a high frequency transistor, the emitter and collector junction areas should be kept as small as possible to reduce the capacitances. The base width should also be small, to reduce the transit time. Also, the doping profile in the base can be tailored to reduce transit time, as we shall see in section 7.2.1. In addition, care should be taken to reduce series resistances associated with each terminal to reduce the parasitic *RC* time constants. In fact, the doping concentration in the extrinsic base regions can be made higher so as to reduce base resistance. In bipolar transistors, due to a phenomenon called *emitter crowding* (discussed later in section 7.2.5), most of the injection of the carriers into the base occurs at the edges of the emitter region. Thus, to maintain sufficient current and reduce the emitter area at the same time, the perimeter-to-area ratio of the emitter must be increased. Several thin emitter stripes, connected electrically by metallization and separated by interspersed base metal contacts are often used. This type of structure, shown in Fig. 7.5, is often referred to as *interdigitated geometry*. Two special types of bipolar transistors used for high frequency operation are the Polysilicon Emitter Bipolar Transistor (PEBT) and the Heterojunction Bipolar Transistor (HBT). We shall discuss about these devices in Section 7.3.

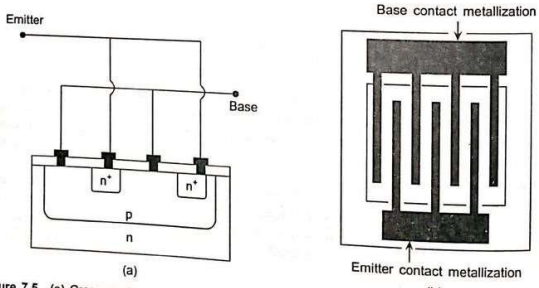


Figure 7.5 (a) Cross-sectional and (b) top view of interdigitated BJT structure showing thin emitter stripes interspersed with multiple base contacts.

7.2 SECOND ORDER EFFECTS IN BJTS

In Chapter 6, we had assumed several ideal conditions in order to simplify the analysis. In this section, we shall throw light on some of the effects which exist in real transistors but have not been considered earlier.

7.2.1 Non-uniform Doping in the Base—Improvement in Base Transit Time

In our discussions so far, we have assumed that the doping concentrations in the BJT do not vary within a particular region of the device. However, as we have seen in the process flow outlined in section 6.12, the base region of the BJT is formed by diffusion or ion-implantation and the doping concentration in the base, as shown in Fig. 6.24, is not uniform. In such a case, the base transit time is greatly modified because of the presence of an in-built electric field. Let us consider a typical impurity profile in the quasi-neutral base region as shown in Fig. 7.6. It shows two distinct sections, one from 0–x_1, where the net doping concentration ($N_A - N_D$) increases with x (region 1) and the other from x_1–W_B, where the net doping concentration decreases with x (region 2). The in-built field in region 1 opposes the diffusion of holes to the left and in region 2, opposes their flow to the right. Thus, the electrons injected from the emitter encounter a retarding field as they enter the base. On the other hand, the electrons in region 2 are subjected to an accelerating field. For most real transistors, the retarding field region forms only a very small fraction of the base. Therefore, the doping concentration in the base can be approximated by the accelerating field region only.

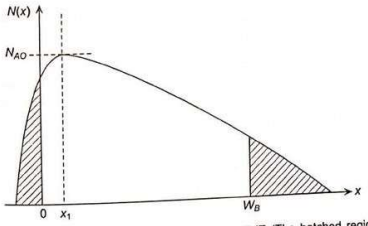


Figure 7.6 Typical doping concentration profile in the base of a BJT (The hatched regions are the depletion regions on either side of the neutral base).

Considering the hole current in the base to be very small, we can write

$$J_p = qp\mu_p \mathscr{E}(x) - qD_p \frac{dp}{dx} \approx 0 \tag{7.28}$$

Let us assume $x_1 \approx 0$ and that the impurity distribution in region 2 is given by $N_A(x) = N_{Ao} \exp[-(x/x_0)]$. Taking $p(x) = N_A(x)$ and solving Eq. (7.28), we get the built-in electric field as

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$$\mathscr{E}(x) = -\frac{V_T}{x_0} \tag{7.29}$$

We can write the expression for the electron current, taking into account both diffusion and drift due to the built-in field as

$$J_n = qD_n \frac{dn}{dx} + qn\mu_n \mathscr{E} = qD_n \left[\frac{dn}{dx} + \frac{n\mathscr{E}}{V_T}\right] \tag{7.30}$$

Substituting Eq. (7.29) in Eq. (7.30) and multiplying both sides by $N_A(x)$, we have

$$N_A(x)\frac{dn}{dx} - n\frac{N_A(x)}{x_0} = \frac{J_n N_A(x)}{qD_n} \tag{7.31}$$

The left hand side of Eq. (7.31) can be expressed as $d/dx\,[N_A(x)\,n(x)]$ and therefore we can write

$$\frac{d}{dx}\left[N_A(x)\,n(x)\right] = \frac{J_n N_A(x)}{qD_n} \tag{7.32}$$

Neglecting recombination in the base so that J_n is a constant, we can write

$$\int_x^{W_B} d\left[N_A(x)\,n(x)\right] = \frac{J_n}{qD_n}\int_x^{W_B} N_A(x)\,dx \tag{7.33}$$

Noting that in the normal active region of operation, $n(W_B) \approx 0$, from Eq. (7.33), we have

$$n(x) = -\frac{J_n}{qD_n N_A(x)}\int_x^{W_B} N_A(x)\,dx = -\frac{J_n}{qD_n}\frac{\displaystyle\int_x^{W_B}\exp\left(-\frac{x}{x_0}\right)dx}{\exp\left(-\frac{x}{x_0}\right)}$$

$$= -\frac{J_n W_B}{qD_n \eta}\left[1 - \exp\left\{-\eta\left(1 - \frac{x}{W_B}\right)\right\}\right] \tag{7.34}$$

where η ($= W_B/x_0$) is defined as the *accelerating field factor*.

The base transit time can now be obtained from Eq. (7.34) as

$$\tau_{tB} = \frac{Q_{nB}}{I_C} = \frac{q\displaystyle\int_0^{W_B} n(x)\,dx}{-J_n} = \frac{W_B}{D_n\eta}\int_0^{W_B}\left[1 - \exp\left\{-\eta\left(1 - \frac{x}{W_B}\right)\right\}\right]dx = \frac{W_B^2}{D_n\eta^2}\left[\eta - 1 + \exp(-\eta)\right] \tag{7.35}$$

In this analysis, we have taken into account an exponentially decaying doping profile function in the base. It is left as a problem for the students to show that any arbitrary doping profile, where the doping concentration reduces from the emitter to the collector end of the base, results in a reduction in transit time from the uniformly doped case.

HELP DESK 7.1

Show that (a) for uniform doping in the base, $\tau_{tB} = W_B^2/2D_n$. (b) for $\eta >> 1$, the base transit time is less than that obtained for uniform doping in the base.

For uniform doping, the impurity concentration everywhere in the base is N_A, that is, η is close to zero. Expressing the exponential term in Eq. (7.35) as a series and considering the first three terms, we obtain,

$$\tau_{tB} = \frac{W_B^2}{D_n\eta^2}\left[\eta - 1 + \left(1 - \eta + \frac{\eta^2}{2}\right)\right] = \frac{W_B^2}{2D_n}$$

For large positive values of η, the exponential term can be neglected. Therefore, we can write,

$$\tau_{tB} \approx \frac{W_B^2}{D_n\eta^2}(\eta - 1) \approx \frac{W_B^2}{D_n\eta}$$

So when $\eta >> 1$ (say 5), the base transit time is less than that obtained for uniform doping in the base.

7.2.2 Variation of β with Collector Current

From our discussion in section 6.6, we have seen that when a transistor is operating in the normal active mode, the base current and the collector current are both proportional to $\exp(V_{BE}/V_T)$. Thus, a plot of ln (I_C) and ln (I_B) versus V_{BE} keeping $V_{BC} = 0$ (popularly known as *Gummel plot*) is expected to be two parallel straight lines. The β of the transistor should therefore be a constant, which depends only on the emitter and base doping concentrations and widths. However for a real transistor, the Gummel plot deviates from linearity and consequently β reduces at both low and high V_{BE}. A typical Gummel plot is shown in Fig. 7.7(a) and the resultant variation of β with I_C is depicted in Fig. 7.7(b).

At low I_C, the reduction in β is caused by the less than expected fall of I_B with reduction in V_{BE}. This is explained by considering the recombination of carriers within the base–emitter junction depletion region as already discussed in section 4.3.4. For small V_{BE}, the base current can be expressed as

$$I_B = I_{BO}\exp\left(\frac{V_{BE}}{n_E V_T}\right) \tag{7.36}$$

where n_E (>1) is the diode ideality factor for the BE junction.

Therefore, for low V_{BE} or low I_C, we have

$$\beta = \frac{I_C}{I_B} \propto \frac{\exp\left(\frac{V_{BE}}{V_T}\right)}{\exp\left(\frac{V_{BE}}{n_E V_T}\right)} = \exp\left[\frac{V_{BE}}{V_T}\left(1 - \frac{1}{n_E}\right)\right] \propto (I_C)^{(1-1/n_E)} \tag{7.37}$$

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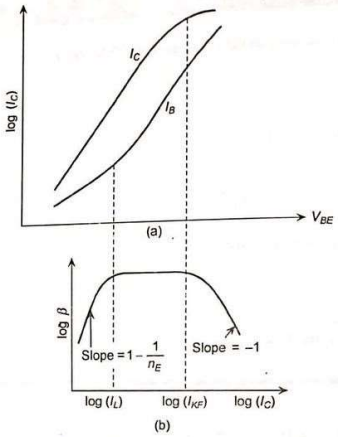


Figure 7.7 (a) Gummel plot showing the variation of ln (I_C) and ln (I_B) with V_{BE} and (b) corresponding variation of β with I_C.

Thus, we see that at low collector currents, β reduces with decrease in I_C.

On the other hand, at high V_{BE}, the collector current is seen to increase at a less than expected rate with increasing V_{BE}. This is caused by the *high level injection* of carriers in the base. In most of our discussions so far, we have assumed a low-level injection, meaning that the majority carrier concentration does not change in any region. However, when a large number of electrons are injected into the base, this assumption no longer remains valid, especially since the base doping concentration is not very high. In order to maintain the charge neutrality, an increased hole concentration is observed in the base. Considering the base to be uniformly doped, we can therefore write

$$p(x) = N_A + n(x) \tag{7.38}$$

Note that Eq. (7.38) always remains valid. However, for low-level injection $n(x) << N_A$, and therefore we take $p = N_A$.

From our discussion on quasi-Fermi levels in section 1.4.1 and noting that the separation between E_{Fn} and E_{Fp} is qV, V being the applied voltage across the junction, the 'pn' product at the edge of the depletion region of the E–B junction can be written as

$$p_{pE}n_{pE} = n_i^2 \exp\left(\frac{E_{Fn} - E_{Fp}}{kT}\right) = n_i^2 \exp\left(\frac{V_{BE}}{V_T}\right) \tag{7.39}$$

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Using Eqs. (7.38) and (7.39) and solving for the electron concentration n_{pE} at E–B junction boundary, we get

$$n_{pE} = \frac{N_A}{2}\left[\sqrt{1 + \frac{4n_i^2}{N_A^2}\exp\left(\frac{V_{BE}}{V_T}\right)} - 1\right] \tag{7.40}$$

An inspection of Eq. (7.40) reveals that while for moderate values of V_{BE}, the electron concentration at E–B junction boundary is approximated to be $(n_i^2/N_A)\exp(V_{BE}/V_T)$, at larger bias it changes to $n_i \exp(V_{BE}/2V_T)$. The collector current I_C, which is proportional to the slope of the electron concentration, thus tends to be proportional to $\exp(V_{BE}/2V_T)$. Therefore for high V_{BE} or high I_C, we have

$$\beta = \frac{I_C}{I_B} \propto \frac{\exp\left(\frac{V_{BE}}{2V_T}\right)}{\exp\left(\frac{V_{BE}}{V_T}\right)} = \exp\left(-\frac{V_{BE}}{2V_T}\right) \propto (I_C)^{-1} \tag{7.41}$$

Thus we see that at high collector currents, β is inversely proportional to I_C.

Another effect of high electron injection into the base is the reduction of emitter efficiency γ. Since the majority carrier concentration in the base increases, the base Gummel number ($GN_B = N_A W_B$) now increases to $\int_0^{W_B} [N_A + n(x)]\,dx$. Consequently γ reduces and β falls further.

In the circuit simulator SPICE, the variation of β with I_C is modelled with the help of the parameters n_E, I_L, and I_{KF} as shown in Fig. 7.7(b), where $I_C < I_L$ and $I_C > I_{KF}$ represent the low current and high current regions respectively.

7.2.3 High Injection in Collector

At high collector currents, the current gain of the BJT drops due to yet another mechanism known as *Kirk effect*. This occurs when there is a heavy injection of electrons across the CB depletion region. Under normal active mode, we have assumed that the minority carrier concentration in the base falls to zero at $x = W_B$ (section 6.4), that is, any electron impinging upon the CB depletion region is quickly swept across. However, this assumption will be violated if any carrier has to flow across this region. A current density J_C requires an electron density at least equal to J_C/qv_{sat}, where v_{sat} is the scattering limited velocity. For moderate V_{BE}, J is small and so the approximation holds good. However, with large V_{BE}, this electron density becomes comparable to the background doping concentration. The Poisson's equation across the depletion region can now be modified as

$$\frac{d\mathscr{E}}{dx} = \frac{1}{\varepsilon_s}\left[qN_D^+ - qN_A^- - \frac{J_C}{v_{sat}}\right] \tag{7.42}$$

The polarity of the electrons adds to the acceptor charge on the base side of the CB depletion region and subtracts from the donors on the collector side. The effect is equivalent to increasing the base doping concentration and reducing the collector doping concentration. Therefore, when the collector current is increased while keeping V_{CB} constant, the depletion region shrinks on the base side resulting in an increase in the neutral base width W_B. Consequently, β decreases and

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the base transit time increases. On the other hand, the effective reduction in the collector doping concentration results in the depletion region extending further into the collector. In an npn transistor with an n^+ buried collector, at very high currents, the depletion region in the collector may extend all the way to the buried collector, which has serious consequences.

7.2.4 Heavy Doping Effects in the Emitter

From our discussion in section 6.5, it can be easily shown that the common emitter current gain (assuming $\alpha_T \approx 1$) is essentially the ratio of the emitter Gummel number to the base Gummel number. So, it seems that a very high doping of emitter will improve the transistor gain. However in reality, the high gain predicted by Eq. (6.43) does not materialize. On the other hand, β becomes significantly less and also depends on temperature. This can be explained by the heavy doping effects, which have not been considered so far. Heavy doping effects are characterized by three mechanisms, namely mobility degradation, Auger recombination, and band gap narrowing.

Mobility degradation

In Chapter 1 section 1.4, we have seen that mobility depends on the time interval between collisions (mean free time) for a carrier moving through a semiconductor. For a heavily doped semiconductor, mobility is primarily determined by ionized impurity scattering. It has been experimentally found that for both silicon and GaAs, carrier mobility decreases drastically once the doping concentration in the semiconductor exceeds 10^{19} cm^{-3}. Since the mobility and the diffusion coefficient are related by Einstein's relationship, the diffusion coefficient D_{pE} also decreases for heavily doped emitter. From the relations in section 6.5 using the narrow base approximation in the emitter, it may appear that the current gain should in fact increase with fall in D_{pE} due to a reduction in I_{pE}. However, this is not actually the case, as described in the next section on Auger recombination.

Auger recombination

Carrier lifetime in a heavily doped semiconductor is also significantly less than that in a lightly doped material. This can be explained by an additional recombination process, which becomes dominant when the doping concentration in the semiconductor is high. This is a three particle, band-to-band recombination process in which the energy and momentum released by the recombination of a hole–electron pair is transferred to a free electron or hole. This is known as *Auger recombination* named after the scientist who first studied this phenomenon. The lifetime associated with Auger recombination process (*Auger lifetime*) is inversely proportional to the square of the doping concentration in the semiconductor. An empirical relationship for the lifetime of holes in the emitter is given [2] as

$$\frac{1}{\tau_p} = 7.8 \times 10^{-13} N_{DE} + 1.8 \times 10^{-31} N_{DE}^2 \text{ s}^{-1} \tag{7.43}$$

where

N_{DE} = donor concentration in the emitter per cubic cm.

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Equation (7.43) signifies that for the emitter doping concentration above 10^{19} cm^{-3}, Auger recombination is the dominant recombination mechanism and the minority carrier lifetime decreases with the square of the doping concentration in the emitter.

The reduction in both diffusion coefficient and lifetime causes the diffusion length for the holes to reduce significantly. Thus, the assumption that $W_E << L_{pE}$ may no longer be valid and the narrow diode approximation no longer holds good. In such situations, the injected hole current into the emitter is given by

$$I_{pE} = \frac{AqD_{pE}n_i^2}{L_{pE}N_{DE}}\exp\left(\frac{V_{BE}}{V_T}\right) = \frac{Aqn_i^2}{N_{DE}}\sqrt{\frac{D_{pE}}{\tau_p}}\exp\left(\frac{V_{BE}}{V_T}\right) \tag{7.44}$$

Due to the reduction in τ_p with increase in emitter doping concentration in the heavy doping range, I_{pE} does not reduce as expected. Therefore, even though N_{DE} is increased, the gain β does not increase.

Band gap narrowing

For low to moderate doping, dopant atoms are sufficiently far away from each other so that their wave functions do not overlap. Thus, the energy levels for the dopants are assumed to be discrete. Also, it is assumed that widely spaced dopants do not affect the perfect periodicity of the silicon crystal lattice. This situation is depicted in the energy band diagram shown in Fig. 7.8(a) for a moderately doped n-type semiconductor.

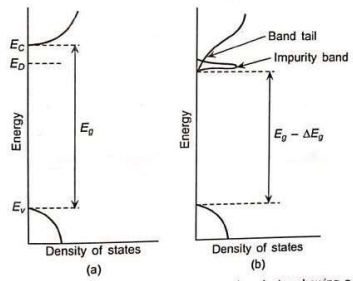


Figure 7.8 Energy band diagram for (a) moderately doped semiconductor showing a single dopant energy level and (b) heavily doped semiconductor showing band gap narrowing effect.

In heavily doped semiconductors, dopant atoms lie close together. So, instead of discrete energy levels, an impurity energy band is created as shown in Fig. 7.8(b) for n^+ material. Also, dopants now disrupt the perfect periodicity of the crystal lattice, giving rise to a band tail. The net result is that the band gap changes from E_g to $(E_g - \Delta E_g)$. For p^+ material also, a similar effect is observed. Only in that case, the band gap narrowing takes place at the valence band edge. An empirical relation for ΔE_g in the heavily doped emitter is given [2] by

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$$\Delta E_g = 18.7 \ln\left(\frac{N_{DE}}{7 \times 10^{17}}\right) \text{meV} \tag{7.45}$$

Thus, for a heavily doped emitter, because of band gap narrowing, the intrinsic carrier concentration in the emitter (n_{iE}) increases to n_i exp ($\Delta E_g/2kT$). While for the base, n_{iB} remains equal to n_i. Assuming $I_C \approx I_{nE}$ and $I_B \approx I_{pE}$, we can now write

$$\beta \approx \frac{I_{nE}}{I_{pE}} = \frac{n'_{pE}}{p'_{nE}} \propto \frac{n_{iB}^2}{n_{iE}^2} \propto \exp\left(-\frac{\Delta E_g}{kT}\right) \tag{7.46}$$

Equation (7.46) shows that β reduces due to band gap narrowing. Also, temperature strongly affects β. As the temperature increases, β increases and consequently the collector current increases. This increase in I_C causes heating and increases the temperature. This further increases β and I_C, and may even cause eventual destruction of the device. This is known as *thermal runaway*.

7.2.5 Emitter Crowding in Bipolar Transistors

Figure 7.9 shows the cross-section of two BJT structures—one with a single base contact and the other with two base contacts. The base resistance in these structures is composed of two parts shown by (a) R_{BX}, the extrinsic base resistance and (b) R_{BI}, the intrinsic base resistance. If V_{BE} is the external voltage applied across the BE terminals, the intrinsic voltage across the BE junction at a distance y from an emitter edge facing the base contact is given by

$$V'_{BE}(y) = V_{BE} - \left[\frac{I_B}{n}(R_{BX} + R_{BI}(y))\right] \tag{7.47}$$

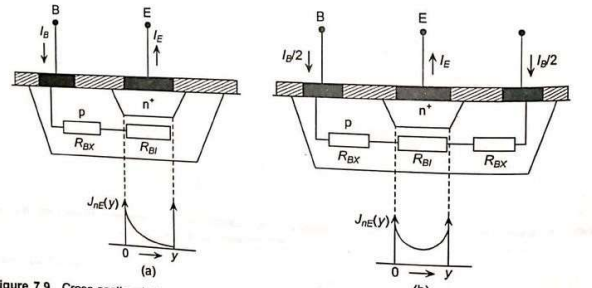


Figure 7.9 Cross-sectional diagram of BJT with (a) one base contact and (b) two base contacts. The variation of emitter current density is also shown.

where
n = number of base contacts and
$R_{BI}(y)$ = intrinsic base resistance for the distance y given by

$$R_{BI}(y) = \frac{y}{\sigma W_B L_E} = \frac{y}{qN_A \mu_p W_B L_E} \tag{7.48}$$

where
σ = conductivity of the base region and
L_E = length of the emitter.

It can be seen from Eqs. (7.47) and (7.48) that a point closer to the emitter edge is more forward biased than a point farther away from it. Since the electron current density $J_{nE}(y)$ is proportional to exp $(V'_{BE}(y)/V_T)$, there is current crowding near the emitter edge as shown in Fig. 7.9. The current is now proportional to the perimeter rather than the area of the emitter. Thus, high injection effects discussed in section 7.2.2 can set in locally at the emitter corners even though the emitter current is not too large. The problem of current crowding can be reduced by having multiple base contacts as well as long thin stripes of the emitter, thereby increasing its perimeter-to-area ratio. Such interlacing of emitter stripes and base contacts in the interdigitated geometry has already been shown in Fig. 7.5.

7.3 NONCONVENTIONAL BJTS

Till now, we have only considered the conventional npn bipolar transistor, where the emitter, base and collector regions are all made of the same material. In this section, we shall discuss some nonconventional BJT structures, where the emitter is made of a dissimilar material. Although the technology for fabrication of these devices is more complicated, their performance, especially at high operating frequencies, is much superior to the conventional BJT.

7.3.1 Polysilicon Emitter Transistor

From the discussion on high frequency transistors in section 7.1, we have seen that the junction depletion capacitances (C_{JE} and C_{JC}) and the base resistance R_B play a crucial role in determining the ac performance of a BJT. The capacitances have two components, namely planar and side wall (peripheral). Let us consider a square emitter having an area a^2 and depth W_E. Since the capacitances are proportional to their respective areas, the side wall capacitance to planar capacitance ratio is proportional to $\{(4aW_E)/a^2\} = (4W_E)/a$. Now this ratio increases when a is reduced, showing a dominance of the side wall capacitance for smaller size emitters. Thus, for effective scaling of C_{JE}, the emitter junction depth (W_E) must be reduced along with the emitter side of dimension a. However, a shallow emitter causes a reduction in the common emitter current gain β. As illustrated in Fig. 7.10, the slope of the minority carrier distribution in emitter becomes steeper when the emitter depth decreases. Consequently, I_{pE} increases and β decreases. This is also reflected in Eq. (6.43), which shows that a reduction in W_E causes a decrease in β.

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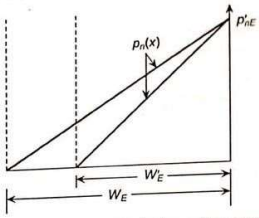


Figure 7.10 Minority carrier concentration distribution in the emitters of two BJTs having two different emitter widths.

This problem can be overcome by using a polysilicon emitter transistor. To fabricate the polysilicon emitter, after opening the emitter window, a layer of polycrystalline silicon is deposited by CVD technique. This polysilicon layer is then patterned and doped by ion-implantation. During subsequent annealing, this polysilicon layer acts as a source of dopant and a shallow emitter is realized. The process steps are outlined in Figs. 7.11(a), (b), and (c). As shown in Fig. 7.11(d), this is followed by oxidation, opening of windows in the oxide, metallization, and then patterning of the metal in order to take contact from the polysilicon emitter. In the polysilicon emitter transistor, the single-crystal region of the emitter, which determines the side wall capacitance, has been made shallow. On the other hand, the slope of the minority carrier distribution, as shown in Fig. 7.12 is less steep as the excess injected hole concentration falls to zero only at the polysilicon–metal interface. Hence, the common emitter current gain β improves. In fact, β is further improved due to the presence of a thin (~10 Å) oxide layer between the single crystal and polysilicon region. This oxide layer suppresses the hole injection from the base to emitter. On the other hand, the oxide layer increases the emitter resistance, degrading the circuit performance, particularly for small geometry devices. Therefore, the trade-off between β and R_E has to be decided carefully by controlling the oxide thickness.

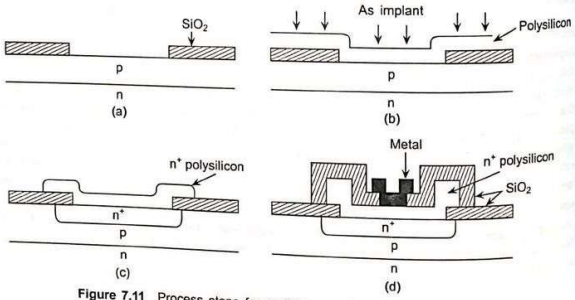


Figure 7.11 Process steps for realizing a polysilicon emitter.

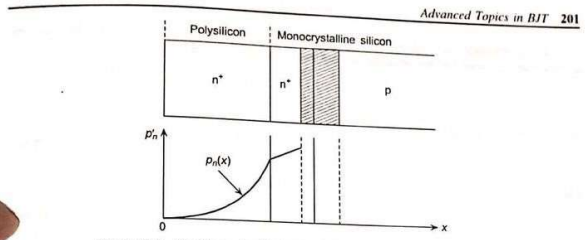


Figure 7.12 Minority carrier distribution in the polysilicon emitter of a BJT.

7.3.2 Heterojunction Bipolar Transistors (HBT)

As discussed in section 7.2.4, we have seen that the current gain of a BJT cannot be increased simply by increasing the emitter doping. One way to improve the current gain β will be to suppress the injection of holes into the emitter, while keeping the electron injection in the base unchanged. This can be achieved if the emitter and base of a transistor are made of two semiconductor materials with different band gaps. Such a junction is called a *heterojunction*. This section details the basic concept of heterojunctions. Consider a heterojuction where material 1 refers to a semiconductor with band gap E_{g1} and material 2 refers to a semiconductor with band gap E_{g2}. Let $E_{g1} > E_{g2}$, as shown in Fig. 7.13(a). In addition, let us assume that the two materials have the same electron affinity, that is, $\chi_1 = \chi_2$. Then, if material 1 is n-type and material 2 is p-type, an unequal barrier is formed at the conduction and valence band edges when the two materials are brought in contact as shown in Fig. 7.13 (b). Therefore, the forces acting on the electrons and holes are different and the potential barrier for holes is greater than the potential barrier for electrons by ΔE_g ($= E_{g1} - E_{g2}$). This is impossible to achieve in a homojunction (where we have the same band gap on both sides of the junction).

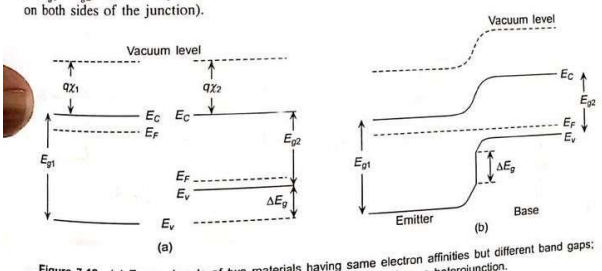


Figure 7.13 (a) Energy bands of two materials having same electron affinities but different band gaps; (b) energy band diagram when these materials form a heterojunction.

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For a heterojunction bipolar transistor, assuming negligible recombination in the base, the approximate relations for the collector current and base current can be written as

$$I_C = \frac{AqD_n n_{i2}^2}{W_B N_A} \exp\left(\frac{V_{BE}}{V_T}\right) \tag{7.49}$$

$$I_B = \frac{AqD_p n_{i1}^2}{W_E N_{DE}} \exp\left(\frac{V_{BE}}{V_T}\right) \tag{7.50}$$

where the subscripts 1 and 2 refer to materials 1 and 2 respectively.

Since $E_{g1} > E_{g2}$,

$$n_{i1} < n_{i2}$$

Assuming $N_{C1} = N_{C2}$ and $N_{V1} = N_{V2}$, we can write

$$n_{i1}^2 = n_{i2}^2 \exp\left(-\frac{\Delta E_g}{kT}\right) \tag{7.51}$$

Thus, from Eqs. (7.49)–(7.51), the current gain β of an HBT (β_{het}) can be written as

$$\beta_{het} = \frac{D_n W_E N_{DE}}{D_p W_B N_A} \exp\left(\frac{\Delta E_g}{kT}\right) = \beta_{hom} \exp\left(\frac{\Delta E_g}{kT}\right) \tag{7.52}$$

where

β_{hom} = current gain for the corresponding homojunction transistor.

However in reality, the situation is not so simple since the electron affinity of the two materials is not same. For example, let us consider an AlGaAs/GaAs heterojunction, which is the most widely used system for HBT applications. Here, AlGaAs has a larger band gap (depending on the percentage of Al, E_g can vary in the range 1.42–2.16 eV) but a smaller electron affinity as shown in Fig. 7.14(a). For such a case the energy band at the junction is shown in Fig. 7.14(b). It can be seen from the figure that there is a *spike and notch* formation at the conduction band edge and a *step* formation at the valence band edge. The spike impedes the flow of electrons from the

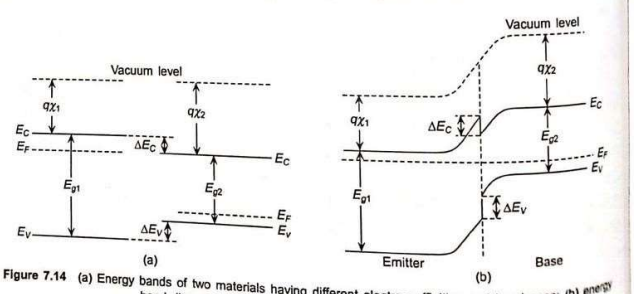


Figure 7.14 (a) Energy bands of two materials having different electron affinities and band gaps; (b) energy band diagram when these materials form a heterojunction.

emitter to the base. The value of the discontinuity in the conduction band (ΔE_c) is equal to the difference in the electron affinity of AlGaAs and GaAs, that is, $\Delta E_c = \chi_1 - \chi_2$. Consequently, the height of the step at the valence band is given by

$$\Delta E_v = \Delta E_g - \Delta E_c$$

Therefore, the improvement in β is less than that predicted by Eq. (7.52). For this case, β is given by

$$\beta_{het} = \beta_{hom} \frac{N_{CE} N_{VE}}{N_{CB} N_{VB}} \exp\left(\frac{\Delta E_v}{kT}\right) \tag{7.53}$$

Another problem is that the injected electrons are collected in the notch in E_c. This increases the recombination in the base and degrades β. This can be countered by using a graded heterojunction, where the percentage of aluminium in AlGaAs is increased slowly as we move away from the junction. Grading smoothens out the conduction band by removing the spike and thereby reduces the energy barrier for electrons in the conduction band. Hence, Eq. (7.52) remains valid.

For an $Al_{0.3}Ga_{0.7}As/GaAs$ heterojunction, $\Delta E_g = 0.374$ eV, and at room temperature $\exp(\Delta E_g/kT) \approx 10^6$. The large improvement in emitter efficiency can be traded for higher base doping to improve the frequency response of the device. We can see from Eq. (7.52) that due to the presence of the term involving ΔE_g, the current gain of an HBT, which has a higher doping in the base than in the emitter, can still be much more than that of a conventional BJT with heavy emitter doping. In fact, in most HBTs, the doping concentration in the base is about two orders of magnitude higher than in the emitter as shown in Fig. 7.15. For a higher doping concentration of the base, the penetration of the depletion layer at the CB junction into the base is less. The 'Early voltage' will therefore be higher, and the base width can be reduced without causing punch-through. A lower base width results in an increase in the value of f_T. A higher base doping also

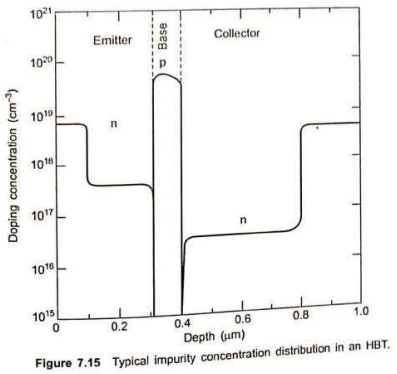


Figure 7.15 Typical impurity concentration distribution in an HBT.

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means a lower base resistance. Thus, the value of f_{max} given by Eq. (7.27) is further increased. Also, since the base doping is high, the high injection effects (discussed in section 7.2.2) are much less. Such a reduction in base resistance also reduces current crowding and therefore increases the current handling capability. Another advantage of HBTs is a lower capacitance across the BE junction due to the lower emitter doping.

Interestingly, it is possible to consider a conventional homojunction BJT having a heavily doped emitter as an *inverse* HBT, if we consider the band gap narrowing effect in the emitter (as discussed in section 7.2.4). In that case the band gap in the emitter is less than that in the base, causing a reduction in the current gain β, while for an actual HBT, the band gap in the emitter is larger, causing an improvement in β. Also, we have seen that band gap narrowing in the emitter of a homojunction BJT causes β to increase with temperature leading to thermal runaway. On the other hand in a HBT, β actually reduces with increase in temperature as can be seen from Eq. (7.52). Thus, the problem of thermal runaway can be averted by using HBT.

The concept of HBT was first proposed by William Shockley in his patent application in 1948. However, the first HBT was realized much later, with the advent of MBE and MOCVD techniques of deposition. Today HBT is a mature technology used for realizing various high speed circuits. The lowest delays in digital circuits have been reported in circuits that use HBTs.

PROBLEMS

P7.1 In an npn bipolar transistor, the base doping is decreasing exponentially as we go away from the base emitter junction and is given by $N_A(x) = N_1 \exp\{-(x/x_0)\}$, where $N_1 = 10^{16}$ cm^{-3} is the concentration at the base–emitter junction and $x_0 = 0.2$ μm.

(a) Find the value of the built-in electric field.

(b) Find the base transit time τ_{tB}.

(c) What would have been the value of τ_{tB} if the base doping was uniform (= N_1)?

(d) Compare and comment on the two values of τ_{tB} obtained in (b) and (c).

[Given $D_n = 20$ cm^2/s, $W_B = 0.4$ μm. Assume $L_n >> W_B$]

P7.2 Show that for any arbitrary doping distribution in the base, where the doping concentration reduces from the emitter side to the collector side of the base, the base transit time is less than that in a uniformly doped base.

P7.3 (a) A constant electric field of –4 KV/cm exists in the nonuniformly doped base of an npn bipolar transistor having a base width of 0.3 μm and a doping concentration of 10^{17} cm^{-3} at the edge of the emitter–base depletion region. What is the doping concentration at the edge of the collector–base depletion region? (Assume that the depletion widths on both sides of the base are negligibly small.)

(b) If the average value of diffusion constant (D_n) in the base is 25 cm^2/s, what is the base transit time for this transistor? Compare it with the base transit time for a transistor having uniform base doping concentration.

P7.4 The band gap narrowing in heavily doped silicon is empirically modelled as $\Delta E_g = 18.7 \ln(N_D/7 \times 10^{17})$ meV, where N_D is the doping concentration (in per cm^3) of the

material. If the emitter of an npn transistor is doped to 10^{20} cm^{-3}, what will be the change in β from its ideal value, if only the band gap narrowing effect is considered? Assume base transport factor = 1.

P7.5 For an npn transistor, the neutral base width $W_B = 2$ μm. The mobility and lifetime of electrons in the base are 1000 cm^2/Vs and 1 μs respectively. Assuming $\gamma = 1$, calculate the

(i) transit time of electrons through the neutral base.

(ii) α-cutoff and β-cutoff frequencies of the transistor.

P7.6 The base width of a bipolar transistor is 0.5 μm and the emitter is a square of dimensions 1 μm × 1 μm. The doping concentration in the base is 10^{17} cm^{-3}. Assuming $\mu_p = 400$ cm^2/Vs and the depletion layer width to be negligible in the base, plot the emitter current density variation along the EB junction for $I_B = 0.1$ mA, if (a) the transistor has a single base contact as depicted in Fig. 7.9(a) and (b) the transistor has two base contacts as depicted in Fig. 7.9(b).

P7.7 In a heterojunction bipolar transistor (HBT), the band gap of the emitter is 1.1 eV and that of the base is 0.8 eV. The doping concentrations are such that in the emitter, the Fermi level is 0.3 eV below E_C and in the base it is 0.5 eV below E_C. If the electron affinity of the emitter is 4.05 eV and that of the base is 4.0 eV, draw the energy band diagram at the emitter–base abrupt junction.

REFERENCES AND SUGGESTED FURTHER READING

[1] Taylor, G.W. and J.G. Simmons, Figure of merit for integrated bipolar transistors, *Solid State Electronics*, Vol. 29, p. 941, 1986.

[2] Alamo, J. Del, S. Swirhun, and R.M. Swanson, Simultaneous measurement of hole lifetime, hole mobility, and band gap narrowing in heavily doped n-type silicon, *IEDM Technical Digest*, p. 290, 1985.

[3] Sze, S.M., *Physics of Semiconductor Devices*, 2nd ed., Wiley, N.Y., 1981.

[4] Roulston, David J., *Bipolar Semiconductor Devices*, McGraw Hill, New York, 1990.

[5] Streetman, B.G. and S. Banerjee, *Solid State Electronic Devices*, 5th ed., Prentice Hall, New Jersey, 2000.

[6] Ashburn, P., *Design and Realization of Bipolar Transistors*, John Wiley and Sons, New York, 1988.

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8 Thyristors

8.1 INTRODUCTION

In the previous chapters, we have already seen that a diode remains off (that is, it passes negligible current) when a reverse bias is applied and turns on under the application of a forward bias. Thus, a diode can be used as a switch, which can be put off or on by reversing the polarity of the voltage applied across it. Similarly, a bipolar junction transistor can be used as a switch by changing the base current to drive the device from cut-off to saturation. However, a number of switching applications require a device that will remain 'off' (that is, in blocking state) under forward bias and will switch to an 'on' (or conducting) state under the application of an external signal. The *thyristor*, also known as *semiconductor controlled rectifier* (SCR), is one such device which remains in a high impedance state under forward bias condition (*forward blocking*) until a switching signal is applied which switches it to a low impedance state (*forward conduction*). Thyristors are important semiconductor power devices, which may have current ratings greater than 1000 A and voltage ratings as large as 10,000 V.

The SCR is basically a three junction p-n-p-n structure, as shown in Fig. 8.1(a), with an anode terminal connected to p_1, a cathode terminal connected to n_2, and a third terminal (gate) connected to the internal p-region (p_2). The p_1-n_1, n_1-p_2, and p_2-n_2 junctions are referred to as J_1, J_2, and J_3 respectively. The doping concentration distributions of the different regions are shown in Fig. 8.1(b). Let us first consider the operation of the two-terminal p-n-p-n device, that is, one with floating gate terminal. Subsequently, we shall discuss the use of the gate to control the operation of the thyristor.

8.2 OPERATION OF THE TWO TERMINAL p-n-p-n DEVICE

The typical current–voltage characteristics of the two terminal p-n-p-n device are shown in Fig. 8.2. As we can see from the figure, there are five distinct regions of operation marked in Fig. 8.2 as (a) forward blocking, (b) negative resistance, (c) forward conducting, (d) reverse blocking, and (e) reverse breakdown. We shall now describe the SCR operation in each of these regions individually.

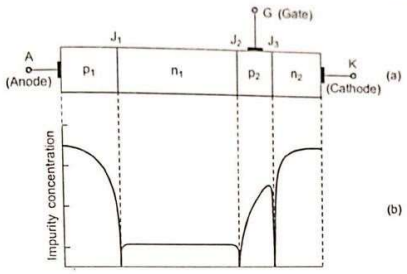


Figure 8.1 (a) Basic structure of a thyristor and (b) the doping concentration distributions in the various regions.

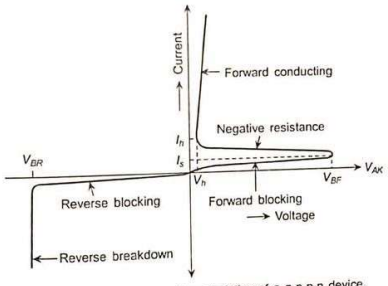


Figure 8.2 Current–voltage characteristics of a p-n-p-n device.

8.2.1 Forward Blocking State

When a positive voltage V_{AK} is applied across the two terminals A (anode) and K (cathode), the junctions J_1 and J_3 get forward biased unlike J_2 which gets reverse biased. Most of the applied voltage is therefore dropped across the junction J_2. We can consider this device as a combination of a pnp (consisting of $p_1n_1p_2$) and an npn (consisting of $n_1p_2n_2$) transistor as shown in Fig. 8.3. Also, the base of the pnp transistor serves as the collector for the npn and the collector of the pnp acts as the base of the npn transistor. Since J_1 and J_3 are forward biased but J_2 is reverse biased, both the transistors are operating in the normal active mode. Therefore, we can write

$$I_{C1} = \alpha_1 I + I_{CBO1} = I_{B2} \tag{8.1}$$

$$I_{C2} = \alpha_2 I + I_{CBO2} = I_{B1} \tag{8.2}$$

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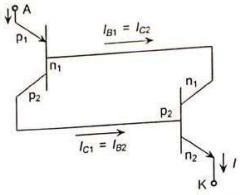


Figure 8.3 Two-transistor equivalent circuit for a p-n-p-n device.

where I refers to the terminal current of the device as well as the emitter currents of the individual bipolar transistors.

In Eqs. (8.1) and (8.2), the subscripts 1 and 2 refer to the pnp and the npn transistors respectively and all the other current symbols have usual meaning. Again, from Fig. 8.3, it is clear that the terminal current I at the anode and the cathode is the sum of I_{C1} and I_{C2}. Therefore, from Eqs. (8.1) and (8.2), we obtain

$$I = I_{C1} + I_{C2} = (\alpha_1 + \alpha_2)I + I_{CBO1} + I_{CBO2} \tag{8.3}$$

and hence,

$$I = \frac{I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)} \tag{8.4}$$

From Eq. (8.4), we see that as long as α_1 and α_2 are small, the current I remains small. The small values of α_1 (and α_2) imply that virtually no holes injected across the forward-biased junction J_1 (or the electrons injected across forward-biased junction J_3) survive to be swept across the reverse-biased junction J_2 by transistor action. Thus, the device is in the forward blocking mode as depicted in Fig. 8.2.

8.2.2 Triggering and Forward Conduction of the p-n-p-n Diode

We have already seen (from our discussion in section 7.2.2) that the current gain varies with collector current in a bipolar transistor. For very low current, the recombination in the EB depletion region dominates and α is small. As the current increases, injection begins to dominate resulting in an increase in α. Also, with increase in V_{AK}, the reverse bias across J_2 increases, resulting in an increase in the width of the depletion layer, which is shown in Fig. 8.4(a). Consequently, α increases due to a reduction in the neutral base width. Also, the avalanche multiplication of carriers becomes significant at higher values of V_{AK}, which effectively increases the value of α further.

As a result of one or all of these effects, $\alpha_1 + \alpha_2$ approaches unity. From Eq. (8.4), we find that this results in a large increase in the terminal current. In physical terms, increase in α_1 and α_2 means that many holes now reach p_2 and many electrons reach n_1. This process is regenerative since a large supply of electrons at n_1 allows a larger injection of holes across J_1 without disturbing the charge neutrality. Similarly, a large supply of holes at p_2 allows a larger injection of electrons across J_3. This large supply of electrons and holes shrinks the depletion region at J_2 as shown in

Fig. 8.4(b) till eventually all the three junctions become forward-biased. This is equivalent to both the pnp and npn transistors switching from the active to the saturation mode of operation. Now, the voltage across the p-n-p-n structure consists of the three forward voltages across J_1, J_2, and J_3. Since two of these voltage drops effectively cancel each other, the actual voltage drop is approximately equal to the drop across one forward-biased p-n junction. The device has now been switched (or triggered) from the forward blocking state to its forward conducting state. The forward voltage required to initiate the switching is usually denoted by V_{BF}.

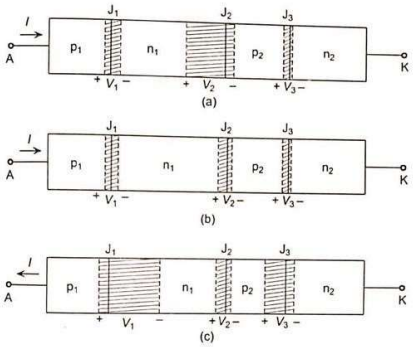


Figure 8.4 The p-n-p-n device in (a) the forward blocking, (b) the forward conducting, and (c) reverse blocking state.

Another process known as *dV/dt* triggering may also take place, if the applied voltage is changed rapidly. With increase in the depletion region width across J_2, electrons are removed from n_1 region and holes from p_2 region. If *dV/dt* is large, this fast rate of charge removal can also increase the current significantly, thereby switching the device to the conducting state at voltages below the steady state triggering voltage V_{BF}.

8.2.3 Reverse Blocking and Breakdown

If the polarity of V_{AK} is reversed, the junctions J_1 and J_3 get reverse-biased while J_2 get forward-biased as shown in Fig. 8.4(c). The current is small due to the two reverse biased junctions. The device is now in its reverse blocking state. As shown in Fig. 8.2, the reverse current remains low till the breakdown voltage V_{BR} is reached. At V_{BR}, the current increases due to avalanche multiplication. For a properly designed device with a long n_1-region, the value of V_{BR} should be close to the avalanche breakdown voltage of junction J_1. For a well-designed device, V_{BR} can be as high as several thousand volts.

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8.3 OPERATION OF A THYRISTOR

We have already discussed the various mechanisms by which the p-n-p-n structure can switch from its forward blocking state to forward conduction state. Essentially this involves increasing the current till the transistor regenerative mechanism takes place and can be achieved simply by increasing V_{AK}. Alternatively, it can be triggered by holding V_{AK} constant and raising the temperature (or shining light) to facilitate generation of more carriers. In a thyristor, this is achieved by applying an external signal to the gate terminal. When the device is biased in its forward blocking state, a small current applied to the gate can initiate the switching to forward conduction. A positive gate current supplies holes to p_2, the base of the npn transistor, which results in increased injection of electrons from n_2. The transistor action thus starts in the npn transistor. These injected electrons reach n_1 after a delay (equal to the transit time through the base). This allows increased hole injection from p_1 and the process becomes self-sustaining. In most thyristors, the gate current is small (a few mA), while the total device current I can be very large (many amperes). Evidently, the switching can be initiated at a lower value of V_{AK} by increasing the gate current as shown by Fig. 8.5.

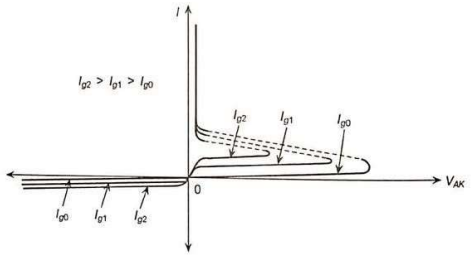


Figure 8.5 Current–voltage characteristics of a thyristor.

Once the device has switched to conducting state, it is not necessary to maintain the gate current, that is, the device operates independently of the gate current. Actually, the gate loses control after the regenerative process has begun. Thus for most devices, a gate current pulse with duration of a few microseconds is sufficient to ensure switching. A simple application of thyristor is to deliver a variable power to a load such as a furnace, or a heater, or simply a light bulb as shown in Fig. 8.6. The amount of power delivered to the load will then depend on the time when the device turns on, that is, on the timing of the gate current pulse. If the timing is adjusted so that the device turns on at the beginning of each power cycle, more power will be delivered to the load. On the other hand, if the pulse is delayed, the device will be 'on' only for a part of the cycle and less power will be delivered. This control mechanism can be utilized to monitor the intensity of a light bulb or to control the temperature in a furnace.

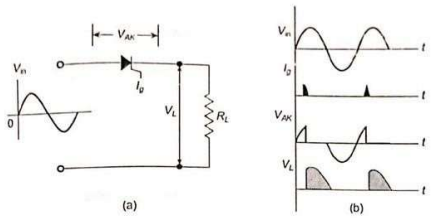


Figure 8.6 Thyristor–controlled power delivery to a load.

In order to switch the device back to its forward blocking state, the current has to be reduced below a critical value, called the *holding current*, I_H. This is usually done by reversing the polarity of V_{AK}. This ensures that the n_1 and p_2 bases are completely emptied of carriers. However as complete removal of carriers can be achieved only by recombination, sufficient time must be allowed. Otherwise, if a forward voltage is applied before all the charge has recombined, the device may again turn on. The turn-off time of a thyristor is usually 5–10 times the carrier lifetime in the base.

In some specially designed thyristors, gate turn-off can be used. Accordingly if the gate voltage is reversed, holes are extracted from the p_2 region. If this rate of hole extraction is sufficiently large, the regenerative action stops and the device can turn off. However, this gate turn-off capability can be utilized only over a limited voltage range for a given device.

8.4 BIDIRECTIONAL SWITCHES

A bidirectional switching device can switch from its blocking state to conducting state for both positive and negative applied voltages and is therefore useful in ac applications. A bidirectional p-n-p-n switch is also known as a *diac* (Diode ac switch) and the basic structure is shown in Fig. 8.7(a). It can be viewed as two p-n-p-n devices with the anode of the first device connected to the cathode of the second and vice versa as shown in Fig. 8.7(b). A diac is a two-terminal device. When a positive voltage V_{AB} is applied across these two terminals, only the p_1-n_1-p_2-n_2 device comes into picture and produces the forward I–V characteristics as depicted in Fig. 8.7(c). Analogously, when the polarity of V_{AB} is reversed, the p_2-n_1-p_1-n_3 device comes into play, and generates the reverse I–V characteristics shown in Fig. 8.7(c). The symmetry of the configuration ensures identical performance for either polarity of applied voltage.

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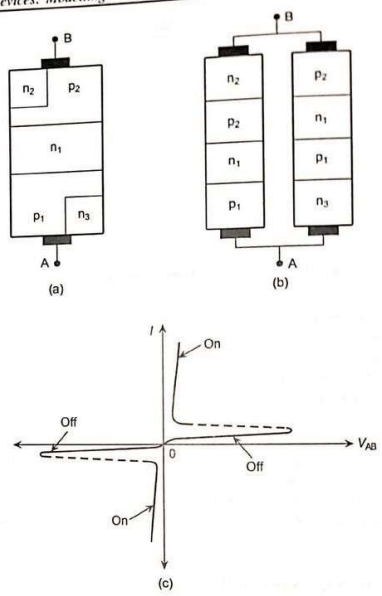


Figure 8.7 (a) Basic structure of a diac; (b) equivalent circuit for diac showing two p-n-p-n devices; and (c) current–voltage characteristics of diac.

A bidirectional three-terminal device is called a *triac*. The basic structure is shown in Fig. 8.8. The I–V characteristics of a triac are similar to that of a diac. However, in a triac the switching can be achieved in either direction by applying a low voltage, low current pulse between the gate and one of the two terminals A and B. This allows control over the breakover voltage as in a thyristor.

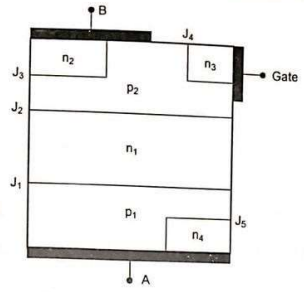


Figure 8.8 Basic structure of triac.

REFERENCES AND SUGGESTED FURTHER READING

[1] Sze, S.M., *Physics of Semiconductor Devices*, 2nd ed., Wiley, N.Y., 1981.

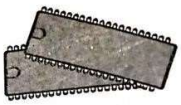
[2] Roulston, David J., *Bipolar Semiconductor Devices*, McGraw Hill, New York, 1990.

[3] Streetman, B.G. and S. Banerjee, *Solid State Electronic Devices*, 5th ed., Prentice Hall, New Jersey, 2000.

[4] Baliga, B. Jayant, *Modern Power Devices*, John Wiley & Sons, New York, 1987.

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9 Junction Field Effect Transistor and Metal-Semiconductor Field Effect Transistor

9.1 INTRODUCTION

In the previous chapters we have considered bipolar devices (p-n junctions and BJTs) where both types of charge carriers (electrons and holes) participate in the conduction process. For example, in a forward-biased p-n junction, electrons are injected from n-region to p-region and holes are injected from p-region to n-region. The total current is actually the sum of these two components. In contrast, we shall now discuss *unipolar* devices, where primarily one type of charge carrier participates in the conduction process. This chapter discusses three such devices, namely metal-semiconductor junctions, junction field effect transistors (JFETs), and metal-semiconductor field effect transistors (MESFETs). The most important unipolar device in VLSI circuits—metal oxide semiconductor field effect transistors (MOSFETs)—is dealt with in the next two chapters.

9.2 METAL-SEMICONDUCTOR JUNCTION

The metal-semiconductor (M-S) junction is the oldest practical semiconductor device. This can either be rectifying or non-rectifying. The rectifying metal-semiconductor junction is also known as *Schottky diode*, while the non-rectifying junction is called an *ohmic contact*. We shall first consider the energy band diagram of a metal-semiconductor junction.

9.2.1 Energy Band Diagram of M-S Junction

Figure 9.1(a) shows the energy band diagram of an isolated metal adjacent to (but not in contact with) an isolated n-type semiconductor. The metal work function (that is, the energy difference between the vacuum level and the Fermi level), $q\phi_m$, is usually different from the semiconductor

work function $q\phi_s$. Figure 9.1 considers the case where $\phi_m > \phi_s$. The electron affinity of the semiconductor (that is, the energy difference between the vacuum level and the conduction level), called χ, is also shown in the figure. When the metal makes intimate contact with the semiconductor, a flow of charges results in aligning of the Fermi levels at thermal equilibrium (refer to section 1.3.9). Also, since the vacuum level represents the potential variation in the M-S system, it must be continuous (a discontinuity in the vacuum level would imply an infinite electric field). These two requirements give rise to a unique energy band diagram for the M-S system as shown in Fig. 9.1(b). For such an ideal system, the barrier height at the junction for the injection of electrons from the metal to the semiconductor conduction band (ϕ_{Bn}) is the difference between the metal work function and the electron affinity of the semiconductor. That is,

$$q\phi_{Bn} = q\phi_m - q\chi_s \tag{9.1}$$

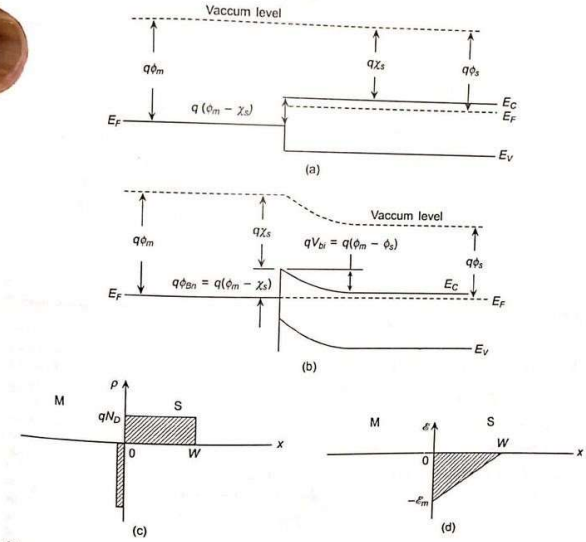


Figure 9.1 (a) Energy band diagram of isolated metal and n-type semiconductor where $\phi_m > \phi_s$, (b) energy band diagram, (c) charge distribution, and (d) electric field distribution at thermal equilibrium for M-S junction of these two materials.

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and the built-in potential is defined as the difference of the metal and semiconductor work functions (refer to section 4.1.1) given by

$$V_{bi} = \phi_m - \phi_s = \phi_{Bn} - (\phi_s - \chi_s) = \phi_{Bn} - V_T \ln\left(\frac{N_C}{N_D}\right) \tag{9.2}$$

where

N_C = effective density of states in the conduction band and

N_D = doping concentration in the n-type semiconductor.

The built-in potential appears across a depletion layer created at the semiconductor surface. In this region, as shown in Fig. 9.1(b), the band bends upwards, reflecting the reduction in the electron concentration in the n-type substrate.

The band diagram for a metal/p-type semiconductor junction, where $\phi_m < \phi_s$ is shown in Fig. 9.2. In this case, the potential barrier for hole injection from the metal to the semiconductor (ϕ_{Bp}) is given by

$$q\phi_{Bp} = E_g - q(\phi_m - \chi_s) \tag{9.3}$$

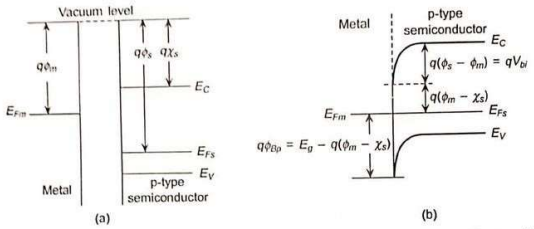


Figure 9.2 (a) Energy band diagram of isolated metal and p-type semiconductor and (b) energy band diagram for a metal/p-type semiconductor junction with $\phi_m < \phi_s$.

and the built-in potential is given by

$$V_{bi} = \phi_s - \phi_m = \phi_{Bp} - \left(\frac{E_g}{q} - \phi_s + \chi_s\right) = \phi_{Bp} - V_T \ln\left(\frac{N_V}{N_A}\right) \tag{9.4}$$

where

E_g = band gap of the semiconductor.

However, in practice this simple theory is not very accurate. Figure 9.3 shows the measured barrier heights for n-type silicon and n-type gallium arsenide with different metals. While the barrier height increases with an increase in ϕ_m, the dependence is not as strong as predicted by Eq. (9.1), particularly for GaAs. This is because, in a practical M-S contact, the discontinuity of the crystal lattice at the semiconductor surface (in contact with metal) produces a large number of surface states distributed in the forbidden energy gap. These surface states, which can either be acceptor-type or donor-type, give rise to charges at the interface. Moreover, it is very difficult to fabricate a perfect metal-semiconductor junction. Usually, there is a thin interfacial layer between

the metal and the semiconductor. In case of metal-silicon junctions, a SiO_2 layer of about 10 Å separates the metal and silicon. Although carriers can tunnel through this thin layer, it creates an additional barrier for carriers flowing across the junction. Therefore, the values of ϕ_{Bn} and ϕ_{Bp} as predicted by Eqs. (9.1) and (9.3) are hardly ever realized. Hence, it is advisable to use measured values of barrier heights, as given in Fig. 9.3, for designing the device.

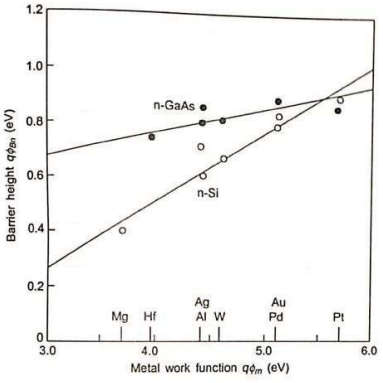


Figure 9.3 Measured barrier heights for different metals [1] on n-Si and n-GaAs.

Figure 9.1(c) shows the charge distribution of a metal n-type semiconductor junction with a negative surface charge on the metal and positive charge on the semiconductor. This charge distribution is identical to a p^+n junction with a corresponding identical field distribution as shown in Fig. 9.1(d). Similarly, the metal p-type semiconductor junction as shown in Fig. 9.2 behaves as an n^+p junction. As in the case of a p-n junction, we can now assume that any applied voltage V appears across the depletion layer. Therefore, the voltage drop in the depletion layer when the M-S junction is biased is given by $(V_{bi} - V)$, where V is positive for a forward-biased junction and negative for a reverse-biased junction. The energy band diagrams for metal/n-type and metal/p-type semiconductor junctions under different biasing conditions are shown in Fig. 9.4.

We can now obtain the expressions for the depletion layer width as well as the electric field in the same manner as used in the case of an abrupt one-sided p^+n junction. For an applied voltage V across this M-S junction, the depletion region width is given by

$$W = \sqrt{\frac{2\varepsilon_s}{qN_D}(V_{bi} - V)} \tag{9.5}$$

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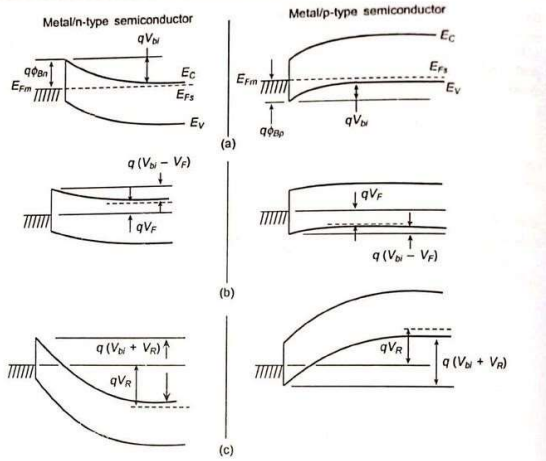


Figure 9.4 Energy band diagrams for metal/n-type and metal/p-type semiconductor junctions under (a) thermal equilibrium, (b) forward bias, and (c) reverse bias.

and the electric field as a function of position is given by

$$|\mathscr{E}(x)| = \mathscr{E}_m - \frac{qN_D x}{\varepsilon_s} \tag{9.6}$$

where

$\mathscr{E}_m$ = peak electric field at $x = 0$ given by

$$\mathscr{E}_m = \frac{2(V_{bi} - V)}{W} \tag{9.7}$$

9.2.2 Current–Voltage Characteristics of M-S Junction

Even though the expressions for depletion region width and electric field distribution of an M-S junction are similar to that of a p^+n junction, the current transport mechanisms are completely different. While the current flow in p-n junction is primarily due to the injection of minority carriers; in an M-S junction, it is mostly due to majority carriers. For Schottky diodes with moderately doped semiconductor ($N_D < 10^{19}$ cm^{-3}), the dominant transport mechanism is thermionic emission of majority carriers over the barrier into the metal. Figure 9.5 illustrates this

mechanism for a metal n-type semiconductor junction. At thermal equilibrium (Fig. 9.5a), electrons from the semiconductor tend to flow into the metal and there is an opposing flow of electrons from the metal into the semiconductor. As both these components are equal and opposite, there is no net current. The electron concentration in the semiconductor at the M-S junction, n_s, is given by

$$n_s = N_C \exp\left[-\frac{(E_C - E_F)|_{x=0}}{kT}\right] = N_C \exp\left[-\frac{\phi_{Bn}}{V_T}\right] = N_D \exp\left[-\frac{V_{bi}}{V_T}\right] \tag{9.8}$$

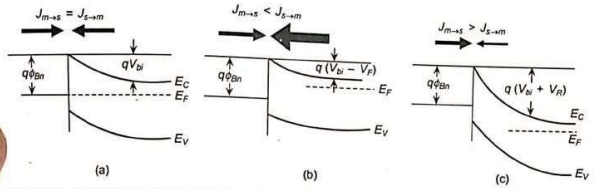


Figure 9.5 Current transport in metal/n-type semiconductor junction at (a) thermal equilibrium, (b) forward bias, and (c) reverse bias.

The electron current density due to flow of electrons from semiconductor to metal ($J_{s\to m}$) is proportional to the density of electrons at the boundary (n_s). Since the electron current density from metal to semiconductor ($J_{m\to s}$) is equal to the electron current density from semiconductor to metal ($J_{s\to m}$) at thermal equilibrium, we can write

$$|J_{m\to s}| = |J_{s\to m}| = C_1 N_C \exp\left[-\frac{\phi_{Bn}}{V_T}\right] \tag{9.9}$$

where

C_1 = proportionality constant.

When a positive voltage V_F is applied to the metal, the potential barrier for electrons in the semiconductor is reduced by qV_F as shown in Fig. 9.5(b). Replacing V_{bi} with $(V_{bi} - V_F)$ in Eq. (9.8), we find that the electron density at the semiconductor surface now increases by $\exp(V_F/V_T)$, implying an increase in the electron current density $J_{s\to m}$ by the same factor. However, since ϕ_{Bn} remains at its equilibrium value, the electron current density from metal to semiconductor $J_{m\to s}$ does not change. Thus, the net current under this forward bias condition can be obtained as the difference of these two current density components given by

$$J = J_{s\to m} - J_{m\to s} = C_1 N_C \exp\left[-\frac{\phi_{Bn}}{V_T}\right]\left[\exp\left(\frac{V_F}{V_T}\right) - 1\right] \tag{9.10}$$

Analogously, when a negative voltage $-V_R$ is applied to the metal, $J_{s\to m}$ reduces by a factor $\exp\{-(V_R/V_T)\}$ while $J_{m\to s}$ does not change, as shown in Fig. 9.5(c) and the net current can be obtained by replacing V_F with $-V_R$ in Eq. (9.10). The term C_1N_C can be expressed as

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$$C_1N_C = A^{**}T^2$$

where

A^{**} = effective Richardson's constant (with unit A/(K^{2}cm^2)) and

T = absolute temperature.

A^{**} depends on the effective mass of the carriers and its value is about 110 for n-Si and 32 for p-Si. The current–voltage characteristics of a metal-semiconductor junction under thermionic emission condition is therefore given by

$$I = I_s\left[\exp\left(\frac{V}{V_T}\right) - 1\right] \tag{9.11}$$

where

$$I_s = AA^{**}T^2 \exp\left(-\frac{\phi_{Bn}}{V_T}\right) \tag{9.12}$$

where

A = area of the M-S junction.

Thus, we see that for the cases discussed in this section, the characteristics of an M-S junction are similar to a p-n junction and the junction has rectifying properties.

Example 9.1

A metal-semiconductor Schottky diode (MS1) fabricated on n-GaAs with $N_D = 10^{16}$ cm^{-3} has a reverse saturation current density $J_s = 5 \times 10^{-8}$ A/cm^2 at 300 K.

(a) Calculate the barrier height and the depletion region width at thermal equilibrium assuming A^{**} for n-GaAs to be 4.4 A/(K^{2}cm^2).

(b) Calculate the minority carrier diffusion current density and show that the Schottky diode current is mostly due to majority carriers (electrons). Assume the hole diffusion coefficient to be 10 cm^2/s and a minority carrier lifetime of 1×10^{-8} s. Also, $N_C = 4.7 \times 10^{17}$ cm^{-3} for GaAs.

Solution: (a) From Eq. (9.12) we can write,

$$\phi_{Bn} = V_T \ln\frac{A^{**}T^2}{J_s} = 0.026 \ln\frac{4.4 \times 300^2}{5 \times 10^{-8}} = 0.772 \text{ V}$$

Again, from Eq. (9.2) we can write the barrier height as,

$$V_{bi} = \phi_{Bn} - V_T \ln\frac{N_C}{N_D}$$

$$= 0.772 - 0.026 \ln\frac{4.7 \times 10^{17}}{10^{16}} = 0.772 - 0.1 = 0.672 \text{ V}$$

Substituting this value of V_{bi} in Eq. (9.5), at zero bias the deptetion region width is given by,

$$W = \sqrt{\frac{2\varepsilon_s}{qN_D}(V_{bi} - V)} = \sqrt{\frac{2 \times 13.1 \times 8.854 \times 10^{-14}}{1.6 \times 10^{-19} \times 10^{16}}(0.672 - 0)} = 3.12 \times 10^{-5} \text{ cm}$$

(b) The diffusion length for holes is given by

$$L_p = \sqrt{D_p\tau_p} = \sqrt{10 \times 10^{-8}} = 3.16 \times 10^{-4} \text{ cm}$$

From Eq. (4.50) we know that for a long base p^+n diode, the minority carrier diffusion current density is expressed as,

$$J_p = \frac{qD_p n_i^2}{L_p N_D} = \frac{1.6 \times 10^{-19} \times 10 \times 3.2 \times 10^{12}}{3.16 \times 10^{-4} \times 10^{16}} = 1.62 \times 10^{-18} \text{ A/cm}^2$$

Comparing the values of J_s and J_p, we can see that $J_s >> J_p$. Hence, it can be concluded that the current in a Schottky diode is mostly due to majority carriers.

9.2.3 Ohmic Contacts

Since metals are used to make contacts with various regions in a semiconductor device, it is also necessary to have M-S junctions which conduct equally in both directions. Such M-S junctions are called *ohmic contacts*. A metal/n-type semiconductor junction may be ohmic when $\phi_m < \phi_s$. The band diagram for such a system under thermal equilibrium is shown in Fig. 9.6. As can be seen from Fig. 9.6(b), the band bending at the semiconductor surface results in accumulation of electrons. These electrons can easily flow from the semiconductor to the metal. Also, the barrier for electron flow from metal to semiconductor is quite small, and can be easily overcome by a small applied voltage. In the case of metal p-type semiconductor junctions, ohmic contacts may be formed when $\phi_m > \phi_s$.

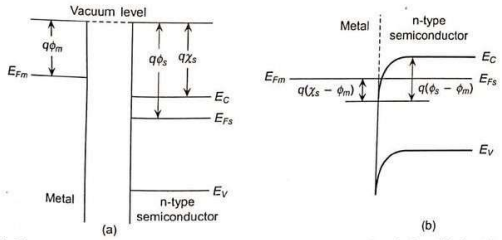


Figure 9.6 Energy band diagram for (a) an isolated metal and n-type semiconductor with $\phi_m < \phi_s$ and (b) a junction of these two materials.

Even when the barrier height is large, ohmic contacts may be formed when the semiconductor is heavily doped. In this case, the depletion layer width becomes narrow and tunnelling current may become the dominant component instead of thermionic current. The tunnelling current [2] can be expressed as

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$$I \propto \exp\left[\frac{-C_2(\phi_{Bn} - V)}{\sqrt{N_D}}\right] \tag{9.13}$$

where

C_2 = constant.

From Eqs. (9.11)–(9.13), we can see that when the semiconductor is moderately doped and the barrier height is large, the metal-semiconductor junction behaves like a rectifier. On the other hand, if the semiconductor is heavily doped and/or the barrier height is small, the current can flow in either direction, that is, the M-S junction behaves like an ohmic contact.

For example, aluminium forms an ohmic contact with p-type silicon, but a rectifying (Schottky) contact with n-type silicon. To form an ohmic contact to n-type silicon using aluminium, the n-region must be heavily doped.

9.3 JUNCTION FIELD EFFECT TRANSISTOR

The term 'field effect' describes the change in conductance of a sample that occurs as a result of an applied field, normal to the sample surface. To understand how the conductance modulation is achieved, let us consider an n-type silicon sample with cross-sectional area A and length L. The conductance of this sample is given by

$$G = \frac{qn\mu_n A}{L} \tag{9.14}$$

Now, the value of this conductance can be changed either by changing A and L, or by changing the electron concentration n. In a junction field effect transistor (JFET), reverse-biased p-n junctions are used to change the cross sectional area of the conducting region (channel). A metal semiconductor field effect transistor (MESFET) is similar to JFET except that instead of p-n junction, a rectifying metal-semiconductor junction is used to modulate the width of the channel. In a metal oxide semiconductor field effect transistor (MOSFET), the carrier concentration in the channel is varied by applying an electric field. Modulation in conductance by any of the above methods causes a variation in the channel current.

9.3.1 Basic JFET Structure and Principle of Operation

The basic structure of an n-channel JFET is shown in Fig. 9.7. The JFET is a three-terminal device. It consists of an n-type semiconductor sandwiched between two p^+ regions (gate). The two ohmic contacts to the n-region are called the *source* and *drain*. Let us assume that the channel is uniformly doped and has a doping concentration N_D while the p^+ regions are much more heavily doped compared to the channel. Thus, the depletion region at the gate-to-channel (p^+n) junction exists primarily in the channel. During operation of the JFET, as the gate–channel junction is increasingly reverse-biased, the width of the depletion region increases. This causes the channel thickness to reduce and consequently the channel conductance decreases. Hence, essentially JFET can be considered as a voltage controlled resistor. We shall now discuss the operation of a JFET in more detail.

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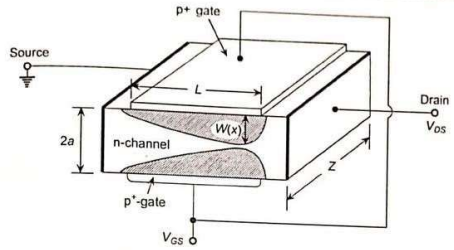


Figure 9.7 Basic structure of a JFET.

Step 1: $V_{GS} = 0$ and $V_{DS} = 0$. Under these conditions, the channel resistance can be expressed as

$$R_0 = \frac{L}{2qN_D\mu_n Z(a - W_0)} \tag{9.15}$$

where

L = channel length,
Z = channel width,
$2a$ = total channel thickness,
μ_n = electron mobility,
q = electronic charge, and
W_0 = zero-bias depletion width at the p^+n junction given by

$$W_0 = \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_D}} \tag{9.16}$$

The depletion region width is constant throughout the length of the channel as shown in Fig. 9.8(a).

Step 2: $V_{DS} > 0$. Let us now apply a small positive voltage between the drain and the source, V_{DS}. Under the influence of the electric field generated between the source and the drain, electrons flow from the source to the drain. The direction of the conventional current is from drain to source and it is termed the drain current I_D. It may be noted that only electrons (majority carriers in the channel) take part in this conduction mechanism. So long as the applied V_{DS} is small compared to the built-in potential (V_{bi}) of the p^+n junction, we can assume that the depletion region width remains approximately constant throughout the channel at its zero-bias value, W_0. The channel resistance is nearly constant ($= R_0$) in this region and I_D increases linearly with V_{DS}.

Step 3: Further increase in V_{DS}. As V_{DS} is increased, the potential increases from the source towards the drain, and the drain end of the channel gets more reverse-biased than the source end. The depletion layer width is now no longer a constant throughout the entire length of the channel,

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but becomes position-dependent as depicted in Fig. 9.8(b). The depletion region width can be expressed as a function of position in the following manner:

$$W(x) = \sqrt{\frac{2\varepsilon_s(V_{bi} + V(x))}{qN_D}} \tag{9.17}$$

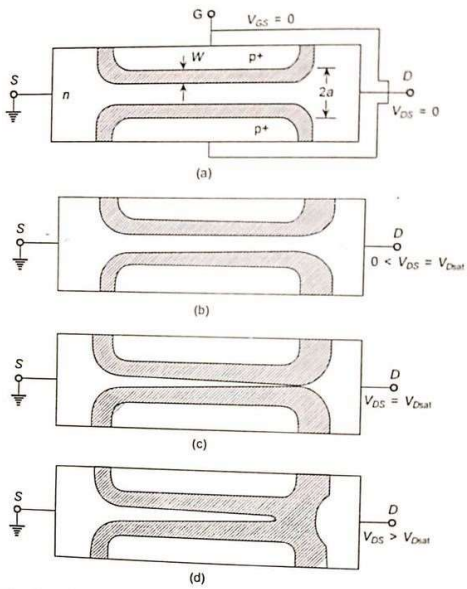


Figure 9.8 The gate–channel depletion region widths in a JFET at (a) $V_{DS} = 0$ V, (b) $0 < V_{DS} < V_{Dsat}$, (c) $V_{DS} = V_{Dsat}$, and (d) $V_{DS} > V_{Dsat}$.

where $V(x)$ and $W(x)$ are respectively the potential drop and the corresponding depletion width in the channel at a distance x from the source end.

Now, the elemental resistance (dR) associated with an incremental length of the channel (dx) is given by

$$dR = \frac{dx}{2qN_D\mu_n Z(a - W(x))} \tag{9.18}$$

Integrating Eq. (9.18) over the entire length of the channel, we obtain the channel resistance as

$$R = \frac{1}{2q\mu_n N_D Z}\int_0^L \frac{dx}{a - W(x)} \tag{9.19}$$

Since the depletion region width $W(x)$ is greater than W_0 at all points in the channel except at the source end, it follows that R is larger than R_0. So, the channel resistance increases with increase in V_{DS}, and the current falls below its initial linear rise with V_{DS} corresponding to a constant channel resistance ($= R_0$).

Step 4: $V_{DS} = V_{Dsat}$. As V_{DS} is further increased, the depletion regions eventually merge, thereby 'pinching off' the channel. Since the depletion width is maximum at the drain end, this has to occur first at the drain end as shown in Fig. 9.8(c). Even though the channel ceases to exist (that is, its thickness has fallen to zero) at the drain end, the drain current is sustained as the electrons are swept across the depletion region by the very high field present in the *pinch-off* region. The situation is somewhat similar to the case of a reverse-biased collector–base junction of a BJT that is operating in normal active mode.

The drain voltage at which pinch-off occurs (V_{Dsat}) can be obtained by putting $W(x) = a$ and $V(x) = V_{Dsat}$ in Eq. (9.17). This is called the *saturation drain voltage* and its value is given by

$$V_{Dsat} = \frac{qN_D a^2}{2\varepsilon_s} - V_{bi} = V_p - V_{bi} \tag{9.20}$$

where

V_p = pinch-off voltage and it is the voltage across the depletion region at the pinch-off condition ($W = a$). The pinch-off voltage V_p is expressed by the relation

$$V_p = \frac{qN_D a^2}{2\varepsilon_s} \tag{9.21}$$

The regions of operation where $V_{DS} < V_{Dsat}$ and $V_{DS} > V_{Dsat}$ are called the *linear* and *saturation* regions respectively.

Step 5: $V_{DS} > V_{Dsat}$. If V_{DS} is increased beyond V_{Dsat}, all the excess voltage ($V_{DS} - V_{Dsat}$) is dropped across the pinch-off region. This causes the pinch-off point to shift towards the source as shown in Fig. 9.8(d). However, the voltage at the new pinch-off point is still V_{Dsat} and as a consequence, the drain current remains almost constant. A small increase in current is observed, which is due to the effective reduction in the channel length as the pinch-off point moves towards the source.

Step 6: $V_{GS} \neq 0$. Steps 1 through 5 are carried out keeping $V_{GS} = 0$. Forward biasing the gate–channel junction is usually avoided in order to ensure that the gate current is negligibly small. On the other hand if V_{GS} is made negative, the depletion region widens at all points in the channel and the channel resistance increases. Thus, for any given value of V_{DS}, the drain current is now less than its corresponding value for $V_{GS} = 0$ V.

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For $V_{DS} = 0$ V, when the magnitude of V_{GS} ($V_{GS} < 0$) is large enough so that the voltage across the depletion regions is equal to V_p, the depletion regions merge. This value of V_{GS} is called the *threshold voltage* (V_{th}). The threshold voltage is defined as the gate-to-source voltage which causes the depletion region width W to become equal to a. Since the voltage drop across the depletion region is the sum of V_{bi} and the applied reverse bias, it can easily be shown that

$$V_{th} = V_{bi} - V_p \tag{9.22}$$

It can be seen from Eqs. (9.21) and (9.22) that V_p is positive ($V_p > 0$), while V_{th} is usually a negative ($V_{th} < 0$) voltage. For an applied gate voltage V_{GS}, the expression for V_{Dsat} can now be obtained by replacing V_{bi} by $(V_{bi} - V_{GS})$ in Eq. (9.20). Also using Eq. (9.22), we have

$$V_{Dsat} = V_p - (V_{bi} - V_{GS}) = V_{GS} - V_{th} \tag{9.23}$$

9.3.2 The *I–V* Characteristics of JFETs

Let us at first consider the linear region of operation of the JFET. From the JFET structure shown in Fig. 9.9(a), in the absence of any diffusion current, the drain current I_D can be expressed as a drift current of electrons in the following form

$$I_D = A(x)qnv_d(x) = 2[a - W(x)]ZqN_Dv_d(x) \tag{9.24}$$

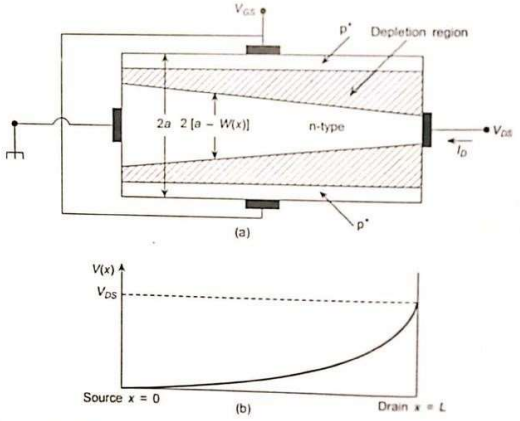


Figure 9.9 (a) The JFET in the linear region of operation; (b) the potential variation along the JFET channel.

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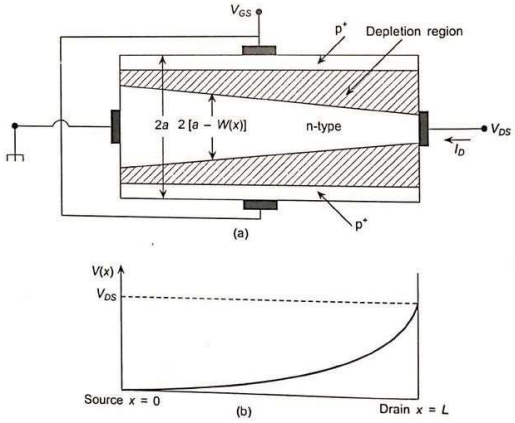


Figure 9.9 (a) The JFET in the linear region of operation; (b) the potential variation along the JFET channel.

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where

$A(x) = 2[a - W(x)]Z$ is the cross-sectional area of current flow,

$n = N_D$ is the electron concentration in the channel, and

$v_d(x)$ = electron drift velocity given by

$$v_d(x) = \mu_n \frac{dV}{dx} \tag{9.25}$$

Here,

μ_n = electron mobility in the channel.

The drain current I_D must be constant throughout the channel in the absence of recombination. Since the channel area reduces from the source to the drain end, this must be compensated by an increase in the drift velocity. If we consider Eq. (9.25), this implies an increase in the electric field towards the drain end as shown in Fig. 9.9(b) by the increase in the slope of the $V(x)$ curve.

Now, using Eqs. (9.24) and (9.25), we can write

$$I_D = 2qN_D\mu_n Z[a - W(x)]\frac{dV}{dx} \tag{9.26}$$

Substituting the expression for $W(x)$ from Eq. (9.17) while replacing V_{bi} with $V_{bi} - V_{GS}$, we can write

$$I_D \int_0^L dx = 2qN_D\mu_n Z \int_0^{V_{DS}} \left[a - \sqrt{\frac{2\varepsilon_s(V_{bi} - V_{GS} + V(x))}{qN_D}}\right] dV \tag{9.27}$$

Integrating Eq. (9.27) and using Eq. (9.21), we have

$$I_D = \frac{2qN_D\mu_n Za}{L}\left[V_{DS} - \frac{2}{3}V_p\left\{\left(\frac{V_{bi} - V_{GS} + V_{DS}}{V_p}\right)^{3/2} - \left(\frac{V_{bi} - V_{GS}}{V_p}\right)^{3/2}\right\}\right] \tag{9.28}$$

When the applied drain voltage is very small, that is, $V_{DS} << (V_{bi} - V_{GS})$, using Eq. (9.28) the drain current can be approximated as

$$I_D = \frac{2qN_D\mu_n ZaV_{DS}}{L}\left\{1 - \left(\frac{V_{bi} - V_{GS}}{V_p}\right)^{1/2}\right\} \tag{9.29}$$

From Eq. (9.29), we see that

$$I_D = 0 \qquad \text{when } V_p = V_{bi} - V_{GS}$$

or

$$V_{GS} = V_{bi} - V_p = V_{th}$$

The drain current in the saturation region can be obtained from Eqs. (9.22), (9.23), and (9.28) by substituting $V_{DS} = V_{Dsat}$ and is calculated to be

$$I_{Dsat} = \frac{2qN_D\mu_n Za}{L}\left[\frac{V_p}{3} + \left\{\frac{2}{3}\left(\frac{V_{bi} - V_{GS}}{V_p}\right)^{1/2} - 1\right\}(V_{bi} - V_{GS})\right] \tag{9.30}$$

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However in practice, the current in a JFET in the saturation region differs from that given by Eq. (9.30). Figure 9.10 shows the actual *I–V* characteristics of an n-channel JFET (solid lines). The theoretical plots are also included in the same figure for comparison (dotted lines). It can be seen that for $V_{DS} > V_{Dsat}$, I_D is not really a constant as predicted but increases slowly with the drain voltage. This is due to the reduction in the effective channel length. As already pointed out, with increase in V_{DS} beyond V_{Dsat}, the pinch-off point shifts towards the source. Thus, the distance between source and pinch-off point is now less than the channel length L. Therefore, although the voltage at the new pinch-off point is fixed at V_{Dsat}, the current increases due to this reduction in the effective channel length.

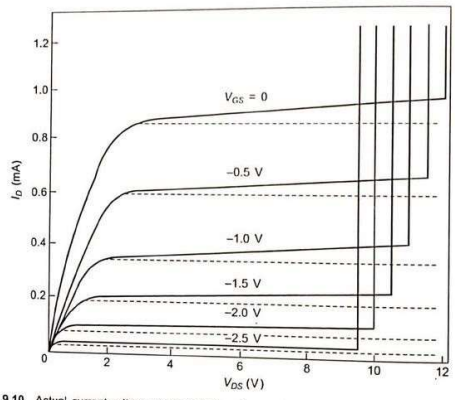


Figure 9.10 Actual current-voltage characteristics of an n-channel JFET (solid lines) and theoretical plots showing constant current in the saturation region (dotted lines).

For high values of V_{DS}, avalanche multiplication of carriers in the pinch-off region also causes an increase in the drain current. The electric field in this region is high and increases rapidly with an increase in V_{DS}. This can cause secondary electron–hole pairs to be generated resulting in the increase of drain current by a factor M called the avalanche multiplication factor. This effect is similar to the collector current multiplication that occurs in the CB depletion region of a BJT. Finally at a particular value of V_{DS}, the avalanche breakdown of the gate–channel p-n junction occurs and the current increases abruptly as shown in Fig. 9.10. At this point, the reverse-bias voltage across the gate–channel junction at the drain end must equal the breakdown voltage V_{BR} of the junction. That is,

$$V_{DS} - V_{GS} = V_{BR} \tag{9.31}$$

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An interesting point about JFET is that the drain current decreases with temperature unlike the collector current in a BJT where thermal runaway may occur. This is because the carrier mobility decreases with increase in temperature. The pinch-off voltage V_p is nearly independent of temperature once all the dopants are ionized, and V_{bi} decreases slowly with increasing temperature. However, the temperature dependence of mobility is the most important factor in determining the temperature sensitivity of JFET current.

9.3.3 Small Signal Parameters of JFETs

In the previous section, we have seen that the drain current is dependent on the gate and drain voltages. Now, when small signal drain and gate voltages, v_d and v_g, respectively are superimposed on the corresponding dc values, V_{DS} and V_{GS}, the resultant drain current is given by $i_D = (I_D + i_d)$, where i_d is the small signal component. The small signal drain current can now be expressed as

$$i_d = \left.\frac{\partial I_D}{\partial V_{DS}}\right|_{V_{GS}} v_d + \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS}} v_g = g_d v_d + g_m v_g \tag{9.32}$$

where

$g_d = \left.\frac{\partial I_D}{\partial V_{DS}}\right|_{V_{GS}}$ is the *channel conductance* (or *drain conductance*) and $g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS}}$ is the *transconductance*.

For small values of V_{DS}, g_d can be directly evaluated by differentiating Eq. (9.29) as

$$g_d = G_0\left[1 - \left(\frac{V_{bi} - V_{GS}}{V_p}\right)^{1/2}\right] \tag{9.33}$$

where

$G_0 = 2qN_D\mu_n(Za/L)$.

In the saturation region, g_d is close to zero as the drain current is almost constant at I_{Dsat} and is independent of V_{DS}. By differentiating Eq. (9.28) with respect to V_{GS}, we get

$$g_m = G_0\left[\left(\frac{V_{bi} - V_{GS} + V_{DS}}{V_p}\right)^{1/2} - \left(\frac{V_{bi} - V_{GS}}{V_p}\right)^{1/2}\right] \tag{9.34}$$

For small values of V_{DS} ($<< V_{bi} - V_{GS}$), the above expression reduces to

$$g_m = \frac{G_0 V_{DS}}{2\sqrt{V_p(V_{bi} - V_{GS})}} \tag{9.35}$$

In the saturation region, g_m can be obtained by considering $V_{DS} = V_{Dsat}$ and substituting the expression for V_{Dsat} from Eq. (9.23) into Eq. (9.34). It can be shown that in this region g_m has the same value as g_d has in the linear region which is given by Eq. (9.33). Since electrons have higher mobility than holes, an n-channel JFET has higher transconductance and better high-frequency response than a p-channel JFET. Therefore, n-channel JFETs are preferred for most applications.

9.4 THE MESFETs

In principle, a MESFET is identical to JFET except that here a rectifying metal-semiconductor junction is used instead of a p-n junction for the gate electrode. MESFETs are usually fabricated on III–V compound semiconductors, for example, GaAs and InP because of their high electron mobilities and also because of the absence of a viable MOSFET in these material systems.

9.4.1 MESFET Structure

Figure 9.11 shows the structure of a MESFET on GaAs. The substrate is semi-insulating (SI), that is, its resistivity is very high ($>10^8$ ohm-cm). This is usually achieved by doping GaAs with chromium or by using undoped material. On this SI substrate, a thin layer of lightly doped n-GaAs is grown by epitaxy to form the channel. Instead of epitaxy, this n-layer can also be formed by ion-implantation. Ohmic contacts on the source and drain and Schottky contact on the gate are formed by depositing suitable metals, for example, Au–Ge for source/drain and Al for gate. To facilitate ohmic contact formation, n^+ implantation may also be carried out in the source/drain regions prior to metal deposition. Since no diffusion is involved, geometrical tolerance is very high and very small MESFETs can be made. Consequently, these devices are used in high frequency circuits.

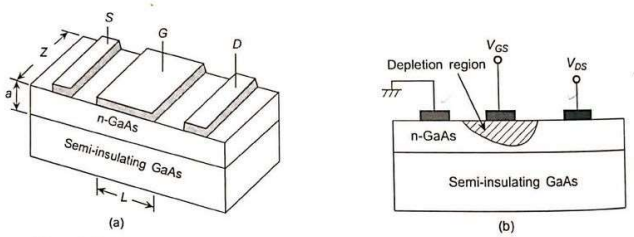


Figure 9.11 (a) Perspective view and (b) cross-sectional view of a basic MESFET structure.

The operation of the MESFET is quite similar to the JFET discussed in section 9.3. However, the usually small geometry of MESFETs necessitates a variety of modifications to be made to the simple theory of JFET operation in order that it may be used for MESFETs. The most common effect occurs due to the increase in the electric field ($\mathscr{E}$) with reduction in the channel length. In this case, the assumption of a constant mobility is no longer valid and an approximation for the drift velocity given by $v_d = \mu_n\mathscr{E}/[1 + (\mu_n\mathscr{E}/v_{sat})]$ is often made. From this expression, we find that at high electric fields at the drain end of the channel, the electron drift velocity becomes constant ($= v_{sat}$) and current saturation takes place because the carrier velocity reaches a saturation value and **NOT** because the depletion region pinches off the channel. The saturation drain current can then be simply expressed as

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$$I_{Dsat} = qnv_{sat}Zh \tag{9.36}$$

where

h = channel thickness at the drain end when velocity saturation occurs.

Most devices, however, operate with characteristics intermediate between the constant mobility and constant velocity regimes.

In our discussion so far, we have considered a JFET and MESFET structure which is normally 'on', where a considerable I_D exists even when $V_{GS} = 0$ V. In other words, V_{th} is negative ($V_{th} < 0$) for these devices. These are called *depletion type* devices. However, for high speed, low power application, the normally 'off' structure, where V_{th} is positive ($V_{th} > 0$) is preferred. These are called *enhancement type* devices. Enhancement type MESFETs can be realized by using a very thin n-GaAs channel, such that the entire channel is depleted at zero bias, implying $V_p < V_{bi}$. Now from Eq. (9.22), it is clear that this will result in a positive V_{th}.

Example 9.2

Two GaAs MESFETs, MES1 and MES2 use metal-semiconductor Schottky diodes similar to MS1 in Example 9.1 as their gates. The channel thicknesses for MES1 and MES2 are 0.2 μm and 0.4 μm respectively. Calculate the threshold voltages for both MESFETs.

Solution: From Example 9.1, we already know that the value of V_{bi} for this Schottky gate is 0.672 V. From Eq. (9.21), for a MESFET with channel thickness a, we can write

$$V_p = \frac{qN_Da^2}{2\varepsilon_s}$$

Thus for MES1,

$$V_p = \frac{1.6 \times 10^{-19} \times 10^{16} \times (0.2 \times 10^{-4})^2}{2 \times 13.1 \times 8.854 \times 10^{-14}} = 0.276 \text{ V}$$

From Eq. (9.22) we get,

$$V_{th} = V_{bi} - V_p$$
$$= 0.672 - 0.276 = 0.396 \text{ V} \quad \text{for MES1}$$

Similarly for MES2,

$$V_p = \frac{1.6 \times 10^{-19} \times 10^{16} \times (0.4 \times 10^{-4})^2}{2 \times 13.1 \times 8.854 \times 10^{-14}} = 1.104 \text{ V}$$

Using Eq. (9.22) we have,

$$V_{th} = V_{bi} - V_p$$
$$= 0.672 - 1.104 = -0.432 \text{ V} \quad \text{for MES2}$$

Therefore, we see that MES1 is a normally 'off' (enhancement type) device while MES2 is a normally 'on' (depletion type) device.

9.4.2 The Heterojunction FETs

The performance of a MESFET can be further improved by using a heterojunction. Here in the basic scheme, an undoped GaAs layer is first grown on the SI substrate followed by an n-doped

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AlGaAs layer. The (triangular) potential well created at the AlGaAs–GaAs heterojunction causes the electrons from the doped AlGaAs layer to accumulate into this well as shown in Fig. 9.12. These accumulated electrons form a two-dimensional electron gas (often referred to as 2-DEG) which is confined to a very thin region at the heterojunction. Very high sheet carrier density (10^{12} cm^{-2}) can be obtained for heterojunctions. Also, since the electrons are now confined in the undoped GaAs layer (and not in the doped AlGaAs layer), the scattering due to impurity ions is minimal and the mobility is limited by lattice scattering alone. As a result, mobility > 250,000 cm^2/Vs at 77 K can be achieved. This device is therefore called a *high electron mobility transistor* (HEMT). Also, since the electrons are separated from the dopant atoms, this is referred to as *modulation doped field effect transistor* (MODFET). *Two-dimensional electron gas field effect transistor* (TEGFET) is another name given to this device. The main advantage of an HEMT is its ability to achieve a high electron density in a very thin layer (<10 nm) close to the gate while eliminating impurity scattering. This results in extremely high cut-off frequency and high transconductance.

Although we have discussed only about an AlGaAs/GaAS HEMT, other material systems such as InGaAsP/InP and even Si/Si–Ge have also been used in practice to fabricate HEMTs.

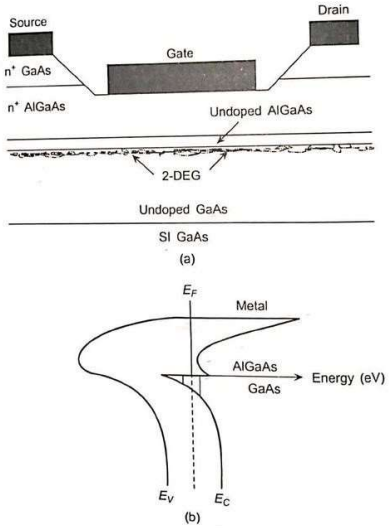


Figure 9.12 (a) Basic structure and (b) energy band diagram of an AlGaAs/GaAs HEMT.

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PROBLEMS

P9.1 Compare the ratio of hole current to electron current for Schottky diodes fabricated on n-type silicon with a doping concentration of 10^{16} cm^{-3} using (a) Au (ϕ_{Bn} = 0.8 V) and (b) Al(ϕ_{Bn} = 0.6 V). Also compare the reverse saturation current densities of the two Schottky diodes.

P9.2 Show that for a JFET, g_m in the saturation region is equal to the g_d in the linear region for the same gate voltage.

P9.3 For an n-channel JFET with $N_D = 10^{16}$ cm^{-3}, $N_A = 10^{19}$ cm^{-3}, $a = 1$ μm, $L = 5$ μm, $Z = 20$ μm, and $\mu_n = 1200$ cm^2/Vs, find (a) pinch-off voltage, (b) threshold voltage, (c) V_{Dsat} at $V_{GS} = 0$ V, (d) I_{Dsat} at $V_{GS} = 0$ V and (e) g_m in the saturation region at $V_{GS} = 0$ V.

P9.4 An n-channel GaAs MESFET has the following parameters: $\phi_{Bn} = 0.9$ V, $N_D = 10^{17}$ cm^{-3}, $a = 0.3$ μm. Calculate the threshold voltage assuming $N_C = 4.7 \times 10^{17}$ cm^{-3}.

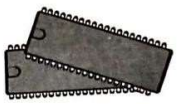
REFERENCES AND SUGGESTED FURTHER READING

[1] Cowley, A.M. and S.M. Sze, Surface states and barrier height of metal-semiconductor system, *Journal of Applied Physics*, Vol. 36, p. 3212, 1965.

[2] Sze, S.M., *Physics of Semiconductor Devices*, 2nd ed., Wiley, N.Y., 1981.

[3] Streetman, B.G. and S. Banerjee, *Solid State Electronic Devices*, 5th ed., Prentice Hall, New Jersey, 2000.

[4] Shur, Michael, *Physics of Semiconductor Devices*, Prentice Hall, New Jersey, 1990.



10 MOSFETs

10.1 INTRODUCTION

The rapid strides of the semiconductor industry in recent years are due to its ability to incorporate more and more devices operating at higher and higher speeds in an integrated circuit (IC). Metal oxide semiconductor field effect transistor (MOSFET) circuits occupy less silicon area and consume less power than their bipolar counterparts, making them the ideal choice for very large scale integrated (VLSI) circuits. In fact, the MOSFET is by far the most widely used semiconductor device today and is used in fabrication of central processing unit (CPU) and memories in computers, digital signal processors (DSPs) for communication purposes, and ICs for a variety of other applications.

In this chapter, we shall discuss the operating principles of the MOSFETs and develop various models for their current-voltage (I–V) characteristics. Figure 10.1 shows the cross-sectional view of a MOSFET, which is a four-terminal device. The terminals are the source (S), the drain (D), the gate (G), and the bulk (B). As we can see, at the heart of the device is a metal oxide semiconductor (MOS) structure, from which the MOSFET derives its name. For proper understanding of the device operation, we shall at first take up the two terminal MOS diode for discussion, followed by the MOSFET by including the additional source and drain terminals.

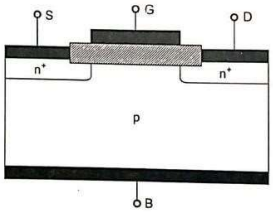


Figure 10.1 Cross-sectional view of a four-terminal MOSFET.

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10.2 MOS DIODE

The MOS diode (Fig. 10.2) consists of an insulating layer (usually silicon dioxide) sandwiched between a metal and a semiconductor (usually silicon). Interestingly, the device consists of all the three different classes of solids, namely metal, semiconductor, and insulator. We have already seen the nature of the energy band diagram of each of these materials (section 1.2.2). However, we shall now discuss the energy band diagram of the composite MOS structure. For this purpose, the vacuum level is taken as the reference for drawing the relative positions of the energy levels of the different materials. We have already discussed in section 1.3.10 that a metal is characterised by its *work function* ($q\phi_m$ in eV), which is the energy required to raise an electron from its Fermi energy level to the vacuum level. For semiconductors and insulators, the *electron affinity* ($q\chi$ in eV), which is the energy required to raise an electron from the bottom of the conduction band to the vacuum level, is a characteristic property of the material.

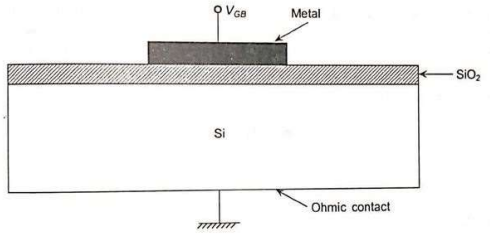


Figure 10.2 Cross-sectional view of a two-terminal MOS diode.

The work function of the semiconductor material (ϕ_s) is dependent on the doping concentration and can be calculated from the position of the Fermi level with respect to the bottom of the conduction band. Thus, for a p-type semiconductor, ϕ_s can be expressed as

$$\phi_s = \chi_s + \frac{(E_C - E_F)}{q} = \chi_s + \frac{E_g}{2q} + \phi_B \tag{10.1}$$

where

χ_s = electron affinity,

E_g = band gap of the semiconductor,

q = electronic charge (= 1.6×10^{-19} C), and

ϕ_B is the energy difference between the Fermi level and the intrinsic level E_i given by

$$\phi_B = V_T \ln\left(\frac{N_A}{n_i}\right) \tag{10.2}$$

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On the other hand, the electron concentration at the semiconductor surface (n_{ps}) is given by

$$n_{ps} = n_i \exp\left(\frac{\psi_s - \phi_B}{V_T}\right) = n_{po} \exp\left(\frac{\psi_s}{V_T}\right) \tag{10.7}$$

where n_{po} is the equilibrium electron concentration in bulk silicon given by

$$n_{po} = \frac{n_i^2}{p_{po}} = n_i \exp\left(\frac{-\phi_B}{V_T}\right)$$

So, we find that with the application of a positive gate voltage, the majority carrier concentration is reduced, while the minority carrier concentration increases at the semiconductor surface. It may also be noted that due to the variation of $(E_i - E_F)$ near the semiconductor surface, the hole (electron) concentration is minimum (maximum) at the surface and gradually increases (reduces) towards the thermal equilibrium value in the bulk of the semiconductor.

Example 10.2

In the device MD1 (refer to Ex. 10.1), calculate the hole concentration p_{ps}, electronconcentr -ation n_{ps}, and the charge density ρ at the semiconductor surface for (i) $\psi_s = 100$ mV and (ii) $\psi_s = 200$ mV.

Solution: (i) For $\psi_s = 100$ mV

Using Eq. (10.6),

$$p_{ps} = p_{po} \exp\left(\frac{-\psi_s}{V_T}\right) = 10^{16} \exp\left(-\frac{100}{26}\right) = 2.1 \times 10^{14} \text{ cm}^{-3}$$

Using Eq. (10.7),

$$n_{ps} = \frac{n_i^2}{p_{po}} \exp\left(\frac{\psi_s}{V_T}\right) = \frac{(1.5 \times 10^{10})^2}{10^{16}} \exp\left(\frac{100}{26}\right) = 1 \times 10^6 \text{ cm}^{-3}$$

Now,

$$\rho = q(N_D^+ - N_A^- + p - n) = 1.6 \times 10^{-19} (-1 \times 10^{16} + 2.1 \times 10^{14} - 1 \times 10^6)$$
$$= 1.6 \times 10^{-19} (-0.979 \times 10^{16}) = -1.566 \times 10^{-3} \text{ C/cm}^3 \approx -qN_A$$

(ii) For $\psi_s = 200$ mV

Using Eq. (10.6),

$$p_{ps} = p_{po} \exp\left(\frac{-\psi_s}{V_T}\right) = 10^{16} \exp\left(-\frac{200}{26}\right) = 4.56 \times 10^{12} \text{ cm}^{-3}$$

Using Eq. (10.7),

$$n_{ps} = \frac{n_i^2}{p_{po}} \exp\left(\frac{\psi_s}{V_T}\right) = \frac{(1.5 \times 10^{10})^2}{10^{16}} \exp\left(\frac{200}{26}\right) = 4.93 \times 10^7 \text{ cm}^{-3}$$

Now,

$$\rho = q(N_D^+ - N_A^- + p - n) = 1.6 \times 10^{-19} (-1 \times 10^{16} + 4.56 \times 10^{12} - 4.93 \times 10^7)$$
$$= 1.6 \times 10^{-19} (-0.9995 \times 10^{16}) = -1.599 \times 10^{-3} \text{ C/cm}^3 \approx -qN_A$$

So we see from Example 10.2 that even when ψ_s is as small as 100 mV, the value of charge density $\rho \approx -qN_A$ at the semiconductor surface since the electron and hole, concentrations are very much less than the acceptor impurity concentration. With increasing ψ_s, the value of ρ further approaches $-qN_A$. Therefore, we can say that due to the application of a positive voltage at the gate of the MOS capacitor, negative charges are induced in the semiconductor as the holes are repelled from the surface and the acceptor ions are uncovered. A depletion (space charge) layer is created at the semiconductor surface where the charge density is given by

$$\rho = -qN_A \tag{10.8}$$

The neglect of charges due to the carriers, as in Eq. (10.8), is often referred to as the *depletion approximation*. It may be mentioned here that this approximation is strictly not true at the edge of the depletion region where the hole concentration is comparable to the acceptor ion concentration. But in order to calculate the depletion width, for the sake of simplicity, we shall assume that Eq. (10.8) holds in the entire depletion layer. We have already derived a relation for the potential drop across a depletion layer of thickness W having a constant charge concentration [refer to Eq. (4.18)]. Using the same relation, the depletion layer width W for a drop ψ_s in the semiconductor, can be written as

$$W = \sqrt{\frac{2\varepsilon_s \psi_s}{qN_A}} \tag{10.9}$$

Therefore, as the semiconductor surface potential increases, the depletion layer width W increases. From the depletion approximation, the total charge density per unit area in the semiconductor, Q_s is approximately equal to the number of acceptor ions per unit area in the depletion layer. This is often referred to as the *bulk charge density* Q_B, and is given by

$$Q_s \approx Q_B = -qN_A W = -qN_A \sqrt{\frac{2\varepsilon_s \psi_s}{qN_A}} = -\sqrt{2qN_A \varepsilon_s \psi_s} \tag{10.10}$$

From the above equation we see that the semiconductor charge also increases with ψ_s.

We shall now evaluate the potential drop across the oxide ψ_{ox} corresponding to a drop of ψ_s in the semiconductor. For this purpose we shall use Gauss law, which is written as

$$\oint_s D \cdot ds = \int_{vol} \rho \cdot dv \tag{10.11}$$

where the electric displacement vector D is related to the electric field $\mathscr{E}$ through the relation $D = \varepsilon\mathscr{E}$ under static conditions.

Gauss law states that the total normal electric flux coming out of a closed surface equals the charge enclosed by the surface. Let us consider Fig. 10.5, which shows an imaginary Gaussian surface enclosing part of the MOS capacitor. The electrical field lines originate from the positive charges in the gate and terminate on the negative charges in the semiconductor. Deep inside the semiconductor, the electric field goes to zero. Hence, there are no flux lines coming out of the bottom of the Gaussian surface. Also, there are no flux lines coming out of the sides, since the electric field lines run parallel to them. The only flux lines intersecting the Gaussian surface are at the top. If $\mathscr{E}_{ox}$ is the electric field in the oxide (it is a constant since there is no charge in the oxide), and assuming unit surface area, from Eq. (10.11) we have

$$\varepsilon_{ox} \mathscr{E}_{ox} = -Q_s \tag{10.12}$$

Ideal MOS diode at zero bias: Figure 10.3(a) shows the band diagram of the ideal MOS device at zero gate-to-bulk bias, that is, $V_{GB} = V_G - V_B = 0$. The vacuum level is flat since there is no voltage dropped in the device. Also since $\phi_{ms} = \phi_m - \phi_s = 0$, the Fermi energy levels in the metal (gate) and semiconductor coincide. The notations in the band diagram in Fig. 10.3(a) can be modified and represented as in Fig. 10.3(b). In Fig. 10.3(b), $q\phi'_m$ is the *metal-oxide barrier energy*, which is defined as the energy required to move an electron from the Fermi level of the metal to the conduction band of the oxide, where

$$\phi'_m = \phi_m - \chi_{SiO_2} \tag{10.3}$$

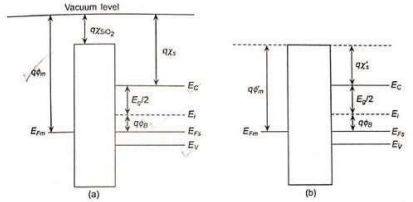


Figure 10.3 Band diagram of an ideal MOS diode at $V_{GB} = 0$ V drawn using (a) ϕ_m, χ_{SiO_2}, χ_s and (b) ϕ'_m, χ'_s.

Similarly, the *silicon-oxide barrier energy* is the energy required to move an electron from the silicon conduction band to the conduction band of the oxide and is given by $q\chi'_s$ where

$$\chi'_s = \chi_s - \chi_{SiO_2} \tag{10.4}$$

It must be noted that ϕ'_m and χ'_s are constant for a particular material system and do not change with the application of bias. As it is more convenient, we shall use the constants ϕ'_m and χ'_s to draw the band diagrams in this chapter.

Ideal MOS diode at $V_{GB} > 0$: Figure 10.4 shows the band diagram of an ideal MOS diode with applied bias $V_{GB} > 0$ V. The applied voltage V_{GB} is the sum of two potential drops—ψ_{ox}, which is dropped in the oxide and ψ_s, which is dropped in silicon. This is reflected in the figure with a band bending ψ_{ox} in the oxide and ψ_s in silicon. A point to be noted in the figure is that the metal Fermi level E_{Fm} is at a lower energy level compared to the Fermi level in the semiconductor E_{Fs} by a value qV_{GB}. This is because the metal is at a higher potential with respect to the semiconductor by V_{GB} volts. (Note: The energy band diagram reflects the electronic potential which is the negative of electrostatic potential.) We shall show that this is indeed true. Adding up the energies in the band diagram, we have

$$E_{Fm} + q\phi'_m + q\psi_{ox} = E_{Fs} + q\phi_B + \frac{E_g}{2} - q\psi_s + q\chi'_s = E_{Fs} - q\psi_s + q\phi'_s$$

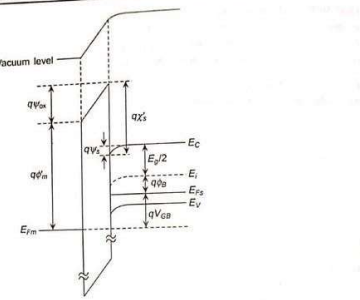


Figure 10.4 Band diagram of an ideal MOS diode at $V_{GB} > 0$ V.

For the ideal MOS diode, we have $q\phi'_m = q\phi'_s$ and since $V_{GB} = \psi_{ox} + \psi_s$, we obtain

$$E_{Fs} - E_{Fm} = qV_{GB} \tag{10.5}$$

Therefore, for drawing the energy band diagram of the MOS structure, the difference in the energy levels of the semiconductor and metal Fermi levels must be made equal to the applied bias V_{GB}.

Another point to be noted is that the Fermi level has been drawn as a straight line in the semiconductor. This is because there can be no current flowing through the oxide and hence there is no steady state current in the semiconductor substrate. As we have already seen in section 1.3.9, a flat Fermi level implies zero current. The vacuum level is also drawn in Fig. 10.4. The vacuum level shows the variation of the electronic potential in the device. Since the electric field, which is the derivative of the electronic potential, is a finite quantity, we find that the vacuum level is indeed continuous, although there may be discontinuities in the band edges at the interface of two dissimilar materials.

We also see in the band diagram, that due to the potential drop of ψ_s in the semiconductor, the intrinsic Fermi level (E_i) is bent towards the Fermi level (E_F) at the semiconductor surface, or in other words, the energy difference $(E_i - E_F)$ is reduced. At the semiconductor surface, $(E_i - E_F)_s = q(\phi_B - \psi_s)$. The hole concentration at the semiconductor surface (p_{ps}) is therefore given by

$$p_{ps} = n_i \exp\left(\frac{\phi_B - \psi_s}{V_T}\right) = p_{po} \exp\left(\frac{-\psi_s}{V_T}\right) \tag{10.6}$$

where p_{po} is the equilibrium hole concentration in the p-type bulk silicon and is given by

$$p_{po} \approx N_A = n_i \exp\left(\frac{\phi_B}{V_T}\right)$$

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Figure 10.5 A MOS structure showing the charges and electric field lines at $V_{GB} > 0$ V. The electric field lines cut the imaginary Gaussian surface only at the top.

If the thickness of the oxide is t_{ox}, then for a drop of ψ_{ox} across it, we have

$$\mathscr{E}_{ox} = \frac{\psi_{ox}}{t_{ox}}$$

Hence, from Eq. (10.12), we have

$$\psi_{ox} = -Q_s \frac{t_{ox}}{\varepsilon_{ox}} = \frac{Q_s}{C_{ox}} \tag{10.13}$$

where

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$$ is the *oxide capacitance per unit area.*

Since the applied voltage V_{GB} is the sum of the potentials dropped in the oxide and semiconductor, we may write using Eqs. (10.10) and (10.13)

$$V_{GB} = \psi_s + \psi_{ox} = \psi_s + \frac{qN_A W}{C_{ox}} = \psi_s + \frac{\sqrt{2\varepsilon_s N_A \psi_s}}{C_{ox}} \tag{10.14}$$

From Eq. (10.14), we see that as ψ_s increases, V_{GB} increases. Conversely, as V_{GB} increases, both ψ_{ox} and ψ_s increase. With the increase in ψ_s, the band is bent further, and when $\psi_s = \phi_B$, the Fermi level at the surface is at the intrinsic Fermi level. At this point, $p_{ps} = n_{ps} = n_i$. As ψ_s increases above ϕ_B, the electron concentration increases to above n_i and the hole concentration reduces to below n_i. Thus, the electrons become the majority carriers and the semiconductor surface is said to be *inverted*. Even then, both electron and hole concentrations are much less than the acceptor impurity concentration and the depletion layer charge is given by Eq. (10.3), that is, $\rho = -qN_A$. Also, the depletion layer goes on widening with increasing ψ_s. When $\psi_s = 2\phi_B$, from Eqs. (10.2), (10.6), and (10.7) we have

$$W_m = \sqrt{\frac{2\varepsilon_s (2\phi_B)}{qN_A}} \quad \text{and} \quad n_{ps} = N_A$$

That is, when $\psi_s = 2\phi_B$, the electron concentration at the surface becomes equal to the hole concentration in the bulk of the p-type semiconductor substrate. This point is referred to as the onset of strong inversion. Figure 10.6 shows the band diagram and the charge distribution in the semiconductor at the onset of strong inversion. We see that the semiconductor charge Q_s is actually the sum of two components, namely the bulk charge ($Q_B = -qN_A W_m$, which is due to uncovering of

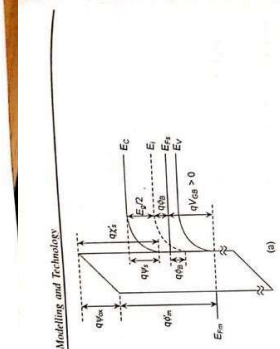


Figure 10.6 Band diagram and charge distribution in the semiconductor at the onset of strong inversion.

the acceptor impurities in the depletion region and the electron charge $\left(Q_n = -q\int_0^W n(x)\,dx\right)$, which is due to the electron concentration at the semiconductor surface. For $\psi_s \le 2\phi_B$, we have assumed that $Q_s \approx Q_B$. This is because, in this range $Q_n << Q_B$. Even when $\psi_s = 2\phi_B$, although $n_{ps} = N_A$, for most of the depletion region $n(x) << N_A$, and so the assumption $Q_s \approx Q_B$ is valid. But when $\psi_s > 2\phi_B$, Q_n becomes comparable to Q_B. We have already seen that Q_B is proportional to $\sqrt{\psi_s}$, while the electron concentration at the semiconductor surface is proportional to $\exp(\psi_s/V_T)$. So a small change in ψ_s results in a large increase in n_{ps}. In fact, at room temperature, n_{ps} increases by an order of magnitude for every 60 mV increase in ψ_s, since $\exp(60/26) \approx 10$. A large increase in n_{ps} results in a correspondingly large increase in Q_n and ψ_{ox} as can be seen from Eq. (10.13). So, in strong inversion a small increase in ψ_s results in a large increase in ψ_{ox}. Since $V_{GB} = \psi_s + \psi_{ox}$, once strong inversion is reached, further increase in V_{GB} is absorbed mostly by ψ_{ox} while ψ_s remains almost equal to $2\phi_B$. The depletion layer width therefore saturates at a maximum value of W_{max}. Substituting $\psi_s = 2\phi_B$ in Eq. (10.9) we have

$$W_{max} = \sqrt{\frac{2\varepsilon_s (2\phi_B)}{qN_A}} \tag{10.15}$$

Physically, the sheet of electrons in the inversion layer at the semiconductor surface can be thought of as screening the semiconductor interior from changes in gate potential.

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The value of V_{GB} that must be applied to just create a condition of strong inversion is known as the *threshold voltage* (V_{th}) *of the MOS diode*, which is one of the most important parameters of any MOS device. The V_{th} value for an ideal MOS diode can now be obtained by substituting $\psi_s = 2\phi_B$ in Eq. (10.14), so that

$$(V_{th})_{ideal} = 2\phi_B + \frac{\sqrt{2qN_A\varepsilon_s (2\phi_B)}}{C_{ox}} \tag{10.16}$$

Example 10.3

In the device MD1, if the gate oxide thickness (t_{ox}) is 1000Å, calculate (i) the maximum depletion width (W_{max}) and (ii) the threshold voltage (V_{th}). [ε_r = 11.9 for Si and ε_r = 3.9 for SiO_2].

Solution:

For device MD1, $\phi_B = 0.35$ V (from Example 10.1).

(i) From Eq. (10.15),

$$W_{max} = \sqrt{\frac{2\varepsilon_s (2\phi_B)}{qN_A}} = \sqrt{\frac{2 \times 11.9 \times 8.85 \times 10^{-14} \times 0.7}{1.6 \times 10^{-19} \times 10^{16}}} = 0.304 \text{ μm}$$

(ii) $C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.85 \times 10^{-14}}{1000 \times 10^{-8}} = 3.45 \times 10^{-8}$ F/cm^2.

From Eq. (10.16),

$$(V_{th})_{ideal} = 2 \times 0.35 + \frac{\sqrt{2 \times 1.6 \times 10^{-19} \times 10^{16} \times 11.9 \times 8.85 \times 10^{-14} \times 0.7}}{3.45 \times 10^{-8}} = 2.11 \text{ V}.$$

Alternately at $V_{GB} = (V_{th})_{ideal}$,

$$\psi_{ox} = \frac{qN_A W_{max}}{C_{ox}}$$

$$= \frac{1.6 \times 10^{-19} \times 10^{16} \times 0.304 \times 10^{-4}}{3.45 \times 10^{-8}} = 1.41 \text{ V}$$

Therefore,

$$(V_{th})_{ideal} = \psi_{ox} + \psi_s = 1.41 + 0.7 = 2.11 \text{ V}$$

HELP DESK / 10.1

How does V_{th} depend on N_A and t_{ox}?

The threshold voltage V_{th} increases with increase in N_A since both ψ_s ($= 2\phi_B$) and V_{ox} ($= \sqrt{2qN_A\varepsilon_s(2\phi_B)}/C_{ox}$) at threshold increase with an increase in N_A. With increase in t_{ox}, the value of C_{ox} decreases and consequently ψ_{ox} increases. Hence, V_{th} also increases with increase in t_{ox}.

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Ideal MOS diode at $V_{GB} < 0$: Figure 10.7 shows the band diagram of an ideal MOS diode with applied bias $V_{GB} < 0$. It may be noted in the figure that E_{Fm} has moved up by qV_{GB} with respect to the semiconductor by V_{GB} volts. E_{Fs}, since the metal is now at a lower potential with respect to the semiconductor by V_{GB} volts. Also, the figure shows that V_{GB} is the sum of two quantities—ψ_{ox} dropped in the oxide and ψ_s dropped in silicon. In this case, the potential drop ψ_s has resulted in the valence band edge coming closer to the Fermi level, which means that the hole concentration at the semiconductor surface has increased. This is known as the *accumulation condition*.

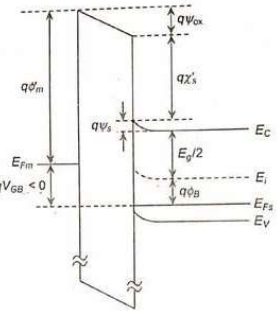


Figure 10.7 Band diagram of an ideal MOS diode at $V_{GB} < 0$ V.

In short, when $V_{GB} = 0$, $\psi_s = 0$ and the bands are flat giving rise to the *flat band condition*. The application of positive voltage on metal induces negative charges in the semiconductor. Initially this is due to holes being repelled from the semiconductor surface. On further increase of V_{GB}, the negative charges are due to the minority carrier electrons, which are attracted to the semiconductor surface and remain there, as they cannot cross the insulator. When $0 < \psi_s \le 2\phi_B$, the semiconductor surface is depleted of carriers giving rise to the *depletion condition*. The region of operation $\phi_B < \psi_s \le 2\phi_B$ is also referred to as the *weak inversion* condition, since the electron concentration at the semiconductor surface is now more than the concentration of holes, which are the majority carriers in the silicon bulk. *Strong inversion* is said to exist when $\psi_s \ge 2\phi_B$. On the other hand, application of negative voltage at the gate induces positive charges at the semiconductor surface by attracting holes (majority carriers) to the semiconductor surface. This region of operation, when $\psi_s < 0$ is known as the *accumulation* condition, since the majority carriers accumulate at the semiconductor surface.

In this section, we have only considered MOS diodes fabricated on p-type semiconductor substrates. For MOS diodes fabricated on n-type semiconductor substrates, the operation is identical except that the polarity of voltages must be interchanged. For example, a positive applied voltage results in accumulation condition and the threshold voltage is negative in these devices.

10.2.2 Operation of MOS Diode with $\phi_{ms} \neq 0$, $Q_{ox} = 0$

Figure 10.8(a) shows the energy band diagram of a MOS diode with $\phi_{ms} \neq 0$ and $Q_{ox} = 0$ biased at $V_{GB} = 0$ V. When the gate and the semiconductor are short-circuited, there is an initial flow of charge till the Fermi energy levels are aligned, that is, $E_{Fs} - E_{Fm} = 0$. As we see in Fig. 10.8(a), for devices with $\phi_{ms} = \phi_m - \phi_s < 0$, the semiconductor surface becomes negatively charged. So in such a device, the semiconductor surface may be inverted even when $V_{GB} = 0$ V. On the other hand, if $\phi_{ms} > 0$, there would be accumulation of holes at the semiconductor surface when $V_{GB} = 0$. We can see from Fig. 10.8(a) that

$$q\phi'_m + q\psi_{ox} = q\phi_B + \frac{E_g}{2} - q\psi_s + q\chi'_s = q\phi'_s - q\psi_s$$

or

$$\phi_{ms} = \phi'_m - \phi'_s = -(\psi_{ox} + \psi_s) \tag{10.17}$$

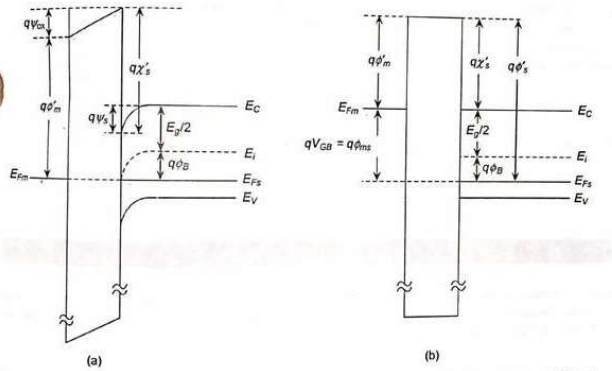


Figure 10.8 Energy band diagram of a MOS diode with $\phi_{ms} \neq 0$ and $Q_{ox} = 0$ biased at (a) $V_{GB} = 0$ V and (b) $V_{GB} = V_{FB}$.

So, the sum of the potential drops across the oxide and semiconductor is equal to the difference between the work functions of the semiconductor and metal. The fact that there is a potential drop in the semiconductor and oxide when $V_{GB} = 0$ is similar to the presence of the built-in potential in a p-n junction, which is also equal to the work function difference between two regions of the device.

In modern MOS technology, n^+ doped polysilicon is often used as the gate material. In this case, the electron affinities of both the gate material and the semiconductor are identical. So the work function difference ϕ_{ms} is actually the difference of the Fermi energy level with respect to the

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conduction band edge in the n^+ polysilicon gate and the silicon substrate. In n^+ polysilicon, we may assume that the Fermi level coincides with the conduction band edge. So, for an n^+ polysilicon gate MOS diode fabricated on a p-type silicon substrate, we have

$$\phi_{ms} = -\frac{E_g}{2q} - \phi_B \tag{10.18}$$

If $\phi_{ms} < 0$, it is possible to revert back to the flat band condition, that is, $\psi_s = 0$, by applying $V_{GB} < 0$ V. By drawing the energy band diagram using Eq. (10.5) as shown in Fig. 10.8(b), we find that the flat band condition is achieved if $V_{GB} = \phi_{ms}$. Thus, when $\phi_{ms} \neq 0$,

$$V_{FB} = \phi_{ms} \tag{10.19}$$

where V_{FB} is the gate voltage with respect to bulk required to achieve flat-band condition and is called the *flat-band voltage*.

As shown in Fig. 10.8(b), when $V_{GB} = V_{FB}$, then $\psi_{ox} = \psi_s = 0$. Now, if $V_{GB} < V_{FB}$, then $\psi_s < 0$ giving rise to the accumulation condition. Also, to have $\psi_s > 0$, that is, for the depletion or inversion conditions, $V_{GB} > V_{FB}$. So, to get the identical condition in the semiconductor as in the ideal diode, an extra voltage of V_{FB} must be applied. In other words, wherever we have V_{GB} in the analysis of ideal diode, we may use $V_{GB} - V_{FB}$, or we may write

$$V_{GB} - V_{FB} = \psi_s - \frac{Q_s}{C_{ox}} \tag{10.20}$$

Using the condition that $\psi_s = 2\phi_B$ when $V_{GB} = V_{th}$, we have from Eqs. (10.10) and (10.20),

$$V_{th} = V_{FB} + 2\phi_B + \frac{\sqrt{2qN_A\varepsilon_s(2\phi_B)}}{C_{ox}} = V_{FB} + (V_{th})_{ideal} \tag{10.21}$$

HELP DESK 10.2

In Fig. 10.8(a), we see that when $V_{GB} = 0$ V, there is a potential drop across the oxide as well as the semiconductor, that is, $\psi_{ox} \neq 0$ and $\psi_s \neq 0$. Is it not possible to account for the work function difference by ψ_{ox} alone, so that $\psi_s = 0$, and the flat band condition exists?

No, it is not possible to manage the work function difference by ψ_{ox} alone. From Gauss law [Eq. (10.11)], we know that in the absence of any charge in the oxide, any electric field in the oxide will result in a corresponding electric field in the semiconductor. Thus if $\mathscr{E}_{ox} \neq 0$, an electric field in the semiconductor must exist so that $\psi_s \neq 0$.

Example 10.4

The MOS diodes MD2 and MD3 are identical to the ideal MOS diode MD1, except that MD2 uses aluminium as the gate metal and MD3 uses n^+ polysilicon. Calculate the threshold voltages of MD2 and MD3. [For Al, $\phi_m = 4.1$ V.]

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Solution:

For device MD1, $\phi_B = 0.35$ V and $\phi_s = 4.95$ V (Example 10.1) and $(V_{th})_{ideal} = 2.11$ V (Example 10.3).

(i) For MD2, $V_{FB} = \phi_{ms} = (\phi_m - \phi_s) = 4.1 - 4.95 = -0.85$ V

Using Eq. (10.21),

$$V_{th} = V_{FB} + (V_{th})_{ideal} = -0.85 + 2.11 = 1.26 \text{ V}$$

(ii) For MD3, using Eq. (10.18),

$$\phi_{ms} = -\frac{E_g}{2q} - \phi_B = -0.55 - 0.35 = -0.9 \text{ V}$$

Therefore,

$$V_{th} = V_{FB} + (V_{th})_{ideal} = -0.9 + 2.11 = 1.21 \text{ V}$$

10.2.3 Operation of MOS Diode with $\phi_{ms} \neq 0$, $Q_{ox} \neq 0$

In addition to the metal-semiconductor work function difference, another non-ideality is the presence of traps at the semiconductor-oxide interface and charges in the oxide. The different types of traps and charges and their location are shown in Fig. 10.9.

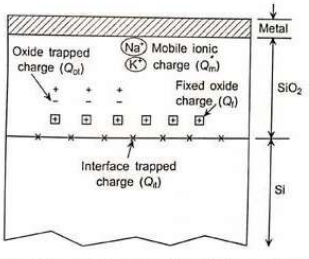


Figure 10.9 The different types of traps and charges and their location [1].

A silicon surface is a discontinuity in an otherwise periodic crystal lattice. From quantum mechanics, it can be shown that the discontinuity at a clean semiconductor surface gives rise to a number of allowed states in the forbidden gap. These surface states are due to the unsatisfied covalent bonds at the surface (also known as 'dangling bonds'). The density of the surface states is equal to the density of the dangling bonds, i.e. about 10^{15} cm^{-2}. On oxidation of the surface layer, most of the surface atoms are bound to oxygen resulting in a sharp reduction of the density of states at the interface of silicon and its oxide to about 10^{11}–10^{12} per cm^2. In a silicon surface of <100> orientation, the interface state density is about an order of magnitude smaller than in <111>. Thus, all MOS devices are fabricated on silicon substrates which have <100> orientation. A low temperature (450°C) hydrogen anneal is often used to reduce the density of states further by the formation of Si-H bonds at the interface. These *interface states* reside within 25 Å of the Si-SiO_2

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interface, and play an important role in determining device behaviour. The interface states are distributed nonuniformly throughout the forbidden gap. Therefore, they are quantified as interface trap density (D_{it}) with a unit of number of states per (cm^2-eV). A typical distribution of D_{it}, as shown in Fig. 10.10, consists of two high peaks near the band edges and a minimum near the middle of the band gap. The levels at the middle of the gap are the most important for device operation and good processing can produce devices with minimum D_{it} of $\approx 10^{10}$ per (cm^2-eV). Depending on the oxidation conditions and substrate crystal orientation, these states can either be donor-like or acceptor-like in behaviour. A donor-like state is neutral when occupied and positively charged when empty, while an acceptor-like state is negatively charged when occupied and neutral when empty. Therefore, these states give rise to surface charges, called *interface-trapped charges* (Q_{it}). Q_{it} is negative for acceptor-like states and positive for donor-like states.

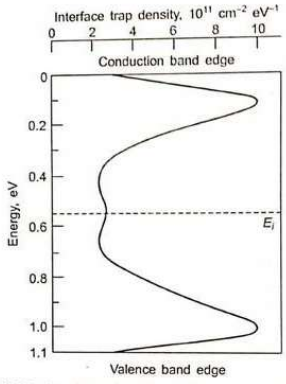


Figure 10.10 A typical distribution of D_{it} in the band gap of silicon [2].

When the silicon surface layer is converted to SiO_2 by a process of oxidation, there exists a thin interfacial layer of about 50 Å having composition SiO_x, where $0 < x < 2$. The excess silicon ions in this layer generate a net immobile positive charge. This charge Q_f, called the *fixed oxide charge* is not dependent on bias. It can be considered as a constant sheet of positive charge at the Si–SiO_2 interface. Typical values of the number of charge centres per unit area, that is, Q_f/q are 10^{10} per cm^2 for <100> silicon and 5×10^{10} per cm^2 for <111> silicon.

The *mobile ionic charges* Q_m, which exist due to contamination by alkali metal ions, cause reliability problems in MOS devices. They move back and forth in the oxide as the bias is changed, thereby altering the device parameters. Hence, care must be taken to eliminate the mobile ions during device fabrication.

When the device is subjected to high energy ionizing radiation or ion bombardment, charges may be trapped in the oxide. These trapped charges are called *oxide trapped charges* Q_{ot}. These traps can be distributed throughout the oxide layer.

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Effect of Q_f, Q_m, and Q_{ot}

Let us consider a MOS diode having an oxide thickness of t_{ox} and a sheet of positive charge of density Q_{ox} cm^{-2} located at a distance x_0 from the metal-SiO_2 interface as shown in Fig. 10.11(a). Let us assume $\phi_{ms} = 0$. Since, the device should be neutral overall, that is, $Q_G + Q_{ox} + Q_s = 0$, negative charges will be induced in the gate metal and semiconductor. Figure 10.11(a) also shows the electric field distribution $\mathscr{E}$ obtained by applying Gauss' law, the potential distribution obtained by integrating $\mathscr{E}$, and the corresponding band diagram. To obtain flat band condition, a negative voltage (V_{FB}) must now be applied to the metal making Q_G more negative, so that when $Q_G = -Q_{ox}$, $Q_s = 0$. Again, the charge, electric field, and potential distributions as well as the band diagram under flat band condition, (that is, $V_{GB} = V_{FB}$) are shown in Fig. 10.11(b). From Fig. 10.11(b), we have

$$V_{FB} = \frac{E_{Fs} - E_{Fm}}{q} = -\mathscr{E}_0 x_0 = -\frac{Q_{ox}}{\varepsilon_{ox}} x_0 = -\frac{Q_{ox}}{C_{ox}} \frac{x_0}{t_{ox}} \tag{10.22}$$

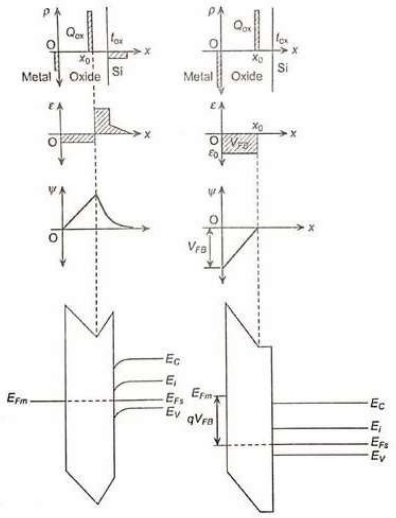


Figure 10.11 The charge, electric field, potential variation and band diagram at (a) $V_{GB} = 0$ V and (b) $V_{GB} = V_{FB}$ for a MOS diode having an oxide thickness of t_{ox} and a sheet of positive charge of density Q_{ox} per cm^2 located at a distance x_0 from the metal-SiO_2 interface.

Equation (10.22) shows that the effect of the oxide charges on V_{FB} depends on the location of these charges. When they are located at the metal-semiconductor interface, that is, at $x_0 = 0$, we get $V_{FB} = 0$. The maximum influence of these charges occurs when they are located at the oxide-semiconductor interface, that is, at $x_0 = t_{ox}$. Since the fixed oxide charges are always located at the Si–SiO_2 interface, the flat band voltage due to a fixed oxide charge density of Q_f per cm^2 can be obtained by substituting $x_0 = t_{ox}$ in Eq. (10.22), so that we have

$$V_{FB} = -\frac{Q_f}{C_{ox}} \tag{10.23}$$

The mobile ionic charges and oxide-trapped charges can however be distributed throughout the oxide layer. Let $\rho(x)$ be the density of charges in the charge sheet located at a distance x from the metal-SiO_2 interface. The flat band voltage V_{FB} can now be calculated by adding the effects of the individual charge sheets in the oxide. So, from Eq. (10.22), by replacing Q_{ox} by $\rho(x)$ and integrating in the limits 0–t_{ox}, we have

$$V_{FB} = -\int_0^{t_{ox}} \frac{\rho(x)}{C_{ox}} \frac{x}{t_{ox}} dx = -\frac{1}{C_{ox}} \left[\frac{1}{t_{ox}} \int_0^{t_{ox}} x\, \rho(x) dx \right] \tag{10.24}$$

Let $\rho_m(x)$ and $\rho_{ot}(x)$ represent the distributions of the mobile ionic charges and oxide-trapped charges respectively. We now define the terms Q_m and Q_{ot} as

$$Q_m = \frac{1}{t_{ox}} \int_0^{t_{ox}} x \rho_m(x)\, dx \tag{10.25a}$$

$$Q_{ot} = \frac{1}{t_{ox}} \int_0^{t_{ox}} x \rho_{ot}(x)\, dx \tag{10.25b}$$

Now, for a MOS diode with $\phi_{ms} \neq 0$, and non-zero values of Q_f, Q_m, and Q_{ot}, using Eqs. (10.19), (10.23), (10.24), and (10.25), we have

$$V_{FB} = \phi_{ms} - \frac{Q_f + Q_m + Q_{ot}}{C_{ox}} \tag{10.26}$$

It should be noted that Q_m and Q_{ot} do not represent the total mobile and oxide-trapped charges, respectively, but an equivalent charge, which when located at the Si–SiO_2 interface would have the same effect as the actual charge distribution. Equation (10.26) is a general expression for the flat band voltage, and the threshold voltage can be calculated by substituting the value of V_{FB} in Eq. (10.21).

The effect of Q_{it} is more complicated and its effect on the *C–V* characteristics is taken up in the next section.

Example 10.5

The MOS diode MD4 is similar to MD3, except that MD4 has a fixed oxide charge density given by $Q_f/q = 8 \times 10^{10}$ cm^{-2}, while $Q_m = Q_{ot} = 0$. Calculate the threshold voltage.

Solution:

For MD3, $C_{ox} = 3.45 \times 10^{-8}$ F/cm^2, $(V_{th})_{ideal} = 2.11$ V (Example 10.3), $\phi_{ms} = -0.9$ V (Example 10.4)

Using Eq. (10.26),

$$V_{FB} = -0.9 - \frac{1.6 \times 10^{-19} \times 8 \times 10^{10}}{3.45 \times 10^{-8}} = -1.27 \text{ V}$$

Therefore,

$$V_{th} = V_{FB} + (V_{th})_{ideal} = -1.27 + 2.11 = 0.84 \text{ V}$$

Example 10.6

A MOS diode MD5 having a threshold voltage of 1 V is to be fabricated. Just like MD3, MD5 has n^+ polysilicon gate. Its substrate doping concentration is similar to MD1 and MD3, which is $N_A = 1 \times 10^{16}$ cm^{-3}. It has a fixed oxide charge density given by $Q_f/q = 8 \times 10^{10}$ cm^{-2} and $Q_m = Q_{ot} = 0$. Calculate the gate oxide thickness t_{ox} required to achieve a $V_{th} = 1$ V.

Solution:

For MD1, $\phi_B = 0.35$ V and $W_{max} = 0.304$ μm (Example 10.3) and for MD3, $\phi_{ms} = -0.9$ V (Example 10.4)

Using Eqs. (10.21), (10.26), and (10.15)

$$1 = -0.9 - \frac{1.6 \times 10^{-19} \times 8 \times 10^{10}}{C_{ox}} + (2 \times 0.35) + \frac{1.6 \times 10^{-19} \times 10^{16} \times 0.304 \times 10^{-4}}{C_{ox}}$$

or

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = 3 \times 10^{-8} \text{ F/cm}^2$$

Therefore,

$$t_{ox} = \frac{3.9 \times 8.85 \times 10^{-14}}{3 \times 10^{-8}} = 1.15 \times 10^{-5} \text{ cm} = 1150 \text{ Å}$$

10.2.4 *C–V* Characteristics of the MOS Diode (Capacitor)

We cannot use the current–voltage (*I–V*) characteristics to characterize a MOS diode, since no current flows through the diode in the steady state. It can, however, be characterized as a capacitor. Hence, the MOS diode is mostly referred to as a MOS capacitor. The capacitance–voltage or *C–V* characteristics can therefore be used to obtain a lot of information about the device. A typical measurement setup is shown in Fig. 10.12. A dc bias (V_{GB}) is applied to the MOS capacitor. The capacitance meter measures the capacitance by applying a small ac voltage (typically 50 mV) on top of the dc bias and measuring the reactive current. By changing the dc bias using a ramp generator and plotting the capacitance as a function of dc bias, the *C–V* plot is obtained.

C–V plot for $Q_{it} = 0$

In a MOS capacitor, $Q_G + Q_{ox} + Q_s = 0$, since the MOS capacitor should be overall charge neutral. Assuming Q_{ox} is independent of V_{GB}, the small signal capacitance per unit area is given by

$$C = \frac{dQ_G}{dV_{GB}} = -\frac{dQ_s}{dV_{GB}} \tag{10.27}$$

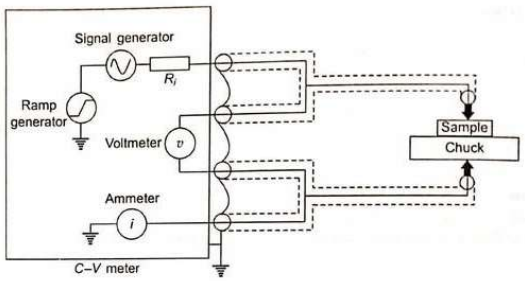


Figure 10.12 A typical setup to measure the capacitance–voltage (*C–V*) characteristics of MOS capacitors.

Hence,

$$\frac{1}{C} = -\frac{dV_{GB}}{dQ_s} = -\frac{d}{dQ_s}\left[V_{FB} + \psi_s - \frac{Q_s}{C_{ox}}\right] = \frac{1}{C_s} + \frac{1}{C_{ox}} \tag{10.28}$$

where

$$C_s = -\frac{dQ_s}{d\psi_s} \text{ and}$$

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$$

So the equivalent circuit of the MOS capacitor as shown in Fig. 10.13 consists of two capacitors, C_s and C_{ox} in series. Of the two capacitors, C_{ox} is a constant for a particular MOS capacitor while C_s is dependent on bias. We shall now discuss the nature of the *C–V* plot in the different regions of operation.

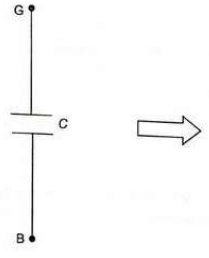


Figure 10.13 Equivalent circuit of the MOS capacitor.

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(a) *The accumulation region:* $V_{GB} < V_{FB}$; $\psi_s < 0$

In the accumulation condition, the hole concentration at the semiconductor surface depends exponentially on ψ_s. This implies that a small change in ψ_s will result in a large increase in Q_s. In other words, C_s will be very large. From Eq. (10.28), if $C_s >> C_{ox}$, $C = C_{ox}$. Physically, the result can be explained by noting that the high density of charge carriers in the accumulation layer makes the semiconductor surface behave almost like a metal plate. Thus, in the accumulation region, the MOS capacitor is electrically equivalent to a capacitor with two metal plates separated by a distance t_{ox}.

(b) *The depletion region:* $V_{FB} < V_{GB} < V_{th}$; $0 < \psi_s \le 2\phi_B$

In the depletion condition, the semiconductor charge is composed mostly of the uncovered impurity acceptor ions and is given by Eq. (10.10). Therefore, we have

$$C_s = -\frac{dQ_s}{d\psi_s} = \sqrt{\frac{qN_A\varepsilon_s}{2\psi_s}} = \varepsilon_s\sqrt{\frac{qN_A}{2\varepsilon_s\psi_s}} = \frac{\varepsilon_s}{W} \tag{10.29}$$

Thus, the capacitance is the same as that of a pn junction depletion width W or a capacitor having two metal plates separated by a distance W. Hence, from Eqs. (10.28) and (10.29), we have

$$\frac{1}{C} = \frac{1}{C_{ox}} + \frac{W}{\varepsilon_s} \tag{10.30}$$

Now, as V_{GB} increases, W increases and therefore C reduces. At threshold voltage, that is, at $V_{GB} = V_{th}$, the depletion width is maximum or $W = W_{max}$, and the depletion region stops expanding further. Therefore, the capacitance per unit area reaches a minimum value (C_{min}) at $V_{GB} = V_{th}$ given by

$$\frac{1}{C_{min}} = \frac{1}{C_{ox}} + \frac{W_{max}}{\varepsilon_s} \tag{10.31}$$

The variation of capacitance with applied voltage is shown in Fig. 10.14.

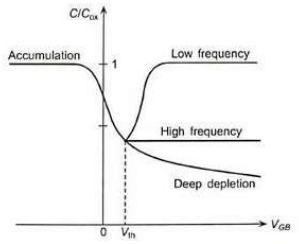


Figure 10.14 The C–V characteristics of MOS capacitors showing the measurements at low frequency (LF), high frequency (HF), and deep depletion (DD) conditions.

Example 10.7

If the device MD1 has an area A of 1×10^{-4} cm^2, what is its (i) maximum capacitance (C'_{max}), (ii) minimum capacitance (C'_{min})? [$\varepsilon_r = 11.9$ for Si]

Solution: For device MD1,

$$W_{max} = 0.304\ \mu\text{m},\ C_{ox} = 3.45 \times 10^{-8}\ \text{F/cm}^2 \text{ (from Example 10.3).}$$

Therefore,

(i) $C'_{max} = AC_{ox} = 1 \times 10^{-4} \times 3.45 \times 10^{-8} = 3.45 \times 10^{-12}$ F = 3.45 pF.

(ii) From Eq. (10.31),

$$\frac{1}{C_{min}} = \frac{1}{3.45 \times 10^{-8}} + \frac{0.304 \times 10^{-4}}{11.9 \times 8.85 \times 10^{-14}}$$

Therefore,

$$C_{min} = 1.73 \times 10^{-8}\ \text{F/cm}^2$$

and

$$C'_{min} = 1 \times 10^{-4} \times 1.73 \times 10^{-8} = 1.73 \times 10^{-12}\ \text{F} = 1.73\ \text{pF}$$

HELP DESK 10.3

How does the maximum to minimum capacitance (C_{max}/C_{min}) ratio depend on N_A and t_{ox}?

Using Eq. (10.31), $\frac{C_{max}}{C_{min}} = \frac{C_{ox}}{C_{min}} = 1 + \frac{C_{ox}W_{max}}{\varepsilon_s}$

The depletion width W_{max} decreases with increase in N_A. Also, with increase in t_{ox}, C_{ox} decreases. Hence, increase in either N_A or t_{ox} results in a decrease in the (C_{max}/C_{min}) ratio.

It is often useful to have an expression showing the variation of C with V_{GB}. Substituting the values of ψ_s and Q_s from Eqs. (10.9) and (10.10) in Eq. (10.20), we have a quadratic equation in W given by

$$V_{GB} - V_{FB} = \frac{qN_AW^2}{2\varepsilon_s} + \frac{qN_AW}{C_{ox}} \tag{10.32}$$

The solution of this quadratic equation is of the form

$$W = \frac{\varepsilon_s}{C_{ox}}\left[\sqrt{1 + \frac{2C_{ox}^2(V_{GB} - V_{FB})}{qN_A\varepsilon_s}} - 1\right] \tag{10.33}$$

Substituting the above expression for W in Eq. (10.30) we have

$$C = \frac{C_{ox}}{\sqrt{1 + \frac{2C_{ox}^2(V_{GB} - V_{FB})}{qN_A\varepsilon_s}}} \tag{10.34}$$

The expression in Eq. (10.34) is often used to extract the doping concentration of the semiconductor substrate N_A from C–V measurements. Equation (10.34) can be rewritten as

$$\frac{1}{C'^2} = \frac{1}{C_{ox}'^2} + \frac{2(V_{GB} - V_{FB})}{A^2qN_A\varepsilon_s} \tag{10.35}$$

where

A = area of the diode and

$C' = AC$ is the actual measured capacitance.

If we now plot the variation of $(1/C'^2)$ versus V_{GB}, a straight line whose slope is equal to $2/(A^2qN_A\varepsilon_s)$ is obtained. This is similar to the case already discussed in section 4.2.1 for an abrupt n$^+$p junction. Figure 10.15 shows the plots for two diodes having different substrate doping concentrations. If the area of the device is known, N_A can easily be calculated from the slope of the line.

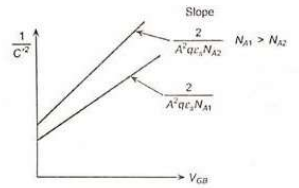


Figure 10.15 $(1/C'^2)$ versus V_{GB} plots for two diodes having different substrate doping concentrations.

A point to be noted is the effect of V_{FB} on the C–V characteristics. From Eq. (10.34), we see that C depends on $(V_{GB} - V_{FB})$, that is, if we have two diodes with identical N_A and C_{ox} but with different flat band voltages, the capacitance per unit area will be the same at the same $(V_{GB} - V_{FB})$. Suppose the capacitance per unit area of an ideal MOS diode (that is, $V_{FB} = 0$ V) at a voltage V_{GB1} is C_1. Then a MOS diode having identical N_A and C_{ox} will have the same capacitance per unit area (C_1) at an applied voltage of $(V_{GB1} + V_{FB})$, since the value of $(V_{GB1} - V_{FB})$ for both the devices is V_{GB1}. This implies that the C–V characteristics of a diode with $V_{FB} \neq 0$ V will experience a parallel shift of V_{FB} with respect to the corresponding ideal diode curve. The C–V characteristics of two such diodes are shown in Fig. 10.16.

(c) *Strong inversion region:* $V_{GB} > V_{th}$; $\psi_s \ge 2\phi_B$

For $V_{GB} > V_{th}$, $Q_s = Q_B + Q_n$ and Q_n is comparable to Q_B. In this region of operation, any increase in Q_s is due to the increase in Q_n, since the depletion width has reached its maximum value, making Q_B constant. So it is expected that C_s is extremely large since Q_n increases exponentially with ψ_s. This is true, provided ψ_s and V_{GB} change slowly. However, electrons in the surface inversion layer cannot respond to rapid change in V_{GB}, and in particular, to the high frequency (~1 MHz) signal used to measure the capacitance. This is because any addition or decay in the surface concentration of electrons, which are minority carriers in the semiconductor bulk, is due to the process of thermal generation or recombination. Unless the minority carrier lifetime is

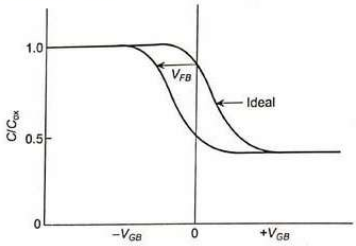


Figure 10.16 The C-V characteristics of diodes with $V_{FB} = 0$ V and $V_{FB} \neq 0$ V.

abnormally small, these generation/recombination processes are extremely slow. For example, in a MOS capacitor fabricated on a high quality silicon substrate, it may take a few seconds for the inversion layer to form after it is biased in the strong inversion condition. (This problem does not exist in a MOSFET, where electrons can be drawn from the n$^+$ doped source region). So we see that depending on the measurement conditions, we may get different C–V plots in the strong inversion region as follows:

(i) *A low frequency signal (few Hertz) is used to measure the capacitance and V_{GB} is changed slowly.* Due to the ac signal on the gate, the semiconductor charge should also vary at the same rate. In this case, the electron concentration in the inversion layer is able to change with the applied small signal. Since C_s is extremely large, the use of Eq. (10.28) shows that $C = C_{ox}$. So the capacitance increases to the maximum value as shown in the LF plot in Fig. 10.14.

(ii) *A high frequency signal (~1 MHz) is used to measure the capacitance and V_{GB} is changed slowly.* As a high frequency ac signal is applied on the gate, the electron concentration in the inversion layer is unable to follow the applied small signal. The depletion width, however, widens and narrows at the same rate as the ac signal. Since the capacitance is now due to variation of the depletion width, it is given by Eq. (10.30), where $W = W_{max}$ is the average depletion width. So the capacitance remains constant at C_{min} given by Eq. (10.31) as shown in the HF plot in Fig. 10.14.

(iii) *V_{GB} is changed quickly (about 10 V/s and higher).* In this case, the inversion layer does not get sufficient time to form. But as V_{GB} is increased, the positive charge on the metal keeps increasing and there must be an equal increase in the negative charge in the semiconductor. To achieve this, the depletion width keeps increasing beyond W_{max} and the capacitance which is given by Eq. (10.28) is less than the value of C_{min} given by Eq. (10.31). This is called the *deep depletion condition* and the *C–V* plot for this condition is also shown in Fig. 10.14. It must be remembered that deep depletion is a non-equilibrium condition. If the voltage sweep is stopped suddenly in this condition, the inversion layer will slowly form due to generation of electrons and the capacitance approaches C_{min}.

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In Figure 10.17, the dashed line shows the plots obtained from the equations presented above, while the solid line shows the nature of the plots obtained experimentally. The deviation is due to the fact that we have used the depletion approximation in our analysis, thereby neglecting the carrier concentrations in the depletion region. However, the hole concentration is significant near flat-band condition and the electron concentration near threshold. This results in the experimental C–V curves being smoothened out in these regions.

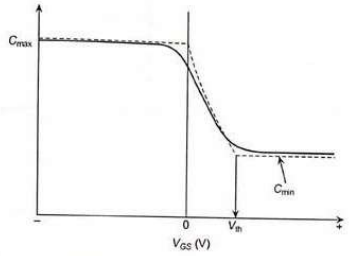


Figure 10.17 Comparison of the nature of C–V plots obtained from our analysis (dashed line) with those obtained experimentally (solid line).

C–V Plot for $Q_{it} \neq 0$

We have already seen that the C–V characteristics of a MOS capacitor are shifted by V_{FB} when compared to the characteristics of the corresponding ideal device (Fig. 10.16). When the fixed oxide charge density is Q_f per cm^2, the resultant shift due to it is equal to $-Q_f/C_{ox}$. The presence of interface trapped charge, Q_{it}, has a similar effect, giving rise to a shift of $-Q_{it}/C_{ox}$. However, there are certain differences. While Q_f is always positive, Q_{it} can be either positive or negative, depending on whether the states are donor-like or acceptor-like. Also Q_f is fixed and does not depend on bias and thereby it results in a parallel shift from the ideal C–V curve. On the other hand, Q_{it} changes with bias and hence the shift from the ideal C–V curve changes continuously from the accumulation to the inversion condition. This can be clearly understood from Fig. 10.18, which shows a p-type silicon substrate with acceptor-like states distributed at the surface. We assume that all states below the Fermi level are filled giving rise to negative charges, while all the states above the Fermi level are empty and therefore neutral. As shown in Fig. 10.18(a), when the silicon surface is in accumulation, some states near the valence band lie under the Fermi level, giving rise to a small amount of negative charges. As we move towards inversion, the bands bend downwards, and more and more states are filled as shown in Fig. 10.18(b). The resulting C–V curve (Fig. 10.19) shows a very small positive shift in accumulation, gradually increasing to a maximum value in inversion. The threshold voltage of the MOS device is increased as the additional positive charge on the gate is balanced by increasing Q_{it} rather than increasing Q_i. This is one of the reasons why it is almost impossible to create an inversion layer in GaAs, which has a large density of acceptor-like surface states.

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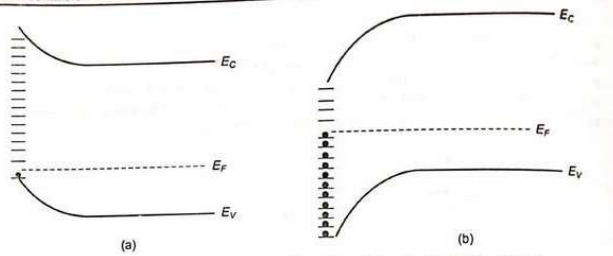


Figure 10.18 Change in charge of acceptor-like surface states due to bending of bands.

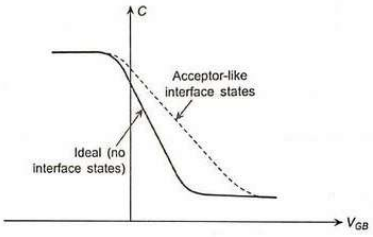


Figure 10.19 Shift in C–V plot due to surface states.

10.3 THE MOSFET

Figure 10.20 shows a perspective view of a four-terminal n-channel MOSFET and the respective symbols for n-channel and p-channel MOSFETs. In addition to the MOS structure, the MOSFET consists of n$^+$ diffused source and drain regions. The space separating the source and drain is called the *channel* of the MOSFET. Figure 10.20 also shows the important parameters of the MOSFET, namely the channel length L, which is the distance between the two p-n junctions, the channel width Z, the gate oxide thickness t_{ox}, the source and drain junction depths r_j and the substrate doping concentration N_A. In this section, we shall use these parameters to develop n-channel MOSFET models. A p-channel MOSFET has p$^+$ doped source and drain regions and an n-type substrate. Although, all the analyses in this chapter are for n-channel MOSFETs, they can also be applied to p-channel MOSFETs by changing the polarity of the terminal voltages. In MOSFETs, the terminal voltages are usually referred to with respect to the source. We shall be following the same convention in this chapter, that is, the model equations will be developed with respect to V_{GS},

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V_{DS}, and V_{BS} which are the gate-to-source, the drain-to-source, and the bulk-to-source voltages respectively. For the normal operation of an n-channel MOSFET, V_{GS} and V_{DS} are positive voltages, while V_{BS} is always negative or zero.

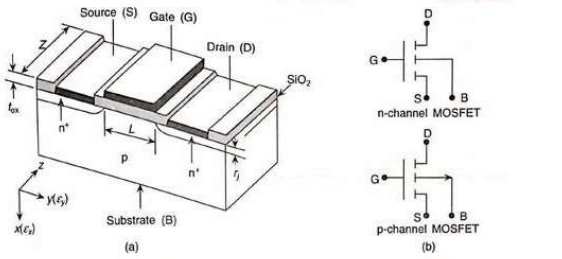


Figure 10.20 (a) Perspective view of a four-terminal n-channel MOSFET and (b) symbols for n-channel and p-channel MOSFETs.

Let us first qualitatively discuss the operation of the MOSFET. The MOSFET consists of two p-n junctions connected back to back. We have already seen in the discussion for a MOS diode, that when the gate-to-bulk voltage exceeds the threshold voltage (that is, $V_{GB} > V_{th}$), an inversion layer is formed. An inversion layer is also formed in the MOSFET channel, when its gate-to-source voltage exceeds its threshold voltage (that is, $V_{GS} > V_{th}$). It may be noted that the threshold voltage for a MOSFET need not be the same as the value for the corresponding MOS diode, as in one case it is referred to as the gate-to-source voltage while in the other it is the gate-to-bulk voltage. When $V_{GS} < V_{th}$, the MOSFET channel is either accumulated or depleted and *there is no conducting path* between the source and drain. When accumulation condition exists in the channel, the only current which can flow is the reverse leakage current of the p-n junction diodes. When the channel region is depleted, a diffusion current exists, with electrons flowing from the source to drain under the influence of a concentration gradient of electrons. This current, called the *subthreshold current*, reduces exponentially as V_{GS} is reduced below V_{th}. The subthreshold current is discussed in detail in the next chapter.

When $V_{GS} > V_{th}$, the silicon surface is inverted and an *n-type conducting channel exists* between the n^+ source and drain regions. Now if a positive voltage is applied to the drain with respect to the source (that is, $V_{DS} > 0$), electrons flow from the source towards the drain under the influence of the electric-field. The *above-threshold current* in MOSFETs is therefore primarily a drift current. The inversion layer behaves as a resistance and as V_{DS} is increased initially from 0 V, the drain current I_D increases linearly. This is shown in Fig. 10.21(a). With increasing V_{DS}, however, the current–voltage (I–V) characteristics deviate from linearity. This is because, with increasing V_{DS} and a fixed V_{GS}, the voltage across the oxide and consequently the electron concentration reduces from the source to the drain end of the channel.

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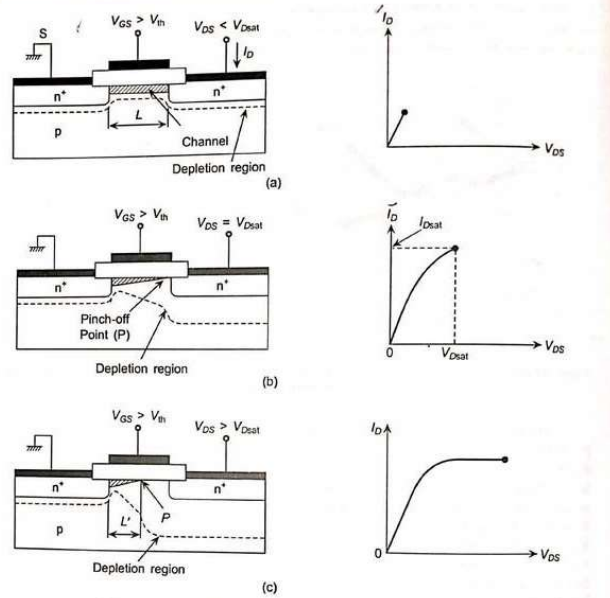


Figure 10.21 Various regions of operation of a MOSFET.

Ultimately, when $V_{DS} = V_{Dsat}$, the voltage across the oxide at the drain end is no longer sufficient to support an inversion layer. This is called the *pinch-off* and is shown in Fig. 10.21(b). Since the voltage across the oxide determines the electron concentration in the semiconductor, V_{Dsat} has a higher value for higher V_{GS}. When $V_{DS} < V_{Dsat}$, the device is said to be in the *linear region* of operation. For $V_{DS} > V_{Dsat}$, the drain current remains almost constant. In this region, called the *saturation region*, the pinch-off point moves towards the source as V_{DS} increases. The voltage at the pinch-off point remains V_{Dsat}, the extra voltage ($V_{DS} - V_{Dsat}$) being now dropped in a depletion layer. Electrons, which reach the pinch-off point, are swept across to the drain by the electric field in the depletion region. This is quite similar to the way the injected carriers are swept across the reverse biased base–collector junction in a bipolar transistor. The current in the saturation region remains almost fixed as the voltage across the inversion layer is fixed at V_{Dsat}. A small increase in current is observed due to the shift of the pinch-off point towards the source end, which effectively reduces the channel length of the MOSFET from L to L' as shown in Fig. 10.21(c).

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The discussion carried out so far accounts for a particular value of $V_{GS} > V_{th}$. With further increase in V_{GS}, the channel conductivity increases, resulting in an increase in I_D for all values of V_{DS}. The complete I–V characteristics of a MOSFET are depicted in Fig. 10.22.

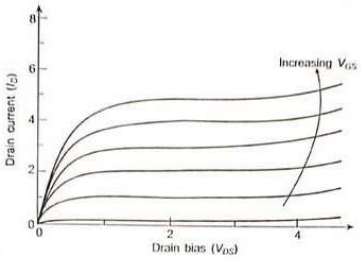


Figure 10.22 I–V characteristics of a MOSFET.

We shall now proceed to develop analytical expressions for the I–V characteristics of MOSFETs. Before taking up the derivation of expressions for the drain current, we shall discuss about the threshold voltage V_{th}, which is one of the most important parameters of any MOSFET.

10.3.1 Threshold Voltage of MOSFET

In a MOSFET, since all voltages are measured with respect to the source, the threshold voltage is measured as a particular value of gate-to-source voltage and not a gate-to-bulk voltage as in a MOS diode. The *threshold voltage* (V_{th}) of a MOSFET is defined as the value of the gate-to-source voltage which is just sufficient to produce a surface inversion layer when $V_{DS} = 0$ V.

Due to the application of a substrate bias with respect to the source and drain, the channel region is no longer in thermal equilibrium. For an n-channel MOSFET, V_{BS} is usually less than or equal to zero, ($V_{BS} \leq 0$), that is, the source and drain junctions are reverse biased, to avoid the flow of a substrate current. We have seen earlier in section 4.3.2 that in such a situation, the Fermi level splits into the electron and hole quasi-Fermi levels separated by an energy (in eV) equal to the applied bias across the p-n junctions (in V). For a reverse bias of V_R, the electron quasi-Fermi level E_{Fn} moves down in the p-region, and at the edge of the depletion region, $(E_F - E_{Fn}) = qV_R$, where E_F is the Fermi level in the bulk.

In a MOSFET, considering the p-n junctions at the source and drain, we have

$$E_F - E_{Fn} = -qV_{BS} \quad \text{and} \quad E_F - E_{Fn} = q(V_{DS} - V_{BS})$$

at the source and drain ends of the channel respectively.
There is therefore a continuous variation in E_{Fn} from the source to the drain end of the channel when $V_{DS} \neq 0$, which accounts for the flow of electrons.

The threshold voltage is, however, defined for $V_{DS} = 0$ V. For this case, the position of E_{Fn} is constant throughout the channel and is given by $E_{Fn} = E_F + qV_{BS}$. The band diagram of the MOS structure in the channel region at the threshold condition is shown in Fig. 10.23. Now at the onset of strong inversion, the electron concentration at the semiconductor surface is equal to the hole concentration in the bulk. That is,

$$n_{ps} = N_A = n_i \exp\left(\frac{E_{Fn} - E_{is}}{kT}\right) \tag{10.36}$$

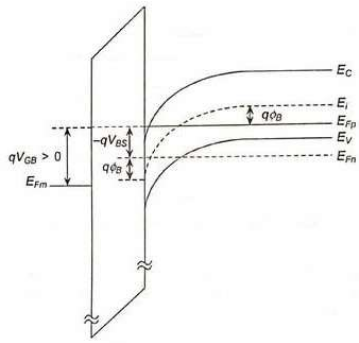


Figure 10.23 Band diagram in the channel region of a MOSFET at threshold condition.

where

E_{is} = intrinsic level at the semiconductor surface, which is related to the intrinsic level at the bulk, E_i, through the relation $E_{is} = E_i - q\psi_s$.

Now, expressing E_{Fn} and E_{is} in Eq. (10.36) in terms of E_F and E_i, we have

$$N_A = n_i \exp\left(\frac{E_F + qV_{BS} - E_i + q\psi_s}{kT}\right)$$

or,

$$E_F - E_i + q(V_{BS} + \psi_s) = kT \ln\left(\frac{N_A}{n_i}\right) = q\phi_B$$

Since $q\phi_B = E_i - E_F$,

at threshold condition, we have

$$\psi_s = 2\phi_B - V_{BS} \tag{10.37}$$

This implies that for $V_{BS} < 0$ V, the potential drop in the semiconductor at threshold is greater than in a MOS capacitor, resulting in an increase in threshold voltage.

Assuming that Q_n is negligible at threshold, so that $Q_s \approx Q_B$, from Eq. (10.20) we have

$$V_{GB} - V_{FB} = \psi_s - \frac{Q_B}{C_{ox}} = 2\phi_B - V_{BS} + \frac{\sqrt{2\varepsilon_s qN_A(2\phi_B - V_{BS})}}{C_{ox}} \tag{10.38}$$

Now, $V_{GS} = V_{GB} + V_{BS}$, and since $V_{GS} = V_{th}$ at threshold, from Eq. (10.38) we have

$$V_{th} = V_{FB} + 2\phi_B + \frac{\sqrt{2\varepsilon_s qN_A(2\phi_B - V_{BS})}}{C_{ox}} \tag{10.39}$$

Equation (10.39) shows that when $V_{BS} = 0$ V, the threshold voltage for the MOSFET reduces to that for the corresponding MOS capacitor given by Eq. (10.21). The effect of V_{BS} on the threshold voltage can now be expressed through the relation

$$\Delta V_{th} = V_{th} - V_{th0} = \gamma(\sqrt{2\phi_B - V_{BS}} - \sqrt{2\phi_B}) \tag{10.40}$$

where

ΔV_{th} = change in threshold voltage due to the application of substrate bias,

V_{th0} = threshold voltage for $V_{BS} = 0$ V given by Eq. (10.21),

and γ is called the *body effect parameter* given by

$$\gamma = \frac{\sqrt{2\varepsilon_s qN_A}}{C_{ox}} \tag{10.41}$$

From Eq. (10.41), we see that the body-effect, which is the change in V_{th} due to a substrate bias, is more pronounced in MOSFETs with higher N_A and t_{ox}.

Example 10.8

The MOS structure in MOSFET MF1 is similar to the MOS diode MD4, that is, it has an n^+ polysilicon gate, the gate oxide thickness (t_{ox}) is 1000 Å and the substrate doping concentration is 1×10^{16} per cm^3. Calculate the threshold voltage of MF1 for (i) $V_{BS} = 0$ V, (ii) $V_{BS} = -2$ V and (iii) $V_{BS} = -5$ V.

Solution: Using Eq. (10.41),

$$\gamma = \frac{\sqrt{2 \times 11.9 \times 8.85 \times 10^{-14} \times 1.6 \times 10^{-19} \times 10^{16}}}{3.45 \times 10^{-8}} = 1.683$$

The threshold voltage V_{th} is calculated using Eq. (10.40), where V_{th0} is the threshold voltage of the corresponding MOS capacitor. From Example (10.5), V_{th} for MD4 is given as 0.84 V. Also, $\phi_B = 0.35$ V for $N_A = 10^{16}$ cm^{-3} (Example 10.1). Therefore from Eq. (10.40),

(i) When $V_{BS} = 0$ V,

$$V_{th} = V_{th0} = 0.84 \text{ V}$$

(ii) When $V_{BS} = -2$ V,

$$V_{th} = 0.84 + 1.683(\sqrt{2.7} - \sqrt{0.7}) = 2.2 \text{ V}$$

(iii) When $V_{BS} = -5$ V,

$$V_{th} = 0.84 + 1.683(\sqrt{5.7} - \sqrt{0.7}) = 3.45 \text{ V}$$

10.3.2 Above-threshold I–V Characteristics of MOSFETs

In this section, we shall extend the analysis for MOS diodes to derive expressions for the drain current I_D of the four-terminal MOSFET as a function of its terminal voltages when $V_{GS} > V_{th}$. In order to simplify the analysis, we shall initially make some assumptions. We shall subsequently discuss the cases where these assumptions are not valid. The assumptions are:

(a) There are no interface states.
(b) The drift component of current is much larger than any diffusion component.
(c) Carrier mobility in the channel is constant.
(d) The channel region is uniformly doped.
(e) The transverse electric field E_x, which is perpendicular to the direction of current flow, is much larger than the longitudinal field E_y in the direction of current flow. This condition is referred to as the *gradual channel approximation* (GCA), and MOSFETs where this approximation is valid are called *long-channel MOSFETs*.

The two-dimensional nature of the device (that is, the presence of two electric fields perpendicular to one another) makes it difficult to obtain simple, yet accurate analytical expressions for the current–voltage characteristics of MOSFETs. The gradual channel approximation is used to simplify the problem. Here, the MOSFET channel is imagined to consist of a large number of MOS capacitors, which fill the space between the source and drain. Since the longitudinal electric field can be ignored, the one-dimensional analysis for MOS capacitors can be used to calculate the charges Q_n and Q_B at the semiconductor surface. These charges vary from capacitor to capacitor, depending on their position with respect to the source and drain. Once the electron distribution along the channel is obtained, the drain current can be calculated by considering the longitudinal electric field through a separate one-dimensional analysis. The gradual channel approximation therefore enables us to reduce a two-dimensional problem into simpler one-dimensional form.

Several analytical models of the above threshold conduction of MOSFETs have been proposed. In this book, we shall discuss three of these models, which are available as Level 1, Level 2, and Level 3 MOSFET models in the popular circuit simulator SPICE.

Bulk charge model (Level 2 in SPICE)

Linear region characteristics. Let us at first derive a relation for the I–V characteristics of a MOSFET in the linear region of operation, that is, when a continuous inversion layer exists from the source to the drain end of the channel. To help us in deriving a relation for the drain current I_{DS}, let us define a quantity $V_{CB}(y)$, which is the *channel voltage with respect to the bulk* at a distance y from the source. When a drain voltage is applied and a current flows from source to drain, V_{CB} also varies continuously from source to drain. At the source, $V_{CB} = -V_{BS}$ and at the drain $V_{CB} = V_{DS} - V_{BS}$. The electron quasi-Fermi level at the semiconductor surface, E_{Fn}, also varies continuously from the source to the drain and is related to V_{CB} by the relaion $E_{Fn}(y) = E_F - qV_{CB}(y)$. This can be understood by considering a p-n junction at the semiconductor surface consisting of the n-type inversion layer and p-type bulk which has a reverse bias of V_{CB}. This results in E_{Fn} moving down by qV_{CB} with respect to E_F.

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In the analysis of the MOS capacitor, it was argued that once strong inversion sets in, the potential at the semiconductor surface should remain fixed at $\psi_s = 2\phi_B$. A similar argument holds for MOSFETs too, but in this case ψ_s varies along the channel. $\psi_s(y)$ is calculated as the surface potential for which the electron concentration at the surface, n_{ps}, equals hole concentration in the bulk, N_A. Using Eq. (10.36), and substituting $E_{Fn}(y) = E_F - qV_{CB}(y)$ and $E_i(y) = E_i - q\psi_s(y)$, it can be shown that the surface potential will be pinned at a value of $\psi_s(y) = 2\phi_B + V_{CB}(y)$, indicating a continuous variation of the surface potential. Figure 10.24 shows the band diagram in a direction normal to surface at a distance y from the source end.

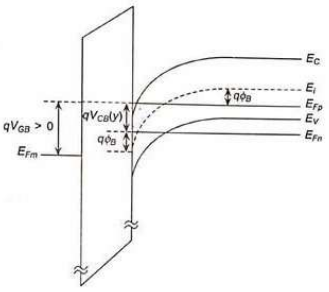


Figure 10.24 Band diagram for a MOSFET in a direction normal to surface at a distance y from the source.

To compute the magnitude of the drain current, it is necessary to determine the amount of mobile electron charge in the channel. Using the relations derived for MOS capacitor and taking $\psi_s(y) = 2\phi_B + V_{CB}(y)$, we shall now calculate the mobile charge density at a distance y from the source. From Eq. (10.20), the total charge density can be calculated as

$$Q_s(y) = -C_{ox}\,[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y)] \tag{10.42}$$

Using Eq. (10.10), the bulk charge density is given by

$$Q_B(y) = -\sqrt{2qN_A\varepsilon_s(2\phi_B + V_{CB}(y))} \tag{10.43}$$

The electron charge density $Q_n(y)$ is now obtained as the difference of the total charge and the bulk charge. Therefore,

$$\begin{aligned} Q_n(y) &= Q_s(y) - Q_B(y) \\ &= -C_{ox}\,[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y)] + \sqrt{2qN_A\varepsilon_s\,[2\phi_B + V_{CB}(y)]} \\ &= -C_{ox}\,[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y) - \gamma\sqrt{2\phi_B + V_{CB}(y)}\,] \end{aligned} \tag{10.44}$$

where γ = body effect parameter given by Eq. (10.41).

From Eq. (10.44), it is clear that since V_{CB} is a function of position along the channel, so is $Q_n(y)$. Let us see why this should be so. As we move from the source to the drain, V_{CB} increases and so does the value of ψ_s required to maintain inversion. Since the gate voltage is fixed but the semiconductor surface potential is more towards the drain, the drop across the oxide ψ_{ox} obviously reduces as we move towards the drain. Since $|Q_s| = C_{ox}\psi_{ox}$, the magnitude of total charge density is less near the drain than near the source. On the other hand, since ψ_s is more towards the drain, the depletion layer width is more and hence the magnitude of the bulk charge density $|Q_B|$ is more. Now since $|Q_s|$ is less and $|Q_B|$ is more and $|Q_n| = |Q_s| - |Q_B|$, the electron concentration will be less towards the drain.

We shall now use the expression for electron charge density to derive an expression for drain current. Assuming that the current flowing in the channel under the influence of the drain voltage is purely drift current, the electron current density at a distance y from the source is given by

$$J_n(x, y) = -qn(x, y)\,\mu_n\,\frac{dV_{CB}(y)}{dy} \tag{10.45}$$

Integrating Eq. (10.45) from the surface at $x = 0$ to the bottom edge of the inversion layer at $x = x_{inv}$ and multiplying it by the channel width Z, the electron current I_n flowing from the source to the drain can be obtained. Therefore,

$$\begin{aligned} I_n(y) &= Z\int_0^{x_{inv}} J_n(x, y)\,dx = -Z\mu_n\,\frac{dV_{CB}(y)}{dy}\int_0^{x_{inv}} qn(x, y)\,dx \\ &= Z\mu_n Q_n(y)\,\frac{dV_{CB}(y)}{dy} \end{aligned} \tag{10.46}$$

In Eq. (10.46), the integral of the electron concentration over the depth of the channel has been taken to be the total number of electrons per unit area in the channel. Now since there is no appreciable hole current in the channel, I_n must be constant all the way from the source to the drain of the channel. If this were not the case, there would be rapid build up of electrons in a portion of the channel, which is not sustainable. As Q_n decreases from source to drain, in order to keep I_n constant, the electric field increases from source to drain and is maximum at the drain end. Since I_n is constant along the channel, we may write

$$\int_0^L I_n(y)\,dy = I_n\int_0^L dy = -I_D L \tag{10.47}$$

Since the drain current I_D is considered to be positive when the current flows into the drain terminal, we have taken $I_D = -I_n$ in Eq. (10.47). Now from Eqs. (10.46) and (10.47), we have

$$I_D = -\frac{Z\mu_n}{L}\int_0^L Q_n\,\frac{dV_{CB}(y)}{dy}\,dy \tag{10.48}$$

Noting that $V_{CB} = -V_{BS}$ at $y = 0$ and $V_{CB} = V_{DS} - V_{BS}$ at $y = L$, we can now rewrite Eq. (10.48) as

$$I_D = -\frac{Z\mu_n}{L}\int_{-V_{BS}}^{V_{DS}-V_{BS}} Q_n(y)\,dV_{CB}(y) \tag{10.49}$$

Now substituting Eq. (10.44) in Eq. (10.49), we have

$$I_D = \frac{Z\mu_n C_{ox}}{L}\int_{-V_{BS}}^{V_{DS}-V_{BS}} \left[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y) - \gamma\sqrt{2\phi_B + V_{CB}(y)}\right]dV_{CB}(y) \tag{10.50}$$

Integrating this expression, drain current is found to be

$$I_D = \frac{Z\mu_n C_{ox}}{L}\left[\left(V_{GS} - V_{FB} - 2\phi_B - \frac{V_{DS}}{2}\right)V_{DS} - \frac{2}{3}\gamma\{(2\phi_B + V_{DS} - V_{BS})^{3/2} - (2\phi_B - V_{BS})^{3/2}\}\right] \tag{10.51}$$

This is referred to as the *bulk charge model* and is available in the Level 2 MOSFET model in SPICE. However, this model has some drawbacks. In deriving Eq. (10.51), it was assumed that when the silicon surface is inverted, its potential is pinned at $(2\phi_B + V_{CB})$. In reality, this is the value when the surface just enters strong inversion. If the electron charge in the channel is substantial, ψ_s must be greater than this value. It is true that ψ_s will never be much larger than $(2\phi_B + V_{CB})$, but it may be more by a few times the thermal voltage V_T. So we may say that bulk charge model consistently underestimates ψ_s. Now, if ψ_s is underestimated, $|Q_B|$ is also underestimated. On the other hand, ψ_{ox} and consequently $|Q_s|$ are overestimated. This implies that $|Q_n| = |Q_s| - |Q_B|$ is definitely overestimated. Therefore, the bulk charge model consistently overestimates the electron concentration in the channel, which means that the drain current is also overestimated. Fortunately, the error involved is within 10%. A more accurate model, which considers the exact surface potential is called *charge sheet model*. However, this model resorts to numerical techniques and is not covered in the present scope of the book.

Saturation region characteristics: In the analysis so far, it has been assumed that an inversion layer exists all the way from the source to the drain. This is true for small V_{DS}. However, as V_{DS} is increased, the energy level E_{Fn} near the drain moves lower. This means that the value of ψ_s required to maintain inversion, should be larger. A higher ψ_s results in higher $|Q_B|$ and lower $|Q_s|$, due to lower ψ_{ox}. Therefore, with increasing V_{DS}, the electron charge density near the drain given by $|Q_n| = |Q_s| - |Q_B|$ reduces. Ultimately, at a particular value of V_{DS}, called V_{Dsat}, Q_n goes to zero. Under this condition, the channel is said to be pinched off near the drain. This might lead one to conclude that the drain current drops to zero. However, this is not true. In fact, once the electron charge at the drain end becomes very small, the electric field parallel to the interface increases, and gradual channel approximation itself fails. Therefore, the pinch-off condition at $V_{DS} = V_{Dsat}$ actually marks the failure of the gradual channel approximation rather than Q_n going to zero. However, for the purpose of modelling, V_{Dsat} is defined as the value of V_{DS} for which the gradual channel approximation gives $Q_n = 0$. Since pinch-off occurs initially at the drain end of the channel when $V_{DS} = V_{Dsat}$, we can now obtain a relation for V_{Dsat} by substituting $Q_n(y) = 0$ and $V_{CB}(y) = V_{Dsat} - V_{BS}$ in Eq. (10.44). Therefore,

$$V_{GB} - V_{FB} - 2\phi_B - V_{Dsat} + V_{BS} = \gamma\sqrt{2\phi_B + V_{Dsat} - V_{BS}} \tag{10.52}$$

Solving Eq. (10.52) as a quadratic equation in V_{Dsat}, we have

$$V_{Dsat} = V_{GS} - V_{FB} - 2\phi_B + \frac{\gamma^2}{2}\left[1 \pm \sqrt{1 + \frac{4(V_{GB} - V_{FB})}{\gamma^2}}\right] \tag{10.53}$$

In order to resolve whether the plus or minus sign is to be taken in the above equation, let us consider the following argument. Since $(V_{GB} - V_{FB})$ equals the sum of the potential drop in the oxide and in the semiconductor, and the voltage drop in the semiconductor at the drain end is equal to $(V_{Dsat} + 2\phi_B - V_{BS})$ when the transistor goes to saturation, we can conclude that

$$V_{GB} - V_{FB} > V_{Dsat} + 2\phi_B - V_{BS}$$

or

$$V_{Dsat} < V_{GB} - V_{FB} - 2\phi_B + V_{BS} = V_{GS} - V_{FB} - 2\phi_B \tag{10.54}$$

In Eq. (10.53), the quantity within brackets is positive if we accept the plus sign and is negative if we accept the minus sign. Thus, Eq. (10.54) can be satisfied only if we retain the negative sign. Therefore, V_{Dsat} is now given by

$$V_{Dsat} = V_{GS} - V_{FB} - 2\phi_B + \frac{\gamma^2}{2}\left[1 - \sqrt{1 + \frac{4(V_{GB} - V_{FB})}{\gamma^2}}\right] \tag{10.55}$$

As already mentioned, when $V_{DS} > V_{Dsat}$, the pinch-off point moves towards the source, leaving a depletion layer between this point and the drain. The voltage $(V_{DS} - V_{Dsat})$ is dropped across this depletion layer, and the high electric field helps to sweep the electrons entering this region to the drain. The current in this region of the characteristics remains almost constant, a small increase resulting from the *channel length modulation*, that is, the reduction of the effective channel length as the pinch-off point moves towards the source. The modelling of the current in the saturation region is detailed in the next chapter.

Square law model (Level 1 in SPICE)

Equation (10.51) is a relatively accurate description of current in the long channel MOSFET. However, this equation is rather cumbersome to use in manual calculations and slow to evaluate when incorporated in circuit simulation packages such as SPICE. A simplified version of Eq. (10.51) is obtained if we assume that $2\phi_B - V_{BS} >> V_{DS}$. Then we have,

$$(2\phi_B - V_{BS} + V_{DS})^{3/2} = (2\phi_B - V_{BS})^{3/2}\left(1 + \frac{V_{DS}}{2\phi_B - V_{BS}}\right)^{3/2} \approx (2\phi_B - V_{BS})^{3/2}\left(1 + \frac{3}{2}\frac{V_{DS}}{2\phi_B - V_{BS}}\right)$$

$$= (2\phi_B - V_{BS})^{3/2} + \frac{3}{2}V_{DS}(2\phi_B - V_{BS})^{1/2} \tag{10.56}$$

Substituting Eq. (10.56) in Eq. (10.51), we have

$$\begin{aligned} I_D &= \frac{Z\mu_n C_{ox}}{L}\left[\left(V_{GS} - V_{FB} - 2\phi_B - \frac{V_{DS}}{2}\right)V_{DS} - \frac{2}{3}\cdot\gamma\cdot\frac{3}{2}\cdot V_{DS}(2\phi_B - V_{BS})^{1/2}\right] \\ &= \frac{Z\mu_n C_{ox}}{L}\left[\left(V_{GS} - V_{FB} - 2\phi_B - \frac{V_{DS}}{2}\right) - \gamma(2\phi_B - V_{BS})^{1/2}\right]V_{DS} \end{aligned} \tag{10.57}$$

Substituting the relation for V_{th} given by Eq. (10.39), we now get a simple relation for the drain current given by

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$$I_D = \frac{Z\mu_n C_{ox}}{L}\left[V_{GS} - V_{th} - \frac{V_{DS}}{2}\right]V_{DS} \tag{10.58}$$

This model is available as Level 1 in SPICE. Now, using the same approximation, that is, $2\phi_B - V_{BS} >> V_{DS}$ in Eq. (10.52), we get an expression for the saturation drain voltage as

$$V_{Dsat} = V_{GB} - V_{FB} - 2\phi_B + V_{BS} - \gamma\sqrt{2\phi_B - V_{BS}} \tag{10.59}$$

Again using the expression for threshold voltage given by Eq. (10.39), we get a very simple relation for V_{Dsat} given by

$$V_{Dsat} = V_{GS} - V_{th} \tag{10.60}$$

Substituting the saturation drain voltage given by Eq. (10.60) in Eq. (10.58), we get an expression for the drain current when the MOSFET goes to saturation, I_{Dsat}, as

$$I_{Dsat} = \frac{Z\mu_n C_{ox}}{2L}\left[V_{GS} - V_{th}\right]^2 \tag{10.61}$$

The *square law model* gets its name from this square law type dependence of the saturation drain current on the gate voltage.

Physically, the square law model can also be derived by assuming that all along the channel the depletion width is constant, and is equal to the actual depletion width at the source end of the channel. Therefore, the bulk charge density Q_B does not depend on the position along the channel and is given by substituting $V_{CB}(y) = -V_{BS}$ in Eq. (10.43). Hence,

$$Q_B = -\sqrt{2qN_A\varepsilon_S(2\phi_B - V_{BS})} \tag{10.62}$$

The electron charge density $Q_n(y)$ is now obtained following the same procedure as in the bulk charge model. Therefore,

$$Q_n(y) = Q_i(y) - Q_B(y)$$

$$= -C_{ox}\left[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y) - \gamma\sqrt{2\phi_B - V_{BS}}\right] \tag{10.63}$$

Using the threshold voltage expression given by Eq. (10.39), we can simplify Eq. (10.63) to

$$Q_n(y) = -C_{ox}\left[V_{GB} - V_{th} - V_{CB}(y)\right] \tag{10.64}$$

Now substituting Eq. (10.64) in Eq. (10.49), we get an expression for the drain current as

$$I_D = \frac{Z\mu_n C_{ox}}{L}\int_{-V_{as}}^{V_{ds}-V_{as}}\left[V_{GB} - V_{th} - V_{CB}(y)\right]dV_{CB}(y) \tag{10.65}$$

The integral gives

$$I_D = \frac{Z\mu_n C_{ox}}{L}\left[V_{GS} - V_{th} - \frac{V_{DS}}{2}\right]V_{DS} \tag{10.66}$$

which is exactly the same as Eq. (10.58).

The square law model consistently overestimates I_D. The depletion width is more at the drain end. The assumption that the depletion width throughout the channel is the same as that at the source end results in an underestimation of $|Q_B|$. Since $|Q_n| = |Q_i| - |Q_B|$, this means that $|Q_n|$ and consequently I_D is overestimated. The overestimation is quite large and may be as high as 20–30%. Only at very low drain voltages, the model is quite accurate as the assumption of a constant depletion width may be reasonably valid.

MOSFET Level 3 model in SPICE

The bulk charge model, although quite accurate, is not really suited for quick calculation of results. On the other hand, the square law model relations are quite simple, but not very accurate. The SPICE Level 3 model was developed to fill the need for a model which is simple, and at the same time quite accurate.

To derive the Level 3 model, we introduce a variable $V_{CS}(y)$, which is the channel-to-source potential. This is related to the channel-to-bulk potential through the relation $V_{CS}(y) = V_{CB}(y) + V_{BS}$. Hence, the relation for the bulk charge density given by Eq. (10.43) can now be expressed in terms of $V_{CS}(y)$ as

$$Q_B(y) = -\sqrt{2qN_A\varepsilon_S(2\phi_B + V_{CS}(y) - V_{BS})} = -\gamma C_{ox}\sqrt{2\phi_B + V_{CS}(y) - V_{BS}} \tag{10.67}$$

In this model, we make the approximation $2\phi_B - V_{BS} >> V_{CS}(y)$. Using this approximation, Eq. (10.67) can be modified as

$$Q_B(y) = -\gamma C_{ox}\sqrt{2\phi_B - V_{BS}}\cdot\sqrt{1 + \frac{V_{CS}(y)}{2\phi_B - V_{BS}}}$$

$$\approx -\gamma C_{ox}\left[\sqrt{2\phi_B - V_{BS}} + \frac{1}{2}\frac{V_{CS}(y)}{\sqrt{2\phi_B - V_{BS}}}\right] \tag{10.68}$$

The electron charge density $Q_n(y)$ is now obtained as the difference of the total charge and the bulk charge. Therefore from Eqs. (10.42) and (10.68), we have

$$Q_n(y) = Q_s(y) - Q_B(y) = -C_{ox}[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y)] + \gamma C_{ox}\left[\sqrt{2\phi_B - V_{BS}} + \frac{1}{2}\frac{V_{CS}(y)}{\sqrt{2\phi_B - V_{BS}}}\right]$$

$$= -C_{ox}\left[V_{GB} - V_{FB} - 2\phi_B - V_{CB}(y) - \gamma\sqrt{2\phi_B - V_{BS}} - \frac{1}{2}\frac{\gamma V_{CS}(y)}{\sqrt{2\phi_B - V_{BS}}}\right] \tag{10.69}$$

Considering the threshold voltage expression given by Eq. (10.39), the above expression can be simplified as

$$Q_n(y) = -C_{ox}\left[V_{GS} - V_{th} - \alpha V_{CS}(y)\right] \tag{10.70}$$

where

$$\alpha = 1 + \frac{1}{2}\frac{\gamma}{\sqrt{2\phi_B - V_{BS}}} \tag{10.71}$$

Now substituting Eq. (10.70) in Eq. (10.49), we get an expression for the drain current as

$$I_D = \frac{Z\mu_n C_{ox}}{L}\int_{-V_{as}}^{V_{ds}-V_{as}}\left[V_{GS} - V_{th} - \alpha V_{CS}(y)\right]dV_{CB}(y) \tag{10.72}$$

Noting that $V_{CS}(y) = V_{CB}(y) + V_{BS}$ and V_{BS} is a constant, we can rewrite Eq. (10.72) as

$$I_D = \frac{Z\mu_n C_{ox}}{L}\int_0^{V_{ds}}\left[V_{GS} - V_{th} - \alpha V_{CS}(y)\right]dV_{CS}(y) \tag{10.73}$$

The drain current in the linear region can therefore be obtained from Eq. (10.73) as

$$I_D = \frac{Z\mu_n C_{ox}}{L}\left[V_{GS} - V_{th} - \alpha\frac{V_{DS}}{2}\right]V_{DS} \tag{10.74}$$

The drain voltage at which pinch-off occurs at the drain end of the channel, that is, V_{Dsat} can be obtained as in earlier models by putting $Q_n = 0$ in Eq. (10.70). Since $V_{CS}(L) = V_{Dsat}$, we have

$$V_{Dsat} = \frac{V_{GS} - V_{th}}{\alpha} \tag{10.75}$$

Substituting the expression for V_{Dsat} from Eq. (10.75) in Eq. (10.74), we get the expression for drain current at saturation as

$$I_{Dsat} = \frac{Z\mu_n C_{ox}}{2\alpha L}\left[V_{GS} - V_{th}\right]^2 \tag{10.76}$$

From Eq. (10.71), we see that the value of the constant α is always greater than unity. Comparing the expressions for V_{Dsat} and I_{Dsat} obtained from the Level 1 and Level 3 models in SPICE, it becomes quite clear that both V_{Dsat} and I_{Dsat} have lower value for the Level 3 model. This is expected as the approximation $2\phi_B - V_{BS} >> V_{CS}(y)$ may be valid for a large portion of the channel and is therefore more accurate than the Level 1 model which assumes $2\phi_B - V_{BS} >> V_{DS}$. The Level 3 model is therefore a far better approximation of the Level 2 model than the Level 1 model.

Example 10.9

The MOSFET MF1 is operated with the substrate shorted to the source, that is, $V_{BS} = 0$ V. The channel length and width of this device is given by $L = 5$ μm and $W = 20$ μm. Calculate V_{Dsat} and I_{Dsat} obtained in SPICE Level 1, Level 2, and Level 3 models for $V_{GS} = 5$ V. Assume $\mu_n = 800$ cm²/Vs.

Solution: For MF1,

$V_{th} = 0.84$ V, $\gamma = 1.683$, and $\phi_B = 0.35$ V (Example 10.8); $C_{ox} = 3.45 \times 10^{-8}$ F/cm² and $V_{FB} = -1.27$ V (Example 10.5)

Level 2:

Using Eq. (10.55),

$$V_{Dsat} = 5 + 1.27 - 2 \times 0.35 + \frac{(1.683)^2}{2}\left[1 - \sqrt{1 + \frac{4(5 + 1.27)}{(1.683)^2}}\right] = 2.54 \text{ V}$$

Substituting $V_D = V_{Dsat} = 2.54$ V in Eq. (10.51), we have

$$I_{Dsat} = \frac{20 \times 10^{-4} \times 800 \times 3.45 \times 10^{-8}}{5 \times 10^{-4}}\left[\left(5 + 1.27 - 0.7 - \frac{2.54}{2}\right)2.54 - \frac{2}{3}(1.683)\left\{(0.7 + 2.54)^{3/2} - (0.7)^{3/2}\right\}\right]$$

$$= 1.104 \times 10^{-4}\left[4.3 \times 2.54 - \frac{2}{3}(1.683)\{5.832 - 0.586\}\right] = 0.556 \text{ mA}$$

Level 1:

Using Eq. (10.60),

$$V_{Dsat} = 5 - 0.84 = 4.16 \text{ V}$$

Using Eq. (10.61),

$$I_{Dsat} = \frac{20 \times 10^{-4} \times 800 \times 3.45 \times 10^{-8}}{2 \times 5 \times 10^{-4}}[5 - 0.84]^2 = 0.955 \text{ mA}$$

Level 3:

Using Eq. (10.71),

$$\alpha = 1 + \frac{1}{2}\frac{(1.683)}{\sqrt{0.7}} = 2.01$$

Using Eq. (10.75),

$$V_{Dsat} = \frac{5 - 0.84}{2.01} = 2.08 \text{ V}$$

Using Eq. (10.76),

$$I_{Dsat} = \frac{20 \times 10^{-4} \times 800 \times 3.45 \times 10^{-8}}{2 \times 5 \times 10^{-4} \times 2.01}[5 - 0.84]^2 = 0.478 \text{ mA}$$

Comparison of the MOSFET models

The Level 1 model uses a lot of approximations and is the simplest of all the MOSFET models. It can only be used for a rough approximation of the circuit performance. On the other hand, the Level 2 model takes much more computation time and may even result in convergence problems. Level 3 model is a good compromise and can be used in most cases, except when highly precise results are required.

10.3.3 Process Flow for a Self-aligned nMOSFET

For the fabrication of an n-channel MOSFET, the starting material is a p-type silicon wafer of (100) orientation. In this section, we shall discuss a self-aligned, polysilicon-gate, oxide-isolated process flow, as shown in Fig. 10.25.

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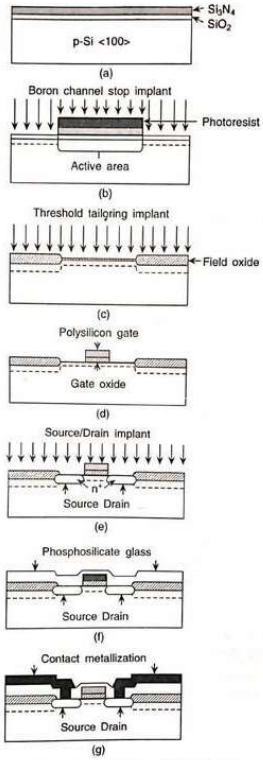


Figure 10.25 Process steps for fabricating a self-aligned MOSFET for integrated circuits.

Step 1: The first step is *isolation*. Oxide isolation is standard in MOSFET technology. In this case, a recessed field oxide is used to isolate adjacent transistors resulting in better planarity. First a thin initial (pad) oxide is grown on the surface, followed by chemical vapour deposition of 0.1 μm thick silicon nitride as shown in Fig. 10.25(a).

Step 2: Nitride deposition is followed by lithography, in which photoresist is used to protect all the areas where active transistors are fabricated. Nitride is etched away from the remaining regions but the photoresist is not removed [Fig. 10.25(b)]. Using this photoresist as a mask, boron *channel stop implant* is carried out. Thus, this implant is automatically aligned to the nitride free area. This prevents inadvertent and unwanted formation of a conducting channel under the field oxide. Photoresist is now removed.

Step 3: Field oxidation is now carried out to realize an oxide thickness of about 1 μm. Since silicon is consumed during oxidation, only about half of this thickness (that is, 0.5 μm) protrudes above the surface while the rest lies below the original surface as shown in Fig. 10.25(c). This results in better planarity of the fabricated device. Oxidation is blocked in the regions protected by the nitride, as the oxidizing species cannot reach the silicon surface in those places. This process is therefore also referred to as *isolation by local oxidation*. Silicon nitride is now removed from the active area using chemical agents such as phosphoric acid. Boron threshold tailoring implant is carried out to adjust the threshold voltage of the MOSFET (this will be discussed thoroughly in Chapter 11).

Step 4: Gate oxidation is now carried out to realize a very thin, high quality, defect-free oxide. This is the most critical step in the MOSFET processing and extreme care must be taken to ensure the high quality of the gate oxide. Polysilicon is now deposited over the entire surface by chemical vapour deposition. Using lithography again, polysilicon is now patterned by wet or dry etching to define the gate [Fig. 10.25(d)]. This step determines the channel length, which is one of the most important parameters in MOSFET. Dimensional control of gate lithography and poly-etching is therefore another key step in the process flow.

Step 5: After the gate patterning, ion-implantation is carried out for source and drain. No lithography is required to undertake this step and hence the process is termed as *self-aligned*. The field oxide and the gate act as masks against implantation and consequently the n^+ source and drain are automatically aligned to the gate [Fig. 10.25(e)].

Step 6: The next step is chemical vapour deposition of phosphosilicate glass [Fig. 10.25(f)]. This is used as a barrier against alkali ions (which may cause instabilities in device performance). This glass is heated to around 1000°C to flow and smoothen the contours of the device. This results in better metal step coverage.

Step 7: Contact windows are now opened in the glass by lithography technique followed by metal deposition and patterning [Fig. 10.25(g)].

Step 8: Finally, a passivation layer (usually silicon nitride) is deposited on the finished device. Again lithography is carried out and only the bondpads are exposed in the process.

PROBLEMS

P10.1 A MOS capacitor is fabricated on p-silicon substrate with a doping concentration of 10^{16} cm^{-3}. If the gate oxide thickness is 500 Å, and n^+ polysilicon is deposited as gate metal, find

(a) The maximum capacitance per unit area.
(b) The minimum capacitance per unit area.
(c) The ideal threshold voltage assuming $V_{FB} = 0$ V.
(d) The threshold voltage taking the work function into account.
(e) If the experimental C–V characteristics of this capacitor show a $V_{FB} = -1$ V, what is the amount of fixed oxide charge?

P10.2 If donor-like interface states are present, how do they affect the high frequency C–V plot of a MOS capacitor on n-Si substrate? Explain with diagrams.

P10.3 (a) An n^+ polysilicon gate MOS capacitor was to be fabricated so that it had a threshold voltage $V_{th} = 1$ V at 300 K. The available silicon wafers were uniformly doped with a concentration of $N_A = 2 \times 10^{16}$ cm^{-3}. It was assumed that the oxide and interface charges would be so small that they would not affect V_{th}. What is the required gate oxide thickness?

(b) After fabrication of the MOS capacitor, V_{th} was measured to be 0.65 V. The capacitor was then subjected to bias heat stress. First, it was heated to 250°C with +30 V applied to the gate, so that all the mobile ions (which are positively charged) moved to the Si–SiO_2 interface. After cooling, V_{th} was measured to be 0.5 V. Again, the capacitor was heated to 250°C, this time with −30 V applied to the gate, so that the ions were attracted to the polysilicon–SiO_2 interface. Now, after cooling, V_{th} was measured to be 0.7 V. Estimate the values of Q_f/q and Q_m/q. Where were all the mobile ions located initially, assuming they were all concentrated in a plane parallel to the interfaces. Assume Q_{it} and Q_{ot} to be negligible.

P10.4 (a) A MOS diode is fabricated on a p-type silicon substrate having a doping concentration of $N_A = 5 \times 10^{15}$ cm^{-3}. The gate oxide thickness t_{ox} is 200 Å. If n^+ polysilicon is used as the gate material for this device and it has a fixed oxide charge density given by $Q_f/q = 1.0 \times 10^{11}$ cm^{-2}, calculate

(i) the maximum depletion width (W_{max}).
(ii) the flat band voltage (V_{FB}).
(iii) the threshold voltage (V_{th}) for this device.

(b) A MOSFET is fabricated on the same substrate with the same gate oxide thickness and fixed oxide charge as in (a) above. This MOSFET also has n^+ polysilicon gate and is operated with $V_{BS} = -2$ V. The channel length and width of this device is given by $L = 1$ μm and $W = 5$ μm. Calculate V_{Dsat} and I_{Dsat} using SPICE Level 3 model for $V_{GS} = 5$ V.

P10.5 Matched p- and n-channel MOSFETs are to be fabricated in the same silicon wafer (that is, $V_{thp} = -V_{thn}$, for $V_{BS} = 0$). To do this, an n-silicon sample with doping concentration $N_D = 10^{15}$ cm^{-3} is taken and a p-well is diffused in it. Identical p- and n-channel MOSFETs are now fabricated in the n- and p-regions respectively of this wafer. The gate oxide thickness is 1000 Å, $Q_f/q = 3 \times 10^{11}$ cm^{-2} and aluminium gate is used. What is the doping concentration of the p-well to obtain matched devices? What are the values of V_{th} for the FETs, if $V_{BS} = 0$?

P10.6 An n-channel MOSFET has the following specifications: channel width $Z = 5$ μm, channel length $L = 1$ μm, gate oxide thickness $t_{ox} = 500$ Å, electron mobility = 800 cm^2/V$_s$, threshold voltage $V_{th} = 1$ V.
Find the drain current flowing through the device for the following conditions:

(i) $V_{GS} = 2$ V, $V_{DS} = 0.5$ V

(ii) $V_{GS} = 2$ V = V_{DS}. Use (a) SPICE level 1 and (b) level 3 model.

P10.7 (a) A n^+ poly-gate n-channel MOSFET has a substrate doping concentration of 10^{16} cm^{-3} and a gate oxide thickness of 1000 Å. What will be the change in its threshold voltage if the source-to-substrate voltage is increased to 2 V from zero?

(b) The above devices are isolated by field oxide in an integrated circuit. For safe operation up to 10 V, what should be the minimum thickness of the field oxide so that the FETs remain isolated?

REFERENCES AND SUGGESTED FURTHER READING

[1] Deal, B.E., Standardized terminology for oxide charges associated with thermally oxidized silicon, *IEEE Trans. Electron Devices*, Vol. ED-27, p. 606, 1980.

[2] Deal, B., The current understanding of charges in the thermally oxidized silicon structure, *Journal of Electrochemical Society*, Vol. 121, no. 6, pp. 198C–205C, June 1974.

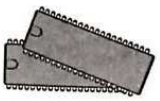
[3] Grove, A.S., *Physics and Technology of Semiconductor Devices*, Wiley, New York, 1967.

[4] Sze, S.M., *Physics of Semiconductor Devices*, 2nd ed., Wiley, N.Y., 1981.

[5] Streetman, B.G. and S. Banerjee, *Solid State Electronic Devices*, 5th ed., Prentice Hall, New Jersey, 2000.

[6] Ong, D.G., *Modern MOS Technology: Processes, Devices and Design*, McGraw Hill Book Company, Singapore, 1984.

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11

Advanced Topics in MOSFETs

11.1 INTRODUCTION

In Chapter 10, we have discussed the basic operation of the MOSFETs and various modelling techniques based on certain assumptions. However, as the device dimensions become smaller, many of these assumptions no longer remain valid. In order to obtain more accurate results, especially for the smaller devices of today, it is necessary to consider some second order effects. These form the subject matter of discussion of the present chapter.

11.2 EFFECT OF GATE AND DRAIN VOLTAGES ON CARRIER MOBILITY IN THE INVERSION LAYER

In all the models discussed in Chapter 10, we have assumed that the mobility in the inversion layer is a constant and is not influenced by either the gate or drain voltage. However, this is strictly not true, and appropriate corrections must be incorporated in the models for greater accuracy. As we have already discussed, the conduction in a MOSFET takes place in a very thin inversion layer adjacent to the Si–SiO_2 interface. The mobility of the carriers depends on the surface roughness, which introduces additional scattering, as well as the presence of interface states which can trap and retard their flow. The carrier mobility in a MOSFET channel is therefore strongly dependent on processing techniques. The mobility for electrons in the inversion layer of an n-channel MOSFET can vary in the range 400–900 cm^2/Vs while for holes in a p-channel MOSFET, the variation can be 100–300 cm^2/Vs. As we can see, these values of mobility on the silicon surface are much lower than the values obtained in bulk silicon.

11.2.1 Effect of Gate Voltage on Carrier Mobility

The surface mobility can be extracted by measuring the drain current of a MOSFET while applying a very small fixed V_{DS} (about 10 mV) and sweeping V_{GS} from close to V_{th} to a large value. Assuming that the square law model is valid, which is in fact a good approximation for small drain

voltages, and taking $V_{DS} \ll 2(V_{GS} - V_{th})$ in Eq. (10.58), we have

$$I_D = \frac{Z\mu_n C_{ox}}{L}(V_{GS} - V_{th})V_{DS} \tag{11.1}$$

Equation (11.1) implies that a plot of I_D versus V_{GS} from our measurements will result in a straight line which intersects the x-axis at $V_{GS} = V_{th}$ and has a slope of $(Z\mu_n C_{ox} V_{DS})/L$. Such an experimental curve is shown in Fig. 11.1. Now if we know the values of the channel length, channel width, and the gate oxide thickness, we can easily obtain the value of μ_n from the slope of the straight line.

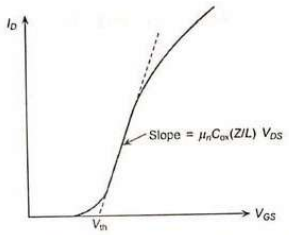


Figure 11.1 Typical variation of I_D with V_{GS} keeping V_{DS} constant.

An interesting observation in Fig. 11.1 is that the straight line deviates from linearity at higher gate voltages. The reduced slope implies a reduction in mobility at higher gate voltages. This decrease in mobility is due to the higher transverse electric field, which is in a direction perpendicular to the actual flow of electrons, resulting in more collisions of the electrons with the surface. In fact, the variation in mobility due to the transverse field ($\mathscr{E}_x$) can be expressed as

$$\mu = \frac{\mu_0}{1 + \alpha_\theta \mathscr{E}_x} \tag{11.2}$$

where

μ_0 = mobility at low gate voltages where the effect of the transverse electric field is negligible and

α_θ = scattering constant.

Now considering $\mathscr{E}_x$ to be proportional to $(V_{GS} - V_{th})$, we may write

$$\mu = \frac{\mu_0}{1 + \theta(V_{GS} - V_{th})} \tag{11.3}$$

where θ is a parameter which can be extracted experimentally by fitting the curve in Fig. 11.1. By using Eq. (11.3) for mobility rather than a fixed value, the overestimation of current at high gate voltages in models discussed earlier can be corrected.

11.2.2 Effect of Drain Voltage on Carrier Mobility

We have already seen in Chapter 1 that the relationship between drift velocity v_d of carriers and the applied electric field $\mathscr{E}$ is linear for low electric fields ($< 3 \times 10^3$ V/cm for electrons in Si), so that the mobility ($\mu = v_d/\mathscr{E}$) is a constant. However, at higher electric fields, the v_d versus $\mathscr{E}$ curve increases sublinearly, finally reaching its saturation value of v_{sat} at very high electric fields ($\sim 5 \times 10^4$ V/cm for electrons in Si). Thus, there is a continuous reduction in mobility as the electric field exceeds a particular value. As the electric field in the channel depends on the drain voltage, the mobility becomes a function of the drain voltage.

Although the electric field at which velocity saturation takes place appears quite large, such fields are easily achieved in modern MOSFETs. Considering $V_{DS} = 2.5$ V, the average electric field in the channel for a MOSFET having $L = 0.5$ μm is 5×10^4 V/cm. We already know that the electric field at the drain end of the channel is much higher than the average value. Thus, the electrons in a large portion of the channel can be travelling at the saturation velocity. In fact, with increasing drain voltage the electron velocity at the drain end may reach v_{sat} even before pinch-off occurs. As the velocity of carriers cannot increase any more, further increase in V_{DS} does not increase the drain current. The saturation voltage (V_{Dsat}) is then decided by which of the two phenomena, namely pinch-off and velocity saturation, occurs at a lower drain voltage.

Now assuming that the current saturation in the MOSFET is due to velocity saturation of carriers and not due to pinch-off, we have $I_{Dsat} = -Zv_{sat}Q_{sat}$, where Q_{sat} is the charge concentration at the drain end of the channel at the point of saturation, that is, when $V_D = V_{Dsat}$, and can be expressed by substituting $V_{CS}(y) = V_{Dsat}$ in Eq. (10.70). Therefore we have,

$$I_{Dsat} = Zv_{sat}\, C_{ox}\left[V_{GS} - V_{th} - \alpha V_{Dsat}\right] \tag{11.4}$$

An expression for I_{Dsat} can also be obtained by substituting $V_{DS} = V_{Dsat}$ in Eqn. (10.74), which is the expression for I_D in the level 3 SPICE model. Equating these two relations for I_{Dsat}, we obtain

$$\frac{Z\mu_n C_{ox}}{L}\left[V_{GS} - V_{th} - \alpha\frac{V_{Dsat}}{2}\right]V_{Dsat} = Zv_{sat}C_{ox}\left[V_{GS} - V_{th} - \alpha V_{Dsat}\right] \tag{11.5a}$$

which on simplification gives

$$\frac{\alpha V_{Dsat}^2}{2} - \left(V_{GS} - V_{th} + \frac{\alpha L v_{sat}}{\mu_n}\right)V_{Dsat} + (V_{GS} - V_{th})\frac{Lv_{sat}}{\mu_n} = 0 \tag{11.5b}$$

Solving the quadratic equation [Eq. (11.5b)], we have

$$V_{Dsat} = \left(\frac{V_{GS} - V_{th}}{\alpha}\right) + \frac{Lv_{sat}}{\mu_n} - \sqrt{\left(\frac{V_{GS} - V_{th}}{\alpha}\right)^2 + \left(\frac{Lv_{sat}}{\mu_n}\right)^2} \tag{11.6}$$

It can be easily shown that V_{Dsat} obtained from this equation is always less than the pinch-off value given by Eq. (10.75), which is $[V_{Dsat} = (V_{GS} - V_{th})/\alpha]$, and approaches this value when L becomes very large. This is to be expected, as higher V_{DS} is required to achieve the necessary electric field for velocity saturation when the channel length is increased. In level 3 SPICE model, Eq. (11.6) is used to calculate V_{Dsat} if v_{sat} is supplied as an input parameter. Else, Eq. (10.75) is used. The

saturation drain current I_{Dsat} can now be obtained by substituting the value of V_{Dsat} obtained from Eq. (11.6) in Eq. (11.4). It is quite obvious from Eq. (11.4) that lower V_{Dsat} results in a higher I_{Dsat}. Thus, reducing L results in lower V_{Dsat} and higher I_{Dsat}.

This level 3 model discussed here considers constant electron mobility before the onset of velocity saturation. Let us now see how this model can be modified considering field dependent mobility instead of constant mobility. We shall assume that the drift velocity v_d of electrons in the inversion layer of an n-channel MOSFET is related to the magnitude of the electric field $\mathscr{E}_y$ by

$$v_d = \frac{\mu_n \mathscr{E}_y}{1 + \dfrac{\mathscr{E}_y}{\mathscr{E}_c}} \tag{11.7}$$

where

μ_n = low field electron mobility and

$\mathscr{E}_c = v_{sat}/\mu_n$ = critical electric field

As we see from Eq. (11.7), when $\mathscr{E}_y \ll \mathscr{E}_c$, $v_d = \mu_n \mathscr{E}_y$ while $v_d = v_{sat}$ for $\mathscr{E}_y \gg \mathscr{E}_c$. Although Eq. (11.7) is a very simplified approximation of the actual drift velocity variation, it is very useful in developing analytical models.

Taking $I_n(y) = ZQ_n(y)v_d(y) = -I_D$ and substituting the expressions for $Q_n(y)$ and $v_d(y)$ from Eqs. (10.70) and (11.7), we have

$$I_D = ZC_{ox}\left[V_{GS} - V_{th} - \alpha V_{CS}(y)\right]\frac{\mu_n \mathscr{E}_y}{1 + \dfrac{\mathscr{E}_y}{\mathscr{E}_c}} \tag{11.8}$$

Since $\mathscr{E}_y = \dfrac{dV_{CS}}{dy}$, from Eq. (11.8), we have

$$I_D \int_0^L \left(1 + \frac{1}{\mathscr{E}_c}\frac{dV_{CS}}{dy}\right)dy = ZC_{ox}\mu_n \int_0^L \left[V_{GS} - V_{th} - \alpha V_{CS}(y)\right]\frac{dV_{CS}}{dy}\,dy$$

Noting that $V_{CS} = 0$ at $y = 0$ and $V_{CS} = V_{DS}$ at $y = L$, after integrating this equation, we have

$$I_D = \frac{Z\mu_n C_{ox}}{L\left(1 + \dfrac{V_{DS}}{L\mathscr{E}_c}\right)}\left[V_{GS} - V_{th} - \alpha\frac{V_{DS}}{2}\right]V_{DS} \tag{11.9}$$

An expression for I_{Dsat} can be obtained by substituting $V_{DS} = V_{Dsat}$ in Eq. (11.9). Following the same procedure used earlier in this section to develop an expression for V_{Dsat}, equating Eq. (11.9) with that for I_{Dsat} with Eq. (11.4), which is obtained assuming velocity saturation, we have

$$\frac{Z\mu_n C_{ox}}{L\left(1 + \dfrac{V_{Dsat}}{L\mathscr{E}_c}\right)}\left[V_{GS} - V_{th} - \alpha\frac{V_{Dsat}}{2}\right]V_{Dsat} = Zv_{sat}\,C_{ox}\left[V_{GS} - V_{th} - \alpha V_{Dsat}\right] \tag{11.10a}$$

On simplifying, this gives

$$\frac{\alpha V_{Dsat}^2}{2} + \alpha L\mathscr{E}_c V_{Dsat} - (V_{GS} - V_{th})\,L\mathscr{E}_c = 0 \tag{11.10b}$$

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Solving the above quadratic equation, we have

$$V_{Dsat} = L\mathscr{E}_c\left[\sqrt{1+\frac{2(V_{GS}-V_{th})}{\alpha L\mathscr{E}_c}}-1\right] = \frac{2(V_{GS}-V_{th})}{\alpha\left[\sqrt{1+\frac{2(V_{GS}-V_{th})}{\alpha L\mathscr{E}_c}}+1\right]} \tag{11.11}$$

Equation (11.11) shows that when $L >> [2(V_{GS} - V_{th})/\alpha\mathscr{E}_c]$, that is, for MOSFETs having long channel lengths, V_{Dsat} reduces to Eq. (10.75). Equation (10.75) was derived assuming current saturation being due to pinch-off. However, at shorter channel lengths, the electric fields in the channel increase, resulting in reduced mobilities and eventually velocity saturation. This further causes current saturation at lower drain voltages. Equation (11.11) is used to calculate V_{Dsat} in MOSFET level 4 model in SPICE, which is popularly known as BSIM.

Example 11.1

Calculate the values of V_{Dsat} and I_{Dsat} when $V_{GS} = 2$ V for an n-channel MOSFET with $V_{th} = 1.0$ V, using (a) level 3 model taking velocity saturation into account and (b) level 4 model, which assumes field dependent mobility. Assume $L = 1$ μm, $Z = 5$ μm, $t_{ox} = 20$ nm, $\mu_n = 500$ cm^2/Vs, $\alpha = 1.05$ and $\mathscr{E}_c = 5 \times 10^4$ V/cm.

Solution: (a) From the given values of $\mathscr{E}_c$ and μ_n, we get $v_{sat} = 500 \times 5 \times 10^4$ cm/s $= 2.5 \times 10^7$ cm/s.

Substituting this value of v_{sat} in Eq. (11.6), we have

$$V_{Dsat} = \left(\frac{2-1}{1.05}\right)+\frac{1\times10^{-4}\times2.5\times10^7}{500}-\sqrt{\left(\frac{2-1}{1.05}\right)^2+\left(\frac{1\times10^{-4}\times2.5\times10^7}{500}\right)^2} = 0.858 \text{ V}$$

Now, C_{ox} can be calculated as $C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{3.9\times8.85\times10^{-14}}{2\times10^{-6}} = 1.72\times10^{-7}$ F/cm^2

Substituting these values of V_{Dsat} and C_{ox} in Eq. (11.4), we get

$$I_{Dsat} = 5\times10^{-4}\times2.5\times10^7\times1.72\times10^{-7}\,[2-1-(1.05\times0.858)] = 2.13\times10^{-4} \text{ A}$$

(b) Using Eq. (11.11), we get

$$V_{Dsat} = 1\times10^{-4}\times5\times10^4\left[\sqrt{1+\frac{2(2-1)}{1.05\times1\times10^{-4}\times5\times10^4}}-1\right] = 0.876 \text{ V}$$

Substituting this value of V_{Dsat} in Eq. (11.4), we get

$$I_{Dsat} = 5\times10^{-4}\times2.5\times10^7\times1.72\times10^{-7}\,[2-1-(1.05\times0.876)] = 1.72\times10^{-4} \text{ A}.$$

A comparison of the V_{Dsat} values obtained from Eqs. (11.6) and (11.11) in Example 11.1 shows that the V_{Dsat} value calculated in level 4 model is higher than that obtained in level 3. This is because in the level 3 model, $v_d = \mu_n\mathscr{E}_y$ is assumed, so that v_{sat} is reached when $\mathscr{E}_y = \mathscr{E}_c = v_{sat}/\mu_n$. On the other hand, in level 4, which uses Eq. (11.7) to calculate v_d, the drift velocity $v_d = (v_{sat}/2)$ when $\mathscr{E}_y = \mathscr{E}_c$, and v_d saturates at electric fields much greater than $\mathscr{E}_c$. Hence in this case, a larger drain voltage is required for velocity saturation. The saturation current I_{Dsat}, on the other hand, is lower in level 4 compared to level 3. This is clear from Eq. (11.4), where a higher value of V_{Dsat} results in lower I_{Dsat}. Since the level 4 model considers both velocity saturation and field dependent mobility, it is more accurate than the level 3 model.

11.3 CHANNEL LENGTH MODULATION

As we have already discussed in section 10.3, when the drain voltage exceeds V_{Dsat}, the pinch-off point (or the velocity saturation point) moves from the drain towards the source, and a depletion layer exists between this point and the drain. This movement is referred to as *channel length modulation*. If the current saturation is due to pinch-off, the same voltage V_{Dsat} is now dropped across a smaller effective channel length L', while the excess drain voltage $(V_{DS} - V_{Dsat})$ is dropped across a depletion layer of width $l_d = (L - L')$. If velocity saturation is the reason for current saturation, a slightly lower V_{Dsat} corresponding to the effective channel length L' is dropped in the region between source ($y = 0$) and the velocity saturation point ($y = L'$). In either case, the drain current increases with increase in drain voltage when $V_{DS} > V_{Dsat}$. The change in drain current is more severe for shorter channel length devices, as in this case the effective change in channel length ($l_d = L - L'$) is a larger fraction of the initial channel length. The drain current in the saturation region can therefore be obtained by substituting L' for L in I_{Dsat}. Since in most cases I_{Dsat} is inversely proportional to L, we have for $V_{DS} > V_{Dsat}$

$$I_D = \frac{I_{Dsat}}{\frac{L'}{L}} = \frac{I_{Dsat}}{\left(\frac{L-l_d}{L}\right)} = \frac{I_{Dsat}}{\left(1-\frac{l_d}{L}\right)} \tag{11.12}$$

where L' and l_d correspond to the applied drain voltage V_{DS}. Empirical equations have been used in level 1 and level 2 models in SPICE using a parameter λ, where λ is given by

$$\lambda = \frac{l_d}{LV_{DS}} \tag{11.13}$$

Substituting Eq. (11.13) in Eq. (11.12), we get the level 2 SPICE relation for I_D in the saturation region given by

$$I_D = \frac{I_{Dsat}}{1-\lambda V_{DS}} \tag{11.14}$$

In long channel devices, assuming $l_d << L$, we can approximate Eq. (11.12) as

$$I_D = \frac{I_{Dsat}}{\left(1-\frac{l_d}{L}\right)} \approx I_{Dsat}\left(1+\frac{l_d}{L}\right) \tag{11.15}$$

Now, substituting Eq. (11.13) in Eq. (11.15), we get the level 1 SPICE relation for I_D in the saturation region given by

$$I_D = I_{Dsat}(1+\lambda V_{DS}) \tag{11.16}$$

A more rigorous approach is used to calculate the drain current in saturation in the level 3 SPICE model. A typical plot for the surface electric field variation in a MOSFET operating in the saturation region is shown in Fig. 11.2. As we have already discussed in section 10.3.2, the electric field in the inversion layer of a MOSFET increases from the source to the drain end. Let the peak electrical field in the inversion layer be given by $\mathscr{E}_p$. From the solution of Poisson's equation, it can be shown that the electric field increases linearly in the depletion layer and is equal to $[\mathscr{E}_p + (qN_Al_d/\varepsilon_s)]$ at the drain end of the depletion layer.

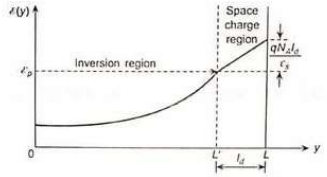


Figure 11.2 Typical plot for surface electric field variation in a MOSFET operating in the saturation region.

Now, since the drop in the depletion layer is equal to the product of the average electric field and the width of the depletion layer, we have

$$\frac{1}{2}\left(\mathscr{E}_p+\mathscr{E}_p+\frac{qN_Al_d}{\varepsilon_s}\right)l_d = V_{DS}-V_{Dsat} \tag{11.17}$$

Solving this quadratic equation, we get an expression for l_d as

$$l_d = \sqrt{\left(\frac{\mathscr{E}_p}{2K}\right)^2+\left(\frac{V_{DS}-V_{Dsat}}{K}\right)}-\frac{\mathscr{E}_p}{2K} \tag{11.18}$$

where

$$K = \frac{qN_A}{2\varepsilon_s}$$

The value of l_d obtained from Eq. (11.18) can now be substituted in Eq. (11.12) to get the drain current corresponding to V_{DS}.

However, in order to calculate l_d from Eq. (11.18), we need to know the value of $\mathscr{E}_p$, which is not easy to evaluate. Several models have been proposed for the evaluation of $\mathscr{E}_p$. In SPICE level 3 model, it is evaluated in the following manner. We can express the drain conductance at the onset of saturation (G_{Dsat}) as

$$G_{Dsat} = \left.\frac{\partial I_D}{\partial V_{DS}}\right|_{V_{DS}=V_{Dsat}} = \frac{\partial I_D}{\partial L}\cdot\frac{\partial L}{\partial V_{DS}} \tag{11.19}$$

Since I_D is inversely proportional to L, at $V_{DS} = V_{Dsat}$, we have

$$\frac{\partial I_D}{\partial L} = -\frac{I_{Dsat}}{L}$$

Assuming an almost constant electric field of $\mathscr{E}_p$ in the depletion layer, we have $[(\partial L/\partial V_{DS}) = -(1/\mathscr{E}_p)]$. Substituting these values in Eq. (11.19), we have

$$G_{Dsat} = \frac{I_{Dsat}}{\mathscr{E}_pL} \quad \text{or} \quad \mathscr{E}_p = \frac{I_{Dsat}}{G_{Dsat}L} \tag{11.20}$$

The drain current in the saturation region can now be obtained using Eqs. (11.20), (11.18), and (11.12).

11.4 MOSFET BREAKDOWN AND PUNCH-THROUGH

As shown in Fig. 11.3, breakdown occurs in a MOSFET when the drain voltage exceeds a certain value. As the drain voltage increases, the peak electric field at the drain end of the channel also increases. When the peak electric field approaches middle of the 10^5 V/cm range, impact ionization takes place at the drain junction leading to a sharp increase in drain current. The high energy electrons, which have gained energy from the electric field, have sufficient kinetic energy to generate secondary electrons and holes by impact ionization. The generated electrons are collected by the drain and add to the drain current, while the holes flow to the substrate contact resulting in a substrate current. This is schematically shown in Fig. 11.4. The substrate current, which produces a voltage drop across the bulk spreading resistance, effectively forward biases the bulk-source junction. This introduces a positive feedback effect by reducing the threshold voltage, thereby increasing the drain current even further. A *lightly doped drain* (LDD) structure has often been used to reduce the peak electric field in a MOSFET. However, this degrades the device performance to some extent. The design of the MOSFET should ultimately take care that the onset of breakdown occurs at voltages much greater than the power supply voltage to be used in the circuit.

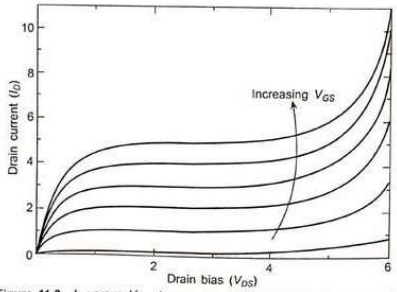


Figure 11.3 I_D versus V_{DS} characteristics of a MOSFET showing breakdown.

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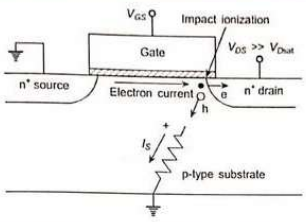


Figure 11.4 Schematic diagram showing the impact ionization at the drain end and consequent flow of secondary holes to the substrate.

There may be sharp increase in drain current at high drain voltages due to another phenomenon, called *punch-through*. This is similar to the base punch-through in BJTs, and occurs when the drain and the source depletion regions overlap. When this happens, the gate loses control and a large drain current can flow even when the gate voltage is below threshold. It is quite obvious that the punch-through voltage (V_{pt}), defined as the drain voltage at which punch-through occurs, is smaller for shorter channel devices.

11.5 SUBTHRESHOLD CURRENT

In all the drain current expressions derived in section 10.3.2 in Chapter 10, I_D becomes zero when $V_{GS} = V_{th}$. All these expressions have been derived assuming the diffusion current to be negligibly small compared to the drift current in the channel. However, when $V_{GS} \approx V_{th}$, the electron concentration in the channel is very small and this approximation no longer remains valid (Remember: The drift current is proportional to the carrier concentration). For $V_{GS} < V_{th}$, or when $(\phi_B - V_{BS}) < \psi_s < (2\phi_B - V_{BS})$, ψ_s being the amount of band bending at the semiconductor surface, the diffusion current is much larger than the drift component. It should, however, be stressed that in the transition from strong inversion to weak inversion, both drift and diffusion components are comparable. Figure 11.5 shows the nature of variation of the drift and diffusion components of drain current with gate voltage in a MOSFET. The drain current when $V_{GS} < V_{th}$ is referred to as *subthreshold* or *weak inversion* current. Considering only the diffusion component of current, the subthreshold current can be expressed as

$$I_D = AqD_n\left[\frac{n(0) - n(L)}{L}\right] \tag{11.21}$$

where

$n(0)$ and $n(L)$ = electron concentrations at the source and drain ends of the channel respectively,

A = cross-sectional area of current flow, and

the other symbols have their usual meanings.

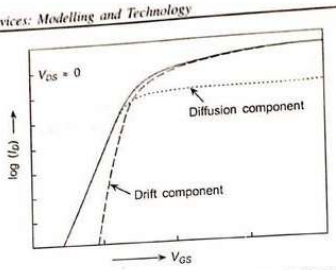


Figure 11.5 Variation of the drift and diffusion components of drain current with V_{GS}. The total drain current is also shown (solid line).

Let us assume that ψ_s is the surface potential corresponding to a particular gate-to-source voltage V_{GS}. In a long-channel MOSFET, in the subthreshold region, the potential across the depletion layer at the semiconductor surface (ψ_s) can be assumed to be almost a constant from the source to the drain end. Therefore, the voltages across the source-to-substrate and drain-to-substrate junctions are $(\psi_s + V_{BS})$ and $(\psi_s + V_{BS} - V_{DS})$ respectively. Using Eq. (4.43b) for p-n junctions, we now have

$$n(0) = \frac{n_i^2}{N_A}\exp\left(\frac{\psi_s + V_{BS}}{V_T}\right) = N_A\exp\left(\frac{\psi_s + V_{BS} - 2\phi_B}{V_T}\right) \tag{11.22}$$

and

$$n(L) = \frac{n_i^2}{N_A}\exp\left(\frac{\psi_s + V_{BS} - V_{DS}}{V_T}\right) = N_A\exp\left(\frac{\psi_s + V_{BS} - V_{DS} - 2\phi_B}{V_T}\right) \tag{11.23}$$

where $n(0)$ and $n(L)$ are electron concentrations at the source and drain ends of the channel respectively.

The cross-sectional area of current flow can be expressed as

$$A = Zx_{ch}$$

where

x_{ch} = thickness of the channel.

Since there is no sharp demarcation for the depth of the channel, a common approximation which is used for analytical modelling is to consider x_{ch} to be the depth where the potential is $(\psi_s - V_T)$. Since the electron concentration falls rapidly as we move below the surface, this is a reasonably good approximation. Since the voltage drop across the depth x_{ch} is V_T, the channel thickness x_{ch} can now be expressed as $x_{ch} = V_T/\mathscr{E}_s$, where $\mathscr{E}_s$ is the surface electric field. Since $\varepsilon_s\mathscr{E}_s = -Q_s$, and Q_s is expressed by Eq. (10.10) ($Q_s \approx Q_B$ in the subthreshold region), we have

$$x_{ch} = \frac{V_T}{\mathscr{E}_S} = -\frac{\varepsilon_s V_T}{Q_s} = \frac{\varepsilon_s V_T}{\sqrt{2\varepsilon_s qN_A\psi_s}} = \frac{\varepsilon_s V_T}{\gamma C_{ox}\sqrt{\psi_s}} \tag{11.24}$$

Now taking $A = Zx_{ch}$ and from Eqs. (11.21) to (11.24), we have

$$I_D = \frac{Z\mu_n qV_T^2\varepsilon_s N_A}{L\gamma C_{ox}\sqrt{\psi_s}}\exp\left(\frac{\psi_s + V_{BS} - 2\phi_B}{V_T}\right)\left[1 - \exp\left(-\frac{V_{DS}}{V_T}\right)\right]$$

$$= \frac{Z\mu_n V_T^2\gamma C_{ox}}{2L\sqrt{\psi_s}}\exp\left(\frac{\psi_s + V_{BS} - 2\phi_B}{V_T}\right)\left[1 - \exp\left(-\frac{V_{DS}}{V_T}\right)\right] \tag{11.25}$$

Equation (11.25) gives a relation for the subthreshold current in terms of the surface potential ψ_s. However, for circuit simulation, we prefer to have a relation in terms of the gate voltage V_{GS}. We will now proceed to develop such a relation. Using Eq. (10.20) for the subthreshold region, we have

$$V_{GB} - V_{FB} = \psi_s - \frac{Q_B}{C_{ox}} \tag{11.26}$$

Therefore,

$$\frac{\partial V_{GS}}{\partial\psi_s} = \frac{\partial V_{GB}}{\partial\psi_s} = 1 - \frac{1}{C_{ox}}\frac{\partial Q_B}{\partial\psi_s} = 1 + \frac{C_D}{C_{ox}} = m \tag{11.27}$$

where

C_D = depletion capacitance given by

$$C_D = -\frac{\partial Q_B}{\partial\psi_s} = \frac{\partial}{\partial\psi_s}\left[\sqrt{2\varepsilon_s qN_A\psi_s}\right] = \frac{\sqrt{2\varepsilon_s qN_A}}{2\sqrt{\psi_s}} = \frac{\gamma C_{ox}}{2\sqrt{\psi_s}} \tag{11.28}$$

We know that when $V_{GS} = V_{th}$, $\psi_s = 2\phi_B - V_{BS}$. If the MOSFET is biased at a gate voltage V_{GS} which is slightly less than V_{th} by ΔV_{GS}, the corresponding surface potential ψ_s will be less than $2\phi_B - V_{BS}$ by $\Delta\psi_s$ (say). Now we may write

$$m = \frac{\partial V_{GS}}{\partial\psi_s} = \frac{\Delta V_{GS}}{\Delta\psi_s} = \frac{V_{GS} - V_{th}}{\psi_s - 2\phi_B + V_{BS}} \tag{11.29}$$

Therefore,

$$\psi_s - 2\phi_B + V_{BS} = \frac{V_{GS} - V_{th}}{m} \tag{11.30}$$

Now substituting Eqs. (11.28) and (11.30) in Eq. (11.25), we have

$$I_D = \frac{Z\mu_n V_T^2 C_D}{L}\exp\left(\frac{V_{GS} - V_{th}}{mV_T}\right)\left[1 - \exp\left(-\frac{V_{DS}}{V_T}\right)\right] \tag{11.31}$$

It can be seen from Eq. (11.31) that the drain current reduces exponentially with decrease in V_{GS} below threshold, and for V_{DS} larger than about $3V_T$, is independent of V_{DS}. The rate at which the current reduces in the subthreshold region also depends on the parameter m. Typically, the value of m lies between 1 and 3. If the interface trap density is significant, the interface trap capacitance C_{it} (= $dQ_{it}/d\psi_s$) is in parallel with the depletion layer capacitance C_D. Therefore, Eq. (11.27) must be modified as

$$m = 1 + \frac{C_D}{C_{ox}} + \frac{C_{it}}{C_{ox}} \tag{11.32}$$

A smooth transition between the above threshold and below threshold characteristics is necessary if the subthreshold current model is to be implemented in a circuit simulator. To achieve this, Eq. (11.31) is modified as

$$I_D = I_{on}\exp\left(\frac{V_{GS} - V_{on}}{mV_T}\right) \tag{11.33}$$

where V_{on} is slightly greater than V_{th} and is given by

$$V_{on} = V_{th} + mV_T \tag{11.34}$$

and I_{on} is the drain current obtained from the above threshold model at $V_{GS} = V_{on}$. Therefore V_{on} is the gate voltage at which the above threshold and below threshold characteristics are combined. This is illustrated using Fig. 11.6. A continuous long channel MOSFET model, available as level 3 model in SPICE, is obtained using Eqs. (10.74), (10.76), and (11.33) as

$$I_D = \begin{cases} I_D = I_{on}\exp\left(\dfrac{V_{GS} - V_{on}}{mV_T}\right) & \text{for } V_{GS} \le V_{on} \\ \dfrac{Z\mu_n C_{ox}}{L}\left[V_{GS} - V_{th} - \alpha\dfrac{V_{DS}}{2}\right]V_{DS} & \text{for } V_{GS} > V_{on};\ V_{DS} \le V_{Dsat} \\ \dfrac{Z\mu_n C_{ox}}{2\alpha L}\left[V_{GS} - V_{th}\right]^2 & \text{for } V_{GS} > V_{on};\ V_{DS} > V_{Dsat} \end{cases} \tag{11.35}$$

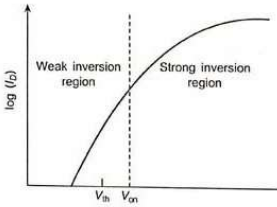


Figure 11.6 Model I_D versus V_{GS} characteristics showing smooth transition above and below V_{th}.

Subthreshold slope: An important parameter for a MOSFET is the *subthreshold slope S*, which is defined as the change in gate voltage required to reduce the subthreshold current by one decade and is expressed in units of Volt (or millivolt) per decade. The value of S should be as small as possible, so that there is a sharp transition between the ON and OFF states of the MOSFET. From the definition of subthreshold slope,

$$S = \frac{dV_{GS}}{d(\log I_D)} = 2.3\left[\frac{dV_{GS}}{d(\ln I_D)}\right] = 2.3\left[\frac{d(\ln I_D)}{dV_{GS}}\right]^{-1} \tag{11.36}$$

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From Eq. (11.33), we have

$$\frac{d(\ln I_D)}{dV_{GS}} = \frac{1}{mV_T} \tag{11.37}$$

Substituting Eq. (11.37) in Eq. (11.36) and from Eq. (11.32), we have

$$S = 2.3mV_T = 2.3V_T\left(1 + \frac{C_D}{C_{ox}} + \frac{C_{it}}{C_{ox}}\right) \tag{11.38}$$

Since the minimum value of m is 1, the minimum possible value of S is 2.3 times V_T, that is, approximately 60 mV per decade at room temperature. Typical values of S lie in the range of 70–100 mV/decade. Except for a small dependence on substrate doping concentration through C_D, S is almost independent of device parameters.

The subthreshold current expressions discussed so far have been developed for long channel MOSFETs. In short channel devices, the surface potential is not constant along the length of the channel and subthreshold currents show strong dependence on V_{DS}. Also, due to the significant influence of the drain voltage, the gate voltage has less control of the surface potential. So a larger change in gate voltage is required to reduce the drain current by the same amount as in a long channel device, implying an increase in subthreshold slope. Figure 11.7 shows the effect of reducing channel length on the subthreshold characteristics of MOSFETs. We see that as the channel length is reduced from 7 μm to 1.5 μm, keeping all other parameters constant, the subthreshold slope increases and also becomes a function of V_{DS}.

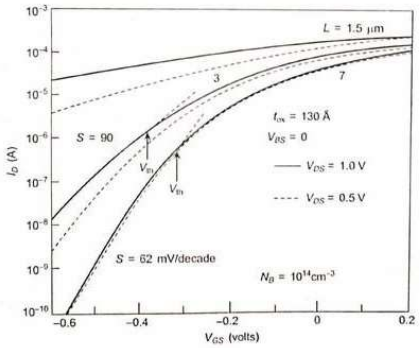


Figure 11.7 Effect of reducing MOSFET channel length on subthreshold slope [1] (source S.M. Sze, copyright © 1981 by John Wiley & Sons, Inc., used by permission).

11.6 MOSFET SCALING

Since the beginning of the use of MOSFETs in ICs, there has been a continuous drive towards reducing the device dimensions. Miniaturization not only results in higher packing density in integrated circuits, but also reduces delays. However, reducing only the channel length and width, without changing the other device parameters, may result in poorer device performance. We have already seen this in an example in Fig. 11.7, where the subthreshold slope increases with reduction in channel length. The major reason for this is that as the channel length is reduced, the longitudinal electric field ($\mathscr{E}_y$) in the direction of current flow increases, and may ultimately become comparable to the transverse electric field ($\mathscr{E}_x$). When this happens, the gradual channel approximation breaks down and the influence of V_{GS} in controlling ψ_s is reduced. The characteristics of the MOSFET also depart from the long channel behaviour, and various short channel effects appear. The design of MOSFETs should therefore be geared towards realizing devices which are dimensionally small, but at the same time behave like long channel devices. The short channel effects can be reduced by keeping $\mathscr{E}_x$ large compared to $\mathscr{E}_y$. This can be achieved by reducing the gate oxide thickness to bring the gate closer to the substrate, thereby increasing $\mathscr{E}_x$. To reduce the effect of V_{DS} on the surface potential ψ_s, the penetration of the source and drain depletion regions under the gate must be reduced. This requires an increase in the substrate doping concentration, as well as reduction in the source and drain junction depths.

A set of guidelines for device scaling is given by the *constant field scaling laws*, where it is proposed that the applied voltages, the gate oxide thickness, and the junction depth be proportionately reduced along with the channel length and width of a MOSFET. Also, the substrate doping concentration is to be increased to reduce the depletion widths. In constant field scaling, since the applied voltages and all the dimensions are reduced by the same factor, the electric fields remain almost constant. The complete constant field scaling laws along with their effect on device and circuit performance are summarized in Table 11.1. Table 11.1 uses α as the scaling factor ($\alpha > 1$) through which all the device dimensions are reduced.

It can also be seen from Table 11.1, that in spite of reduction of channel length, the transverse to longitudinal electric field ratio has remained unaltered. Also the drain depletion width W_D, scales in proportion to the channel length, provided V_{DS} is sufficiently large compared to the built-in potential. This drain depletion width, W_D, is given by

$$W_D = \sqrt{\frac{2\varepsilon_s(V_{DS} + V_{bi})}{qN_A}} \tag{11.39}$$

Thus, the influence of the drain voltage on the surface potential is the same in the reference and the scaled device. All the capacitances reduce by the factor α, since the areas reduce by α^2 and the widths, (that is, t_{ox} in the case of gate capacitance and W_D in the case of drain-to-substrate capacitance), reduce by α. Since the width to length ratio is unaltered and the oxide capacitance per unit area (C_{ox}) increases by α, the drain current reduces by α. Thus, the drain current reduces by the same factor as the drain voltage.

These factors, as given in Table 11.1 have profound impact on circuit performance. The delay involved in charging a capacitance C through a voltage V by a constant current I is given by CV/I. Since C, V, and I all reduce by the factor α in the scaled device, the delay also reduces by the same factor. With both I and V reducing by α, the power dissipation reduces by α^2. The power-

delay product therefore reduces by α^3. It is clear that reduction in the device dimensions using constant field scaling laws results in dramatic improvement in circuit performance. As the device area reduces by α^2, the packing density, that is, the number of devices in a given area, goes up by a factor α^2. In spite of the large increase in packing density, the power dissipation per unit area, which affects the substrate temperature, remains the same.

Table 11.1 Constant field scaling laws and their effect on device and circuit parameters

	MOSFET parameters	Reference device	Scaled device
Scaling laws	Channel length	L	L/α
	Channel width	Z	Z/α
	Gate area, Source/Drain junction area	A	A/α^2
	Gate oxide thickness	t_{ox}	t_{ox}/α
	Source and drain junction depths	x_j	x_j/α
	Substrate doping concentration	N_A	αN_A
	Voltages	V_{GS}, V_{DS}, V_{BS}	$V_{GS}/\alpha, V_{DS}/\alpha$, V_{BS}/α
Effect on device parameters	Electric field ($\mathscr{E}$ = Voltage/distance)	$\mathscr{E}_x$, $\mathscr{E}_y$	$\mathscr{E}_x$, $\mathscr{E}_y$
	Carrier velocity ($v = \mu\mathscr{E}$ or v_{sat})	v	v
	Depletion layer width $W = \sqrt{\frac{2\varepsilon_s V}{qN_A}}$	W	W/α
	Capacitance ($C = A\varepsilon_{ox}/t_{ox}$ or $C = A\varepsilon_S/W$)	C	C/α
	Current ($I_D \propto (Z/L)\, C_{ox}V^2$)	I	I/α
	Channel resistance ($R_{ch} = V/I$)	R_{ch}	R_{ch}
Effect on circuit parameters	Delay time ($t_d \propto CV/I$)	t_d	t_d/α
	Power dissipation ($P = VI$)	P	P/α^2
	Power-delay product (Pt_d)	Pt_d	Pt_d/α^3
	Power density ($P_d = P/A$)	P_d	P_d
	Packing density ($PD \propto 1/A$)	PD	$\alpha^2 PD$

In the present discussion, while considering the effect of scaling on I_D, it has been implicitly assumed that the threshold voltage (V_{th}) scales by the same factor α. However, this is not true. Considering the expression for V_{th} given by Eq. (10.39), we find that the terms V_{FB} and $2\phi_B$ do not change in the scaled device. Since $V_{BS} = 0$ in CMOS circuits, even assuming $(V_{FB} + 2\phi_B)$ to be negligibly small, the threshold voltage only reduces by a factor $\sqrt{\alpha}$ in the scaled device. Also, the subthreshold current does not scale properly. Since the built-in potential of the junctions does not change in the scaled device, it is in fact necessary to increase the substrate doping concentration to higher levels than suggested by the scaling laws in order to avoid short channel effects. These are some of the reasons why the scaling laws cannot strictly be followed.

11.7 NONUNIFORM DOPING IN THE CHANNEL

We have seen in the previous section that for proper scaling, the substrate doping concentration (N_A) has to be increased. However, increasing N_A results in an increase in the body effect parameter (γ) as well as the junction capacitances. To overcome this, in practical devices the substrate doping concentration is kept low, but the doping concentration in the channel region close to the Si–SiO_2 interface is increased by ion implantation. This is known as *threshold tailoring implant*, as the

threshold voltage is finally decided by the implantation parameters. The nature of doping distribution $N_A(x)$ in the implanted channel is shown in Fig. 11.8. It is difficult to model the device using the actual distribution. Therefore, one approximation which is widely used for analytical modelling is to replace the original profile with a step profile as shown in Fig. 11.8 having a step width x_s equal to the sum of the projected range and standard deviation of the implantation. The surface concentration (N_s) in the step profile is given by

$$(N_s - N_B)x_s = \int_0^\infty (N_A(x) - N_B)\,dx = D_I \tag{11.40}$$

where

N_B = background doping concentration before implantation and

D_I = dose of implantation.

For modelling purposes, we shall consider three different cases depending on the thickness x_s discussed as follows:

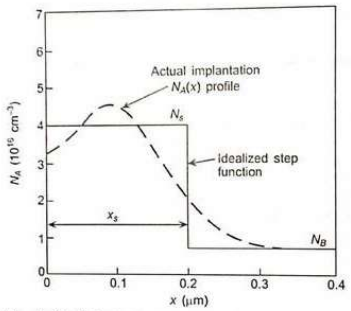


Figure 11.8 Actual threshold tailoring implantation distribution and approximate step distribution curves.

Case 1

x_s is very small. When x_s is very small, the implanted dopants can be considered as a sheet of negative charge (ionized boron atoms) lying very close to the Si–SiO_2 interface. The presence of this charge is equivalent to reducing the fixed oxide charge by an amount qD_I. Therefore, for this case, from Eqs. (10.39) and 10.23), we have

$$V_{th} = V_{FB} + 2\phi_B + \frac{\sqrt{2\varepsilon_s qN_A(2\phi_B - V_{BS})}}{C_{ox}} + \frac{qD_I}{C_{ox}} \tag{11.41}$$

The *threshold voltage shift* (ΔV_{th}) due to ion implantation is therefore given by

$$\Delta V_{th} = V_{th}(D_I) - V_{th}(D_I = 0) = \frac{qD_I}{C_{ox}} \tag{11.42}$$

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Case II

$x_s \geq W_{max}$. When x_s is larger than the maximum depletion width at threshold (W_{max}), then the substrate can be considered to be uniformly doped with a concentration N_s. The threshold voltage can therefore be calculated using Eq. (10.39), where substrate doping concentration is taken to be N_s, which means $N_A = N_s$ and ϕ_B is then calculated corresponding to N_s.

Case III

$W_{max} > x_s$. In this case, the maximum depletion layer width has to be calculated from Poisson's equation for the step doping distribution. We can do that graphically using Fig. 11.9 which shows the electric field variation for this distribution. When the depletion width in the channel reaches its maximum value (W_{max}) at threshold, let the surface potential be ψ_s. There is some controversy regarding the choice of ψ_s in order to evaluate V_{th} for MOSFETs with nonuniformly doped substrates. Some times ($2\phi_B - V_{BS}$) is used [as in Eq. (10.37)], where ϕ_B is calculated from N_B, some have used N_s, while others have used the actual doping concentration at the depletion edge, $N_A(W_{max})$. Furthermore, some others contend that strong inversion occurs when the free carrier density at the surface equals the dopant concentration at the depletion edge, which gives

$$\psi_s = \phi_B + V_T \ln\left[\frac{N_A(W_{max})}{n_i}\right] \tag{11.43}$$

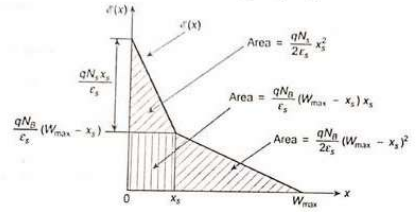


Figure 11.9 Electric field variation for the step implantation distribution.

But it has been shown that the difference in the values of V_{th} obtained for the different values of ψ_s is quite small. Assuming the surface potential at threshold to be $2\phi_B - V_{BS}$ and equating it to the area under the electric field plot in Fig. 11.9, we have

$$2\phi_B - V_{BS} = \frac{qN_B}{2\varepsilon_s}(W_{max} - x_s)^2 + \frac{qN_B}{\varepsilon_s}(W_{max} - x_s)x_s + \frac{qN_s}{2\varepsilon_s}x_s^2$$

$$= \frac{qN_B}{2\varepsilon_s}W_{max}^2 + \frac{qx_s^2}{2\varepsilon_s}(N_s - N_B) \tag{11.44}$$

Therefore, we have

$$W_{max} = \sqrt{\frac{2\varepsilon_s}{qN_B}\left[2\phi_B - V_{BS} - \frac{qx_s^2}{2\varepsilon_s}(N_s - N_B)\right]} = \sqrt{\frac{2\varepsilon_s}{qN_B}\left[2\phi_B - V_{BS} - \frac{qx_sD_I}{2\varepsilon_s}\right]} \tag{11.45}$$

Following Eqs. (11.45) and (10.39), the threshold voltage of the MOSFET is expressed as

$$V_{th} = V_{FB} + 2\phi_B - \frac{Q_B}{C_{ox}} = V_{FB} + 2\phi_B + \frac{q}{C_{ox}}\left[N_BW_{max} + (N_s - N_B)x_s\right]$$

$$= V_{FB} + 2\phi_B + \frac{q}{C_{ox}}\left[N_BW_{max} + D_I\right]$$

$$= V_{FB} + 2\phi_B + \frac{\sqrt{2q\varepsilon_sN_B}}{C_{ox}}\sqrt{2\phi_B - V_{BS} - \frac{qx_sD_I}{2\varepsilon_s}} + \frac{qD_I}{C_{ox}} \tag{11.46}$$

Thus in this case, it can be seen that the change in V_{th} due to ion-implantation is a complicated function of D_I and x_s. Some trial and error may be necessary to find out whether Case II or Case III is applicable for a particular set of implantation conditions, as illustrated in the following example.

Example 11.2

An n-channel MOSFET with n^+ polysilicon gate has a substrate doping concentration of 5×10^{15} cm^{-3} and a gate oxide thickness of 20 nm. A boron implantation is carried out in the channel region for threshold tailoring which can be approximated by a box of width 0.2 μm and a surface concentration of 5×10^{16} cm^{-3}. Neglecting the effect of Q_I, find the values of V_{th} at (a) $V_{BS} = 0$ V and (b) $V_{BS} = -5$ V.

Solution: (a) We first assume that the depletion region lies partly within and partly outside the implantation box (Case III). Therefore, it should be possible to calculate the maximum depletion width using Eq. (11.45). Assuming that the value of ϕ_B refers to the surface concentration N_s, we get

$$\phi_B = 0.026 \ln\left(\frac{5 \times 10^{16}}{1.5 \times 10^{10}}\right) = 0.39 \text{ V}$$

However, substituting this value of ϕ_B in Eq. (11.45) yields a negative value of $\{2\phi_B - V_{BS} - (qx_s^2/2\varepsilon_{Si})(N_s - N_B)\}$ and consequently a non-feasible solution for W_{max} is obtained when $V_{BS} = 0$. This means that our assumption is not justified and the maximum depletion width actually lies completely within the implantation box in this case. To confirm, using $N_s = 5 \times 10^{16}$ cm^{-3}, from Eq. (10.15), we have

$$W_{max} = \sqrt{\frac{2 \times 11.9 \times 8.85 \times 10^{-14} \times 2 \times 0.39}{1.6 \times 10^{-19} \times 5 \times 10^{16}}} = 1.43 \times 10^{-5} \text{ cm} = 0.143 \,\mu\text{m}$$

Therefore, $x_s > W_{max}$ and Case II is valid. We now calculate V_{th} using Eq. (10.39) assuming the doping concentration to be 5×10^{16} cm^{-3}. For this case,

$$V_{FB} = -\left(\frac{E_g}{2} + \phi_B\right) = -0.94 \text{ V}$$

and C_{ox}, as already calculated in Example 11.1 is given by $C_{ox} = 1.72 \times 10^{-7}$ F/cm^2. Therefore, V_{th} is given by

$$V_{th} = -0.94 + 0.78 + \frac{\sqrt{2 \times 11.9 \times 8.854 \times 10^{-14} \times 1.6 \times 10^{-19} \times 5 \times 10^{16} \times 0.78}}{1.72 \times 10^{-7}} = 0.507 \text{ V}$$

(b) For $V_{BS} = -5$ V, the depletion region width is obtained from Eq. (11.45) as

$$W_{max} = \sqrt{\frac{2 \times 11.9 \times 8.854 \times 10^{-14}}{1.6 \times 10^{-19} \times 5 \times 10^{15}}\left[2 \times 0.78 + 5 - \frac{1.6 \times 10^{-19} \times 0.04 \times 10^{-8}}{2 \times 11.9 \times 8.854 \times 10^{-14}}(5 - 0.5) \times 10^{16}\right]}$$

$$= 1.08 \times 10^{-4} \text{ cm}$$

Therefore, $x_s < W_{max}$, and Case III is indeed valid.
Using Eq. (11.46), V_{th} can now be calculated as

$$V_{th} = -0.94 + 0.78 + \frac{1.6 \times 10^{-19}}{1.72 \times 10^{-7}}\left[5 \times 10^{15} \times 1.08 \times 10^{-4} + 4.5 \times 10^{16} \times 0.2 \times 10^{-4}\right] = 1.18 \text{ V}$$

11.8 THRESHOLD VOLTAGE OF SHORT-CHANNEL MOSFETs

As the distance between source and drain of a MOSFET is reduced, the longitudinal electric field (parallel to the Si–SiO_2 interface) may become comparable to the transverse electric field (perpendicular to the Si–SiO_2 interface). In that case, the potential variation in the channel at the threshold condition is not only dependent on the transverse electric field, which is controlled by the gate and substrate bias, but also by the longitudinal electric field, which is controlled by V_{DS}. Unlike in long-channel MOSFETs, where the surface potential is almost constant, there is now a two-dimensional variation of potential in the channel region, and the gradual channel approximation is no longer valid. This effect is well explained with the surface potential plots shown in Fig. 11.10(a). For a long-channel MOSFET (Curve A), the surface potential is practically constant along the channel, showing a broad minimum. For a short-channel MOSFET, on the other hand, the surface potential shows a sharp minimum (Curve B) and the value of the minimum surface potential increases with decreasing channel length for a fixed drain and gate bias. Thus, the proximity of the source and drain reduces potential barrier at the source resulting in an increased channel current. Consequently, the threshold condition for a long-channel MOSFET given by Eq. (10.37), that is, $\psi_s = 2\phi_B - V_{BS}$ must be modified. In a short-channel MOSFET, the *minimum* surface potential at threshold should be $(2\phi_B - V_{BS})$ to ensure that strong inversion exists in the entire channel. Since the minimum surface potential increases due to reduction in channel length, it follows that a lower gate voltage is required in shorter channel MOSFETs to achieve $(\psi_s)_{min} = 2\phi_B - V_{BS}$. Thus, the threshold voltage reduces in the shorter channel device.

One of the important effects in short-channel MOSFETs is *Drain Induced Barrier Lowering* (DIBL). This can also be explained using the potential plots in Fig. 11.10(a). If the drain voltage in a short-channel MOSFET is increased, there is a further increase in the minimum surface potential (Curve C) due to the penetration of field lines from the drain to the source end. Thus we find that DIBL, due to reduction in channel length or increase in drain voltage, reduces the threshold voltage of a MOSFET, since now a lower value of V_{GS} is required to maintain the same potential barrier at the source. The nature of variation of threshold voltage of short-channel MOSFETs with channel

length and drain voltage is shown in Fig. 11.10(b). Reduction in substrate doping concentration aids DIBL as this allows field lines to penetrate from the drain to the source end more easily. On the other hand, reduction in gate oxide thickness reduces DIBL, as the effect of gate voltage becomes more dominating.

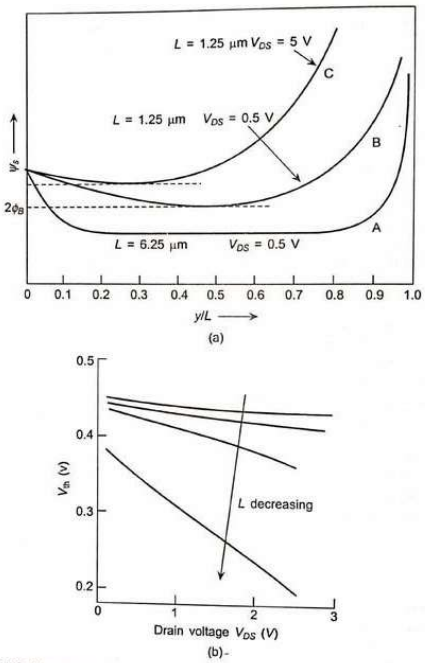


Figure 11.10 (a) Surface potential variation in the MOSFET channel for different channel lengths and V_{DS} values [2] and (b) variation of V_{th} with channel length and V_{DS}.

To obtain the threshold voltages of short-channel MOSFETs, empirical models [3] have widely been used, the most common being

$$V_{th(SC)} = V_{th} - \sigma V_{DS} \tag{11.47}$$

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where V_{th} and $V_{th(SC)}$ are the threshold voltages of long-channel and short-channel MOSFETs respectively and σ is the DIBL factor. In level 3 MOSFET model in the circuit simulator SPICE [4], σ is taken as

$$\sigma = \frac{8.15 \times 10^{-22}\eta}{C_{ox}L^3} \tag{11.48}$$

where η is a parameter used to fit model results with experimental data.

In a more accurate approach to obtain $(V_{th(SC)})$, two-dimensional Poisson's equation may first be solved to obtain the surface potential variation. The threshold voltage is then defined as the gate voltage for which the minimum surface potential is $(2\phi_B - V_{BS})$. Although this procedure leads to very accurate results, it is not suited for circuit simulation, as excessive computation time is required to calculate the threshold voltage for every drain voltage. A particular class of short-channel threshold models, popularly known as *charge sharing models*, has been widely used in circuit simulators. These models are based on the fact that all the field lines terminating in the space charge region under the gate do not originate at the gate, but some of them originate from the source and drain. Thus, as a result of sharing of charge in the depletion region of MOSFETs between the gate and the drain and the source, the amount of charge to be controlled by the gate is reduced. Therefore, only a fraction of the space charge between the source and drain contributes towards the threshold voltage $(V_{th(SC)})$, resulting in lower values of $V_{th(SC)}$ than the long-channel value (V_{th}).

The original charge-sharing model was proposed by Yau [5]. In Yau's model, the channel depletion width W is calculated using a one-dimensional analysis. Assuming that the surface potential at threshold is constant and equal to $(2\phi_B - V_{BS})$, the depletion width at threshold is therefore equal to the maximum depletion width given by

$$W = W_{max} = \sqrt{\frac{2\varepsilon_s(2\phi_B - V_{BS})}{qN_A}} \tag{11.49}$$

In Yau's model it is assumed that the field lines which terminate at the fixed charges inside the trapezoid, shown in Fig. 11.11, originate within the gate, while those which terminate in the hatched region outside the trapezoid, originate in the source or drain diffusion islands. In

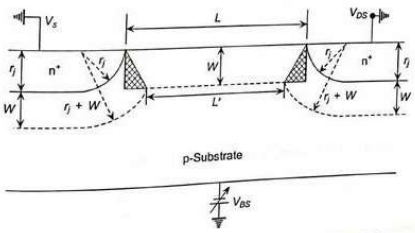


Figure 11.11 Yau's trapezoid for charge sharing in the MOSFET.

Fig. 11.11, cylindrical junctions of depth r_j, and source and drain depletion regions of width W are assumed. Now, from the figure, by geometry we have

$$\frac{L - L'}{2} = \sqrt{(r_j + W)^2 - W^2} - r_j = r_j\left(\sqrt{1 + \frac{2W}{r_j}} - 1\right) \tag{11.50}$$

For short-channel MOSFETs, the long-channel threshold voltage expression given by Eq. (10.39) is modified as

$$\begin{aligned} V_{th(SC)} &= V_{FB} + 2\phi_B + F\frac{\sqrt{2\varepsilon_s qN_A(2\phi_B - V_{BS})}}{C_{ox}} \\ &= V_{FB} + 2\phi_B + F\gamma\sqrt{(2\phi_B - V_{BS})} \\ &= V_{th} - (1 - F)\gamma\sqrt{(2\phi_B - V_{BS})} \end{aligned} \tag{11.51}$$

where

F = charge sharing factor given by the ratio of the area of the trapezoid to the area of the rectangle under the gate and

V_{th} = long-channel threshold voltage value given by Eq. (10.39). The charge sharing factor can therefore be expressed as

$$F = \frac{L + L'}{2L} = 1 - \frac{L - L'}{2L} = 1 - \frac{r_j}{L}\left(\sqrt{1 + \frac{2W}{r_j}} - 1\right) \tag{11.52}$$

As can be seen from Eq. (11.52), F approaches unity for large values of L. Therefore for long-channel devices, the threshold voltage obtained from Eq. (11.51) approaches V_{th}.

Yau's model is, however, an oversimplified version of the true picture. It does not take into account the effect of drain voltage. Also, a fixed value of W is used for all the depletion widths. It does not consider the variation in surface potential and depletion width along the channel. In spite of all these drawbacks, Yau's model still forms the basis of charge sharing models developed later.

A variation of Yau's model is used in SPICE to calculate the threshold voltage of short-channel MOSFETs. The modified trapezoid in this model is shown in Fig. 11.12. In this model, the depletion widths at the source and drain ends are W_S and W_D, where

$$W_S = \sqrt{\frac{2\varepsilon_s(2\phi_B - V_{BS})}{qN_A}} \tag{11.53}$$

$$W_D = \sqrt{\frac{2\varepsilon_s(2\phi_B - V_{BS} + V_{DS})}{qN_A}} \tag{11.54}$$

and the charge sharing factor F given by Eq. (11.52) is modified as

$$F = 1 - \frac{r_j}{2L}\left(\sqrt{1 + \frac{2W_s}{r_j}} + \sqrt{1 + \frac{2W_D}{r_j}} - 2\right) \tag{11.55}$$

The effect of drain voltage is introduced in this model through W_D. The above procedure is used to calculate threshold voltage in the level 2 model in SPICE.

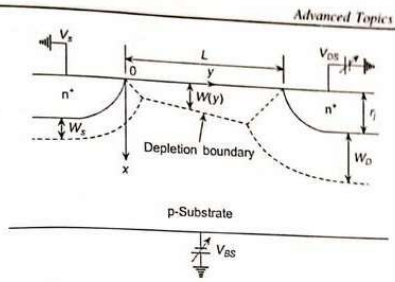


Figure 11.12 Modified trapezoid for charge sharing used in SPICE.

In the level 3 model in SPICE, in addition to the DIBL factor given by Eq. (11.48), a charge sharing factor assuming elliptical source–drain junctions is used. Thus, the threshold voltage expression is

$$V_{th(SC)} = V_{th} - (1 - F)\gamma\sqrt{(2\phi_B - V_{BS})} - \sigma V_{DS} \tag{11.56}$$

11.9 SMALL SIGNAL ANALYSIS

In our discussion of the MOSFETs in the preceding sections, we have only considered dc biasing. Let us now consider the effect of small changes in the applied voltages on the gate and depletion layer charges. Figure 11.13(a) depicts the channel region of a MOSFET under strong inversion.

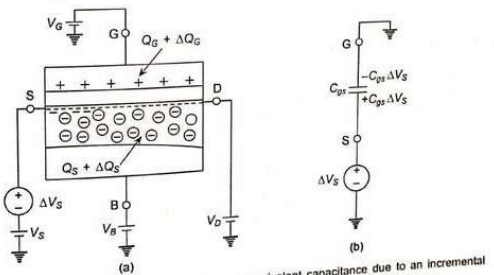


Figure 11.13 (a) MOSFET channel in strong inversion; (b) equivalent capacitance due to an incremental change in V_S, causing an incremental change in gate charge ΔQ_G.

The voltages V_S, V_D, V_G, and V_B are the voltages applied to the source, drain, gate, and body terminals respectively. If now, only the source voltage is changed by an amount ΔV_S as shown in Fig. 11.13(a), the gate charge per unit area will change by an amount ΔQ_G. The relation between ΔQ_G and ΔV_S can be expressed by the small signal equivalent circuit shown in Fig. 11.13(b), where the voltage ΔV_S places a charge $C_{gs}\Delta V_S$ on the bottom plate of the capacitor and $-C_{gs}\Delta V_S$ on the top plate. Hence, if this amount of charge represents ΔQ_G, we have

$$C_{gs} = -\frac{\Delta Q_G}{\Delta V_S} = -\left.\frac{\partial Q_G}{\partial V_S}\right|_{V_G, V_D, V_B = \text{constant}} \tag{11.57}$$

The negative sign signifies that when ΔV_S is positive, ΔQ_G is negative. This is because the potential across the oxide decreases due to ΔV_S and hence the gate charge decreases. It is important to note that C_{gs} is not associated with any real parallel plate capacitor structure. It is simply a fictitious capacitance per unit area which allows the change in gate charge per unit area to be ΔQ_G when the source voltage is changed by ΔV_S.

Analogously, we can define a capacitance per unit area, C_{gd}, associated with a change in the drain voltage and corresponding change in the gate charge. Finally we investigate the effect of changing V_B. This is easier to understand in weak inversion, where Q_n is negligible. Then increasing V_B causes positive charges to flow into the substrate terminal, which will be balanced by an opposite change in Q_G. Thus, the capacitance per unit area associated with a change in the drain and substrate voltage are given by

$$\left.\begin{aligned} C_{gd} &= -\frac{\Delta Q_G}{\Delta V_D} = -\left.\frac{\partial Q_G}{\partial V_D}\right|_{V_G, V_B, V_S = \text{constant}} \\ C_{gb} &= -\frac{\Delta Q_G}{\Delta V_B} = -\left.\frac{\partial Q_G}{\partial V_B}\right|_{V_G, V_D, V_S = \text{constant}} \end{aligned}\right\} \tag{11.58}$$

However, once the device is in strong inversion, the effect of any change in V_B on the gate is screened by the surface charge and hence C_{gb} is zero.

11.9.1 Meyer's Model

We can now draw the Meyer's model [6] of the small signal equivalent circuit for the MOSFET taking into account the capacitances discussed here. This is shown in Fig. 11.14. The three capacitances shown in Fig. 11.14 are defined as

$$\left.\begin{aligned} C'_{GS} &\equiv \frac{\partial Q_{GT}}{\partial V_{GS}}, \quad V_{GD} \text{ and } V_{GB} \text{ constant} \\ C'_{GD} &\equiv \frac{\partial Q_{GT}}{\partial V_{GD}}, \quad V_{GS} \text{ and } V_{GB} \text{ constant} \\ C'_{GB} &\equiv \frac{\partial Q_{GT}}{\partial V_{GB}}, \quad V_{GD} \text{ and } V_{GS} \text{ constant} \end{aligned}\right\} \tag{11.59}$$

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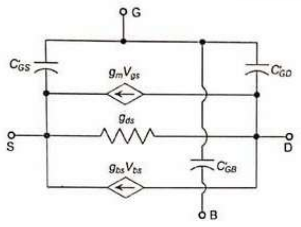


Figure 11.14 Meyer's equivalent MOSFET circuit showing three capacitances.

where

$$Q_{GT} = \text{total gate charge} = Z\int_0^L Q_G dy$$

C'_{GS}, C'_{GD}, and C'_{GB} = total gate-to-source, gate-to-drain, and gate-to-bulk capacitances respectively.

Considering that the MOSFET is operating in the linear region, and noting that $Q_G = -Q_S$, using Eqs. (10.62) and (10.64) we can write

$$Q_{GT} = -Z\int_0^L (Q_n + Q_B)\,dy = ZC_{ox}\int_0^L (V_{GB} - V_{th} - V_{CB})\,dy + ZL\sqrt{2\varepsilon_S qN_A(2\phi_B - V_{BS})} \quad (11.60)$$

The first term in the above expression is the inversion layer charge while the second term is the depletion layer charge.

Considering the square law model, Eqs. (10.46) and (10.64) give

$$I_D = -I_n(y) = -Z\mu_n Q_n(y)\frac{dV_{CB}(y)}{dy} = Z\mu_n C_{ox}\left[V_{GB} - V_{th} - V_{CB}(y)\right]\frac{dV_{CB}}{dy} \quad (11.61)$$

Again, equating the expressions for I_D given by Eqs. (11.61) and (10.58), and noting that $V_{GB} - V_{CB} = V_{GS} - V_{CS}$, it can be seen that

$$\frac{dV_{CB}}{dy} = \frac{dV_{CS}}{dy} = \frac{\left(V_{GS} - V_{th} - \frac{V_{DS}}{2}\right)V_{DS}}{(V_{GS} - V_{th} - V_{CS})L} \quad (11.62)$$

Substituting Eq. (11.62) into Eq. (11.60) and integrating,

$$Q_{GT} = ZLC_{ox}\int_0^{V_{gs}} \frac{(V_{GS} - V_{th} - V_{CS})^2}{\left(V_{GS} - V_{th} - \frac{V_{DS}}{2}\right)V_{DS}}\,dV_{CS} + ZL\sqrt{2\varepsilon_s qN_A(2\phi_B - V_{BS})}$$

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$$= \frac{2C_{ox}ZL}{3\left[2(V_{GS} - V_{th}) - V_{DS}\right]V_{DS}}\left[(V_{GS} - V_{th})^3 - (V_{GS} - V_{th} - V_{DS})^3\right] + ZL\sqrt{2\varepsilon_s qN_A(2\phi_B - V_{BS})}$$

$$= \frac{2}{3}C_{ox}ZL\left[\frac{(V_{GS} - V_{th})^3 - (V_{GS} - V_{th} - V_{DS})^3}{(V_{GS} - V_{th})^2 - (V_{GS} - V_{th} - V_{DS})^2}\right] + ZL\sqrt{2\varepsilon_s qN_A(2\phi_B - V_{BS})} \quad (11.63)$$

The first term in Eq. (11.63) is of the form $k\,(a^3 - b^3)\,(a^2 - b^2)^{-1}$ and can be expressed as $k\,[a + b - ab(a + b)^{-1}]$, where $a = (V_{GS} - V_{th})$; $b = (V_{GS} - V_{th} - V_{DS}) = (V_{GD} - V_{th})$; and $k = [(2C_{ox}ZL/3)]$.

Thus, Q_{GT} can be finally expressed as

$$Q_{GT} = \frac{2}{3}C_{ox}ZL\left[V_{GS} + V_{GD} - 2V_{th} - \frac{(V_{GS} - V_{th})(V_{GD} - V_{th})}{V_{GS} + V_{GD} - 2V_{th}}\right] + ZL\sqrt{2\varepsilon_s qN_A(2\phi_B - V_{BS})} \quad (11.64)$$

The capacitances defined in Eq. (11.59) can now be evaluated as

$$C'_{GS} = \frac{2C_{ox}ZL}{3}\left[1 - \frac{(V_{GD} - V_{th})^2}{(V_{GS} + V_{GD} - 2V_{th})^2}\right] \quad (11.65a)$$

$$C'_{GD} = \frac{2C_{ox}ZL}{3}\left[1 - \frac{(V_{GS} - V_{th})^2}{(V_{GS} + V_{GD} - 2V_{th})^2}\right] \quad (11.65b)$$

As already mentioned, C'_{GB} is zero when the MOSFET is in strong inversion as the change in the substrate voltage will not have any effect on the gate charge. However, in weak inversion, C'_{GB} can be obtained from Eq. (10.34).

It may be noted that in the subthreshold region ($V_{GS} < V_{th}$), C'_{GS} and C'_{GD} are negligible. This is because, varying the source or drain voltage (with respect to gate) can only change the inversion layer charge. However in this regime, the inversion charge itself is very small compared to Q_B, and any change in Q_n is insignificant. The only important capacitance effect comes from C'_{GB}.

In the linear region, assuming $V_{GS} >> V_{DS}$, we can write $V_{GD} \approx V_{GS}$. Therefore, Eq. (11.65), can be rewritten as

$$C'_{GS} = C'_{GD} = \frac{C_{ox}ZL}{2} \quad (11.66)$$

At the onset of saturation, $V_{DS} = V_{GS} - V_{th}$, so that $V_{GD} = V_{th}$. Then Eq. (11.65) reduces to

$$\left.\begin{aligned} C'_{GS} &= \frac{2}{3}C_{ox}ZL \\ C'_{GD} &= 0 \end{aligned}\right\} \quad (11.67)$$

In saturation, since changing drain voltage has virtually no effect on Q_{GT}, $C'_{GD} = 0$. For similar reason, C'_{GS} also remains constant at the value given by Eq. (11.67) in the saturation region. Figure 11.15 shows the variation of the three capacitances as a function of V_{GS}.

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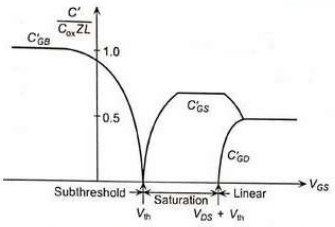


Figure 11.15 Variation of C_{GB}, C_{GS}, and C_{GD} with V_{GS} from Meyer's model.

The simple Meyer's model may not be adequate in some situations. Also, it may give rise to large errors when used for transient analysis. More rigorous models such as Ward's model [7], may become necessary to use in such situations.

11.9.2 Small Signal Equivalent Circuit of MOSFET Amplifier

Let us now consider the small signal equivalent circuit of a MOSFET shown in Fig. 11.16. The small signal drain current can be expressed as

$$i_d = g_m v_{gs} + g_{ds} v_{ds} + g_{bs} v_{bs} \quad (11.68)$$

where g_m, g_{ds}, and g_{bs} are the transconductance, drain conductance, and substrate transconductance of the MOSFET respectively, and are given by the following expressions

$$\left.\begin{aligned} g_m &\equiv \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{ds}, V_{bs} = \text{constant}} \\ g_{ds} &\equiv \left.\frac{\partial I_D}{\partial V_{DS}}\right|_{V_{gs}, V_{bs} = \text{constant}} \\ g_{bs} &\equiv \left.\frac{\partial I_D}{\partial V_{BS}}\right|_{V_{gs}, V_{ds} = \text{constant}} \end{aligned}\right\} \quad (11.69)$$

Here, v_{gs}, v_{ds}, and v_{bs} are the small signal gate-to-source, drain-to-source, and bulk-to-source voltages respectively. Let us consider a MOSFET amplifier circuit shown in Fig. 11.16(a), the small signal equivalent circuit of which is shown in Fig. 11.16(b). Assuming that the MOSFET is biased in the saturation region, $C'_{GD} = C'_{GB} = 0$. From Fig.11.16(b), we see that $i_g = j\omega C_{Gs} v_{gs}$, and $i_d = g_m v_{gs}$. Therefore, the current gain is given by

$$A_i = \frac{i_d}{i_g} = \frac{g_m}{\omega C_{GS}} \quad (11.70)$$

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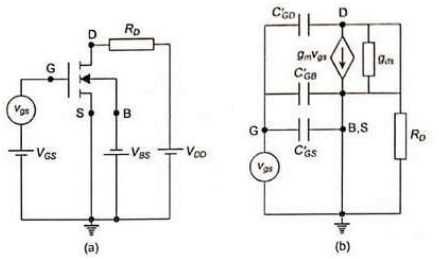


Figure 11.16 (a) A MOSFET amplifier circuit; (b) small signal equivalent circuit for the amplifier.

Now, in the saturation region, from Eq. (10.61), we get

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \frac{Z\mu_n C_{ox}}{L}(V_{GS} - V_{th}) \quad (11.71)$$

Substituting the expressions for C'_{GS} and g_m from Eqs. (11.67) and (11.71) respectively in Eq. (11.70), we obtain

$$A_i = \frac{3\mu_n(V_{GS} - V_{th})}{2\omega L^2} \quad (11.72)$$

We can now define a *cut-off frequency* (f_T) for this amplifier as the frequency at which the current gain becomes unity. From Eq. (11.72), we see that

$$f_T = \frac{3\mu_n(V_{GS} - V_{th})}{4\pi L^2} \quad (11.73)$$

Equation (11.73) illustrates that f_T is inversely proportional to L^2, thus underlining the importance of reduced channel length in modern VLSI technology. This dependence is due to the increase in capacitance and reduction in drain current when L is increased. On the other hand, both drain current and capacitance increase proportionately when Z is increased. Hence, we find that f_T is independent of the channel width.

11.10 OTHER MOSFET CONFIGURATIONS

So far we have discussed the operation of the MOSFETs having the basic structure given in Fig. 10.1. However, other MOSFET configurations are used for enhanced performance or for special applications. Some of these devices are detailed in this section.

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11.10.1 SOI MOSFET

In modern ULSI circuits, power dissipation has become a matter of very serious concern. In order to keep the power dissipated by the whole chip within a reasonable limit, the power dissipation per transistor must be reduced as the number of transistors per chip keeps on increasing. When a complementary MOS (CMOS) inverter circuit switches at a frequency f per second, the power dissipated by it, called the *active power* (P_{active}), can be expressed as

$$P_{active} \propto C_L V_{DD}^2 f \tag{11.74}$$

where

C_L = load capacitance and

V_{DD} = power supply voltage.

On the other hand, in the steady state, the *standby power* ($P_{standby}$) dissipated by the circuit is due to the subthreshold current of the transistor which is 'off'. Therefore, substituting $V_{GS} = 0$ and $V_{DS} >> V_T$ in Eq. (11.31), we have

$$P_{standby} \propto \exp\left[-\frac{V_{th}}{mV_T}\right] \tag{11.75}$$

where

m = subthreshold current parameter given by Eq. (11.32).

For a circuit operating at high frequency, f is large. In order to reduce P_{active}, either V_{DD} or C_L (or both) has to be reduced. However, if V_{DD} is reduced, V_{th} also needs to be scaled down, and consequently $P_{standby}$ increases. Thus, for effective reduction of power dissipation, the load capacitance, which is dependent on the input capacitance of the load transistors must be reduced.

In this context, *Silicon-on-Insulator* (SOI) MOSFET has gained a lot of importance. The basic structure of an SOI MOSFET is depicted in Fig. 11.17. The device is fabricated on a thin film of single crystal silicon, which is separated from the silicon substrate by an oxide layer. Since the source and drain junctions extend all the way to this oxide, the junction capacitance is very low. Also, because of the oxide isolation between the active region and the substrate, parasitic capacitance is minimal. It has been shown [8] that reduction in power dissipation by 66% can be achieved using the

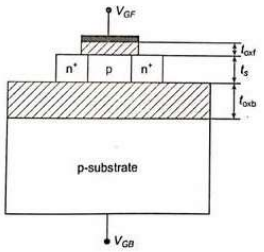


Figure 11.17 Basic configuration of SOI MOSFET.

SOI MOSFET compared to the conventional bulk MOSFETs. Alternately, for the same power dissipation, up to 35% improvement in the operating frequency can be obtained. In addition, SOI MOSFETs are less sensitive to radiation damage and hence, find wide use in space applications. Despite their many advantages, SOI MOSFETs have found application only in niche areas due to the prohibitive cost of the SOI substrates. However, with major strides made in technology, the cost has come down and SOI MOSFETs now find more acceptance in mainstream circuit applications.

An SOI MOSFET can be viewed as a combination of two MOS structures—the front gate, (front gate oxide (FOX) and silicon film constitute the front MOS) and the back gate, (buried oxide (BOX) and silicon film constitute the back MOS). Accordingly, as shown in Fig. 11.17, the SOI MOSFET is provided with two gates, one at the front and the other at the back. Depending on the thickness of the silicon film (which houses the actual device), SOI MOSFETs can be classified into two categories, namely (i) *partially depleted* (PD) and (ii) *fully depleted* (FD). In a PD structure, the sum of the maximum depletion region widths due to front and back MOS structures (W_{max}) is less than the thickness of the silicon film (t_s). By contrast, in a FD structure, the silicon film is fully depleted. In a fully depleted SOI MOSFET, there is charge coupling between the front and back gate and the threshold voltage of the front MOS depends on the back gate bias and vice versa.

Figure 11.18 shows the *I–V* characteristics of a partially depleted SOI MOSFET [9]. It can be seen that for large values of V_{DS}, the drain current does not really saturate but shows sudden periodic increases (commonly known as kinks). This is because, as V_{DS} is increased, some impact ionization may take place at the drain end as already mentioned in section 11.4. In a bulk MOSFET, the holes generated by this process, flow out of the substrate contact. However, in a SOI MOSFET, as there is no substrate contact, these holes accumulate near the source. This can effectively change the potential between the source and the floating silicon film. Consequently, V_{th} reduces and I_D increases, resulting in the *kink effect*. Also, this reduction in V_{th} at high values of V_{DS} increases the 'off current' (subthreshold current) of the PD MOSFET.

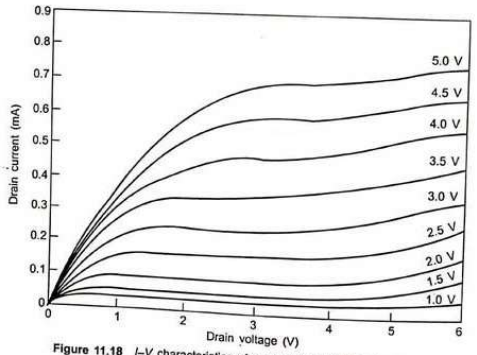


Figure 11.18 *I–V* characteristics of a PD SOI MOSFET showing "kinks".

Providing a contact to the silicon film can eliminate the kinks in the *I–V* characteristics of the PD MOSFET. However, technologically that is not very easy. In order to fully utilize the potential of SOI MOSFET, the FD structure, which uses a thinner silicon film, is preferred. Since the silicon film is completely depleted, there is no floating substrate in the FD SOI MOSFET. Thus the effects that arise due to majority carrier storage are much less and kink effect is reduced. Also, there is an improvement in subthreshold slope. This is because, in an FD structure, the depletion region width is comprised of three components—t_s, t_{BOX}, and W_{sub}, the depletion region in the bulk substrate. Hence, the effective value of the depletion capacitance C_D decreases and the subthreshold slope improves. Figure 11.19 shows the values of the subthreshold slope as a function of the silicon film thickness for SOI devices [10]. It can be verified from this figure that as t_s increases and the device makes a transition from FD to PD, the subthreshold slope degrades significantly. Because of their better subthreshold slope, circuits using FD SOI MOSFETs register lower *standby power* compared to the PD devices.

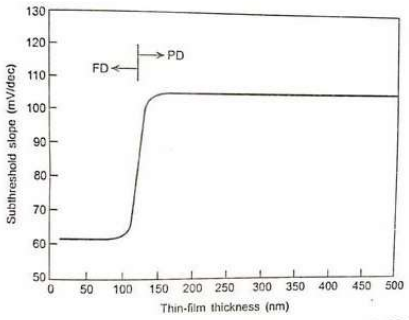


Figure 11.19 Subthreshold slope as a function of silicon film thickness in SOI MOSFET.

11.10.2 Buried Channel MOSFET

It has already been discussed in section 11.7 that the threshold voltage of a MOSFET can be modified by carrying out ion-implantation in the channel region (usually known as threshold tailoring implant). When the type of the implanted impurity is opposite to that of the substrate, a buried channel may be formed. The basic structure of a **buried channel MOSFET** is shown in Fig. 11.20, where arsenic implantation has been carried out in a p-type substrate. As can be seen from Fig. 11.20, there are two depletion regions associated with the buried channel device. One is formed around the n-p junction at $x = x_j$. The other is formed at the oxide–Si interface. When V_{GS} is positive, the n-type silicon surface is in accumulation and there exists a conducting channel between the source and the drain. As V_{GS} is reduced, the depletion region at the oxide–Si interface begins to form.

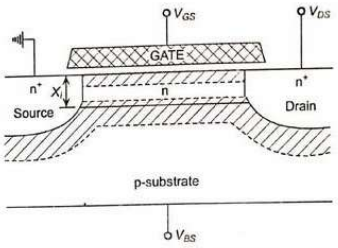


Figure 11.20 Basic structure of a buried channel MOSFET.

The conducting channel lies in between the two depletion regions instead of at the semiconductor surface as in a conventional MOSFET, and hence the name "buried channel". If a sizable current still flows in the device even when V_{GS} is zero, it is called a *normally on* buried channel MOSFET. However, as V_{GS} is reduced further, eventually at a particular value of V_{GS}, called the *pinch-off voltage* (V_P), the two depletion regions merge and pinch off the conducting channel. This is analogous to the operation of a JFET or a MESFET. Depending on the dose and junction depth of the arsenic implantation, V_P can be tailored and be even made zero. A pinch-off voltage $V_P = 0$ signifies that the two depletion regions merge when $V_{GS} = 0$ and the channel is pinched off. In this case, the device is a *normally off* buried channel device. Figure 11.21 shows the output characteristics of a normally on buried channel MOSFET.

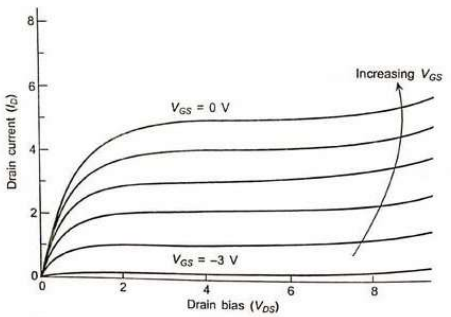


Figure 11.21 Output *I–V* characteristics of a buried channel MOSFET.

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PROBLEMS

P11.1 (a) A n^+ poly-gate n-channel MOSFET has a substrate doping concentration of $N_A = 10^{16}$ cm^{-3}, a gate oxide thickness of 200 Å, and a fixed oxide charge density $Q_f/q = 5 \times 10^{10}$ cm^{-2}. (Neglect Q_{it}). The above device has channel length $L = 1$ µm, width $Z = 5$ µm, electron mobility $\mu_n = 800$ cm^2/Vs and $v_{sat} = 10^7$ cm/s.

(a) Calculate the values of V_{Dsat} and I_{Dsat} at $V_{GS} = 2$ V and $V_{BS} = 0$ V assuming

(i) SPICE level 3 model without considering velocity saturation of electrons in the channel.

(ii) SPICE level 3 model taking into account the effect of velocity saturation.

(iii) SPICE level 4 model assuming field dependent mobility.

(b) Comment on the differences (if any) between the values obtained for the three cases and discuss the reasons.

(c) Calculate the subthreshold slope and drain current of the above MOSFET at $V_{GS} = V_{th}$.

P11.2 If a threshold tailoring boron implant for the above MOSFET is carried out, such that the implanted region can be represented by a box of uniform doping concentration $N_A = 4 \times 10^{16}$ cm^{-3} and thickness $x_s = 0.1$ µm, what will be its new threshold voltage for $V_{BS} = 0$?

P11.3 Using Yau's model, find out the threshold voltage of a MOSFET with $N_A = 10^{16}$ cm^{-3}, $r_j = 2$ µm, $t_{ox} = 200$ Å, and channel length $L = 1.0$ µm. Assume $V_{BS} = 0$ and neglect Q_f.

P11.4 Use the modified Yau's model (SPICE level 2 model) to calculate the threshold voltages of n^+ polysilicon gate n-channel MOSFETs with (i) $L = 10$ µm, $V_{DS} = 0.5$ V, (ii) $L = 1$ µm, $V_{DS} = 0.5$ V, (iii) $L = 10$ µm, $V_{DS} = 5$ V, and (iv) $L = 1$ µm, $V_{DS} = 5$ V. In these devices, assume $N_A = 10^{16}$ cm^{-3}, $r_j = 1$ µm, $t_{ox} = 500$ Å, $V_{BS} = 0$ V, $V_{bi} = 0.8$ V for source and drain junctions and take the oxide and interface charges to be negligibly small.

P11.5 A buried channel MOSFET is fabricated by implanting phosphorus ions in the original substrate of the MOSFET in P.11.2. If the implanted region can be represented by a box of uniform doping concentration $N_D = 8 \times 10^{16}$ cm^{-3} and thickness $x_i = 0.1$ µm, what will be its pinch-off voltage (the voltage at which the channel just ceases to exist) for $V_{BS} = 0$?

REFERENCES AND SUGGESTED FURTHER READING

[1] Sze, S.M., *Physics of Semiconductor Devices*, 2nd ed., Wiley, N.Y., 1981.

[2] Troutman, R.R., VLSI limitations from drain induced barrier lowering, *IEEE Trans. Electron Devices*, Vol. ED-26, pp. 461–468, 1979.

[3] Arora, N., *MOSFET Models for VLSI Circuit Simulation—Theory and Practice*, Springer-Verlag, Wien, 1993.

[4] Antognetti, P. and G. Massobrio, *Semiconductor Modelling with SPICE*, McGraw Hill Book Co., Singapore, 1988.

[5] Yau, L.D., A simple theory to predict the threshold voltage of short-channel IGFET's, *Solid State Electronics*, Vol. 17, pp. 1059–1063, 1974.

[6] Meyer, J.E., MOS models and circuit simulation, *RCA Review*, Vol. 32, pp. 42–63, 1971.

[7] Ward, D.E. and R.W. Dutton, A charge-oriented model for MOS transistor capacitances, *IEEE Journal Solid State Circuits*, Vol. 13, pp. 703–707, 1978.

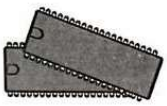
[8] Geppert, L., Solid state semiconductors, [1999] technology analysis and forecast, *IEEE Spectrum*, Volume: 36 Issue: 1, pp. 52–56, Jan 1999.

[9] Chen, S.S. and J.B. Kuo, An analytical CAD kink effect model of partially depleted SOI NMOS devices operating in strong inversion, *Solid State Electronics*, Vol. 41, pp. 447–458, 1997.

[10] Colinge, J.P., Subthreshold slope of thin film SOI MOSFETs, *IEEE Electronic Device Letters*, Vol. 7, pp. 244–246, 1986.

[11] Taur, Y. and T.H. Ning, *Fundamentals of Modern VLSI Devices*, Cambridge University Press, U.K., 1998.

[12] Ong, D.G., *Modern MOS Technology: Processes, Devices and Design*, McGraw Hill Book Company, Singapore, 1984.



Appendix I

Crystal Structure of Silicon

There exist a variety of crystal structures in nature. The simplest crystal structure is a *cubic* lattice where the atoms are located at the eight corners of a cube. In a *face-centred cubic* (FCC) lattice, in addition to these corner atoms, atoms are also located at the centre of the six faces of the cube. In a *zinc blende* crystal lattice, there exist two interpenetrating face centred cubic (FCC) sublattices with one atom of the second sublattice located at 1/4th of the distance along a major diagonal of the first sublattice. GaAs and InP have *zinc blende* crystal structure where one sublattice is made of the group III element (Ga or In) and the other sublattice is made of group V element (As or P). The *diamond* lattice is a degenerate form of the *zinc blende* structure with identical atoms in each sublattice. Silicon belongs to this category of diamond lattice.

A unit cell in the zinc blende crystal structure is shown in Fig. A-1. From this figure, the coordinates of the atoms are found to be

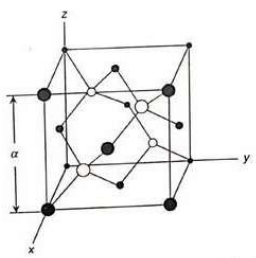


Figure A-1 Zinc blende crystal lattice; black circles represent atoms from the first sublattice and the outlined circle represent atoms from the second sublattice. (Source: S.K. Gandhi, copyright © 1983 by John Wiley & Sons, Inc. Used by permission).

For corner atoms: (0,0,0), (0,1,0), (0,0,1), (0,1,1), (1,0,0), (1,0,1), (1,1,0), (1,1,1)

For face centred atoms: $\left(\frac{1}{2},0,\frac{1}{2}\right)$, $\left(\frac{1}{2},1,\frac{1}{2}\right)$, $\left(\frac{1}{2},\frac{1}{2},0\right)$, $\left(\frac{1}{2},\frac{1}{2},1\right)$, $\left(0,\frac{1}{2},\frac{1}{2}\right)$, $\left(1,\frac{1}{2},\frac{1}{2}\right)$

For atoms from the second sublattice: $\left(\frac{1}{4},\frac{1}{4},\frac{1}{4}\right)$, $\left(\frac{1}{4},\frac{3}{4},\frac{3}{4}\right)$, $\left(\frac{3}{4},\frac{1}{4},\frac{3}{4}\right)$, $\left(\frac{3}{4},\frac{3}{4},\frac{1}{4}\right)$

Each atom of the second sublattice forms a tetrahedron with four nearest neighbours belonging to the first sublattice. The distance between two nearest neighbouring atoms is given by

$$a\sqrt{\left(\frac{1}{4}-0\right)^2+\left(\frac{1}{4}-0\right)^2+\left(\frac{1}{4}-0\right)^2}=\frac{\sqrt{3}}{4}a$$

where

a = edge of the cube.

Assuming the atoms to be hard spheres touching each other, the tetrahedral radius r_0 is evaluated as half the distance between neighbouring atoms. That is,

$$r_0 = \frac{\sqrt{3}}{8}a \tag{A1}$$

The tetrahedral radii of the dopant atoms may be different from that of silicon and are therefore expressed as

$$r_{0_{dopant}} = r_{0_{Si}}(1 \pm \varepsilon) \tag{A2}$$

where

ε = misfit factor.

The introduction of a dopant in the silicon lattice introduces a strain and the amount of strain is indicated by the magnitude of the misfit factor. Also, this is an indication of the maximum dopant amount that can be electronically active. The following Table A-1 lists the misfit factors and the tetrahedral radii for some of the common dopants in silicon.

Table A-1 Tetrahedral radii and misfit factors for silicon and some common dopants

	P	As	Sb	B	Si
r_0 (Å)	1.1	1.18	1.36	0.88	1.18
ε	0.068	0	0.153	0.254	

From this table, it can be seen that phosphorus has a smaller misfit factor in silicon than boron. The presence of large amount of boron in silicon lattice causes strain-induced defects leading to considerable crystal damage. Thus, in practice, a maximum carrier concentration of only about 5×10^{19} cm^{-3} can be maintained in boron-doped silicon structures, while an active carrier concentration of 10^{21} cm^{-3} can be easily achieved in phosphorus-doped silicon structures.

Let us now discuss some of the properties of the crystal planes commonly encountered and made use of in silicon-based IC technology. Any plane is defined by the intercepts it makes on the three axes as

$$hx + ky + lz = 1 \tag{A3}$$

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where h, k, and l are the reciprocal of the intercepts the plane makes on the x, y, and z axes respectively. These are also known as *Miller Indices*. A plane can be simply expressed as (hkl). In Fig. A-2, the plane ABCD is given by (100) while BCEF is (010). However, since all of these have the identical atomic structure and hence belong to the family of planes {100}. Figure A-2 also shows (110) and (111) planes as ADEF and DEG respectively. The separation between two adjacent parallel planes in a lattice unit cell is given by

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \tag{A-4}$$

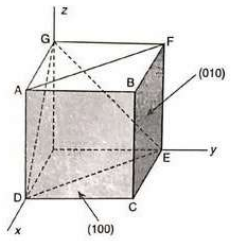


Figure A-2 Major crystal planes in a unit cell.

It can be easily verified from Eq. (A4) that the separation between set of (111) planes is the smallest. Therefore, the crystal growth is easiest while chemical etching is slowest along this plane. Also, the tensile strength and modulus of elasticity are both higher for this plane in silicon and therefore this is the natural cleavage plane for silicon.

The density of atoms in a crystal plane is defined as the number of atoms in a plane divided by plane area. From Fig. A-2, this density can be easily calculated. For example, in (100) plane, the contribution from the four corner atoms is 1/4 each (since each atom is shared by four planes) while the contribution from the face centred atom is 1. Since the plane area is a^2, the density of atoms in (100) plane is $2/a^2$. Similarly, it can be verified that the density of atoms in (111) plane is $4/(\sqrt{3}a^2)$. This larger density of atoms results in a higher fixed oxide charge for (111) silicon surface compared to the (100) plane.

The angle between any two planes ($h_1k_1l_1$ and $h_2k_2l_2$) is given by

$$\cos\theta = \frac{h_1h_2 + l_1l_2 + k_1k_2}{\sqrt{(h_1^2 + k_1^2 + l_1^2)(h_2^2 + l_2^2 + k_2^2)}} \tag{A5}$$

Thus, it can be seen that {111} planes meet {100} planes at an angle of 54.74°. This property of the crystal is widely utilized in etching V-grooves in silicon. If the silicon surface is (100) and anisotropic wet etchant is used, it preferentially etches silicon by exposing {111} planes which

form a V pattern as shown in a cross-sectional view in Fig. A-3. Again {111} planes (in the bulk) meet {110} planes (at the surface) at 90° and this can be exploited to etch deep vertical grooves in silicon. The line describing the intersection of two planes ($h_1k_1l_1$ and $h_2k_2l_2$) is given by $[hkl]$, where

$$\begin{aligned} h &= k_1l_2 - k_2l_1 \\ k &= l_1h_2 - l_2h_1 \\ l &= h_1k_2 - h_2k_1 \end{aligned} \tag{A6}$$

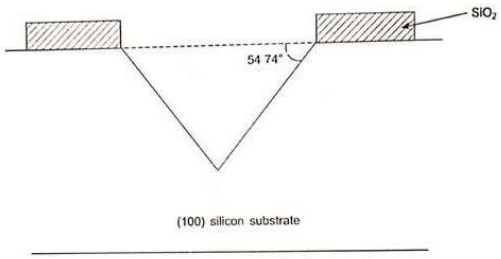


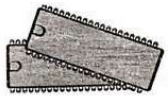
Figure A-3 V-groove etching in (100) silicon showing the exposed {111} side walls.



Appendix II

Properties of Some Important Semiconductors at 300 K

Property	Si	Ge	GaAs	InP
Lattice constant (Å)	5.43	5.64	5.63	5.86
Band gap (eV)	1.12 (I)	0.66 (I)	1.42 (D)	1.35 (D)
Electron mobility (cm^2/eV)	1350	3900	8500	4600
Hole mobility (cm^2/eV)	450	1900	400	150
Dielectric constant	11.9	16.0	13.1	12.4
Effective mass for electrons	0.98	1.64	0.067	0.077
	0.19	0.082		
Effective mass for holes	0.16	0.044	0.082	0.64
	0.49	0.28	0.45	
Effective density of states in conduction band, N_C (cm^{-3})	2.8×10^{19}	1.04×10^{19}	4.7×10^{17}	5.7×10^{17}
Effective density of states in valence band, N_V (cm^{-3})	1.04×10^{19}	6.0×10^{18}	7.0×10^{18}	1.1×10^{19}
Electron affinity, χ(V)	4.05	4.0	4.07	4.38
Intrinsic carrier concentration (cm^{-3})	1.5×10^{10}	2.4×10^{13}	1.79×10^{6}	1.3×10^{7}
Intrinsic resistivity (Ohm-cm)	2.3×10^{5}	47	10^{8}	8.6×10^{7}
Breakdown field (V/cm)	$\sim 3 \times 10^{5}$	$\sim 10^{5}$	$\sim 4 \times 10^{5}$	$\sim 5 \times 10^{5}$

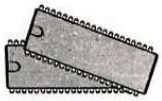


Appendix III

Properties of Some Important Dielectric Materials at 300 K

Properties	SiO_2	Si_3N_4
Density (gm/cm^3)	2.2	3.1
Refractive index	1.46	2.05
Dielectric constant	3.9	7.5
Band gap (eV)	9	~5
Breakdown field strength (V/cm)	10^7	10^7
DC resistivity (Ohm-cm)	$>10^{14}$	$\sim 10^{14}$

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Appendix IV

Values of Some Physical Constants

Quantity	Symbol	Value
Boltzmann constant	k	1.38066×10^{-23} J/K
Electronic charge	q	1.6×10^{-19} C
Planck's constant	h	6.62617×10^{-34} Js
Permittivity in vacuum	ε_0	8.854×10^{-14} F/cm
Speed of light in vacuum	c	3×10^{10} cm/s
Thermal voltage at 300 K	V_T	0.0259 V
Electron rest mass	m_0	9.1×10^{-31} kg
Electron volt	eV	1 eV = 1.6×10^{-19} J
Angstrom unit	Å	1 Å = 0.1 nm = 10^{-4} μm = 10^{-8} cm = 10^{-10} m



Appendix V

List of Symbols

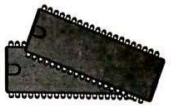
Symbols	Description	Unit
a	Lattice Constant	Å
A	Area	cm^2
A^{**}	Richardson's Constant	$A/(K^2cm^2)$
c	Speed of light in vacuum	cm/s
C	Capacitance	F
C_J	Depletion capacitance of a p-n junction	F
C_d	Diffusion capacitance of a p-n junction	F
C_{JC}	CB junction capacitance in a BJT	F
C_{JE}	EB junction capacitance in a BJT	F
C_{ox}	Oxide capacitance	F/cm^2
C'_{GS}	Gate-to-source capacitance in a MOSFET	F
C'_{GD}	Gate-to-drain capacitance in a MOSFET	F
C'_{GB}	Gate-to-bulk capacitance in a MOSFET	F
D	Diffusion coefficient	cm^2/s
D_n	Diffusion coefficient for electrons	cm^2/s
D_p	Diffusion coefficient for holes	cm^2/s
D_{it}	Interface trap density	number of states/cm^2.eV
E_A	Acceptor energy level	eV
E_C	Conduction band minima	eV
E_D	Donor energy level	eV
E_V	Valence band maxima	eV
E_g	Energy bandgap	eV
E_F	Fermi energy	eV
E_i	Intrinsic energy level	eV
E_{ion}	Ionization energy	eV
$\mathscr{E}$	Electric field	V/cm
f	Frequency	Hz
$f(E)$	Fermi–Dirac distribution function	
$g(E)$	Density of states	$(cm^3\text{–}eV)^{-1}$

Symbols	Description	Unit
g_{cb}	Output conductance in common-base mode	$(ohm)^{-1}$
g_{ce}	Output conductance in common-emitter mode	$(ohm)^{-1}$
g_m	Transconductance	$(ohm)^{-1}$
G	Generation rate of carriers	number/s
h	Planck constant	Js
$h\nu$	Photon energy	eV
I	Current	A
I_0	Reverse saturation current of the diode	A
I_B	Base current	A
I_{BR}	Base recombination current	A
I_C	Collector current	
I_{CBO}	Leakage current in the reverse biased CB junction with the emitter open	A
I_{CEO}	Leakage current flowing between collector and emitter terminals with the base open	A
I_D	Drain current in FET	A
I_{nE}	Minority carrier electron current component at the EB junction	A
I_{pC}	Minority carrier hole current component at the CB junction	A
I_{pE}	Minority carrier hole current component at the EB junction	A
J	Current density	A/cm^2
J_{dr}	Drift current density	A/cm^2
J_{diff}	Diffusion current density	A/cm^2
J_n	Electron current density	A/cm^2
J_p	Hole current density	A/cm^2
k	Boltzmann constant	J/K
L_n	Diffusion length for electrons	cm
L_p	Diffusion length for holes	cm
L	(Channel) length	cm or μm
m_e^*	Effective mass of electron	kg
m_h^*	Effective mass of hole	kg
m_ε	Mass of free electron	kg
M	Current multiplication factor due to impact ionization	
n	Electron concentration	cm^{-3}
n_i	Intrinsic carrier concentration	cm^{-3}
n_n	Electron concentration in n-type semiconductor	cm^{-3}
n_{n0}	Electron concentration in n-type semiconductor at thermal equilibrium	cm^{-3}
n'_n	Excess electron concentration in n-type semiconductor	cm^{-3}
n_p	Electron concentration in p-type semiconductor	cm^{-3}
n_{p0}	Electron concentration in p-type semiconductor at thermal equilibrium	cm^{-3}
n'_p	Excess electron concentration in n-type semiconductor	cm^{-3}
N	Doping concentration	cm^{-3}
N_A	Acceptor concentration	
N_C	Effective density of states in conduction band	

Symbols	Description	Unit
N_D	Donor concentration	cm^{-3}
N_V	Effective density of states in valence band	cm^{-3}
p	Hole concentration	cm^{-3}
p_n	Hole concentration in n-type semiconductor	cm^{-3}
p_{n0}	Hole concentration in n-type semiconductor at thermal equilibrium	cm^{-3}
p'_n	Excess hole concentration in n-type semiconductor	cm^{-3}
p_p	Hole concentration in p-type semiconductor	
p_{p0}	Hole concentration in p-type semiconductor at thermal equilibrium	cm^{-3}
p'_p	Excess hole concentration in p-type semiconductor	cm^{-3}
q	Electronic charge	C
Q	Charge	C
Q_{it}	Interface-trapped charge	C/cm^2
Q_f	Fixed oxide charge	C/cm^2
Q_m	Mobile ionic charges	C/cm^2
Q_{ot}	Oxide trapped charge	C/cm^2
R	Resistance	Ohm
S	Subthreshold slope	mV/decade
t	Time	s
t_{sd}	Storage delay time	s
T	Temperature	K
v_{th}	Thermal velocity	cm/s
v_d	Drift velocity	cm/s
v_{sat}	Saturation velocity	cm/s
V_{xy}	Voltage between x and y terminals	Volt
V_{bi}	Built-in potential	Volt
V_{BR}	Breakdown voltage	Volt
V_{CEO}	Breakdown voltage in common-emitter mode with base open	Volt
V_{CBO}	Breakdown voltage in common-base mode with emitter open	Volt
V_{th}	Threshold voltage	Volt
V_{FB}	Flat band voltage	Volt
V_T	Thermal voltage	Volt
W	(Depletion layer) width	cm or μm
W_B	(Neutral) base width	cm or μm
W_E	Emitter width	cm or μm
W_C	Collector width	cm or μm
Z	Channel width	cm or μm
α	Common-base current gain of BJT	
α_T	Base transport factor	
β	Common-emitter current gain of BJT	
χ	Electron affinity	V
ε_0	Permittivity in vacuum	F/cm

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Symbols	Description	Unit
ε_{ox}	Dielectric constant of SiO_2	F/cm
ε_s	Dielectric constant of Semiconductor	F/cm
γ	Emitter efficiency	
ϕ_m	Metal work function	Volt
ϕ_s	Semiconductor work function	Volt
μ_n	Mobility of electron	cm^2/Vs
μ_p	Mobility of hole	cm^2/Vs
μ_0	Mobility of electron at low gate voltage	cm^2/Vs
ρ	Resistivity	Ohm-cm
σ	Conductivity	Ohm-cm^{-1}
τ_t	Transit time	s
τ_c	Mean time between collisions	s
τ_n	Electron lifetime	s
τ_p	Hole lifetime	s
τ_{tB}	Base transit time	s
τ_{tB}^F	Forward base transit time	s
τ_F	Forward transit time	s
ψ_{ox}	Band bending in the oxide	Volt
ψ_s	Band bending in silicon	Volt



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